

The Selmer converse for elliptic curves at every prime

OpenAI

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Abstract

We prove the Selmer converse in coranks zero and one for every elliptic curve over $\mathbb{Q}$ and every prime p : if the full p -power Selmer group has $\mathbb{Z}_p$ -corank $r \in \{0, 1\}$, then the analytic and Mordell–Weil ranks both equal r , and the entire Tate–Shafarevich group is finite. As an application at the additive prime 3, we prove that for every prime $\ell \equiv 4, 7, 8 \pmod{9}$, the cubic $X^3 + Y^3 = \ell Z^3$ has analytic and Mordell–Weil rank one and finite Tate–Shafarevich group. In particular, every such ℓ is a sum of two rational cubes.

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1 Introduction

Let $A_0/\mathbb{Q}$ be an elliptic curve and let p be a rational prime. Write $\text{Sha}(A_0/\mathbb{Q})$ for the kernel of localization from the Weil–Châtelet group to the product of the local Weil–Châtelet groups. The

full p -power Selmer group, with local Kummer conditions at every place, fits into the exact sequence

$$0 \longrightarrow A_0(\mathbb{Q}) \otimes \mathbb{Q}_p/\mathbb{Z}_p \longrightarrow \mathrm{Sel}_{p^\infty}(A_0/\mathbb{Q}) \longrightarrow \mathrm{Sha}(A_0/\mathbb{Q})[p^\infty] \longrightarrow 0.$$

Thus its corank measures both the Mordell–Weil rank and any divisible p -primary contribution from Sha. Put

$$s_p(A_0) = \mathrm{corank}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(A_0/\mathbb{Q}), \quad a(A_0) = \mathrm{ord}_{s=1} L(A_0, s).$$

The Selmer converse asks whether this arithmetic invariant at a specified prime determines the analytic order of vanishing. We prove the converse in coranks zero and one without restrictions on the curve or the prime.

Theorem 1.1 (The unrestricted low-corank Selmer converse). *For every elliptic curve $A_0/\mathbb{Q}$, every prime p , and $r \in \{0, 1\}$,*

$$s_p(A_0) = r \implies a(A_0) = \mathrm{rank}_{\mathbb{Z}} A_0(\mathbb{Q}) = r, \quad \# \mathrm{Sha}(A_0/\mathbb{Q}) < \infty.$$

There is no restriction on reduction type, complex multiplication, the residual representation, rational torsion, or rational isogenies.

The finiteness conclusion concerns the entire Tate–Shafarevich group, not just its p -primary part. In particular, the Kummer sequence then shows that all prime-power Selmer coranks agree with the rank determined at p . The theorem gives rank equality and finiteness; it does not evaluate the leading coefficient in the refined Birch–Swinnerton-Dyer formula.

A concrete application comes from Sylvester’s cube-sum problem: which rational primes are sums of two rational cubes? Section 10 applies the theorem at $p = 3$ to show that, for every prime $\ell \equiv 4, 7, 8 \pmod{9}$, the cubic $X^3 + Y^3 = \ell Z^3$ has analytic and Mordell–Weil rank one and finite Sha. These cases have also been treated by direct Heegner-point methods, including recent work of Yin and Burungale–Tian [14, 77, 78]. The deduction here illustrates the use of an unrestricted converse at a prime of additive reduction: Satgé’s paired isogeny descents, with their Tate–Shafarevich terms retained, bound the full 3-power Selmer corank by one, and the negative root sign selects analytic rank one.

1.1 The converse problem and its development

The conjecture of Birch and Swinnerton-Dyer grew from computations relating rational points to the behavior of the elliptic L -function near its central point [11]. Coates and Wiles proved an early analytic-to-arithmetic implication for elliptic curves with complex multiplication [30]. Modularity supplies analytic continuation and the functional equation for every elliptic curve over $\mathbb{Q}$ [13]. The height formula of Gross and Zagier and Kolyvagin’s Euler systems give rank equality and finiteness of Sha when the analytic rank is at most one [37, 50]. Kato’s zeta elements provide another fundamental route from a nonzero central value to arithmetic finiteness [43]. The converse reverses this direction: a small Selmer group must force a nonzero central value or derivative. More generally, the p -converse predicts $a(A_0) = s_p(A_0)$; see Keller and Yin [48, Conjecture A].

In rank zero, Skinner–Urban’s main-conjecture theorem gives the Iwasawa divisibility opposite to the one obtained from Kato’s Euler system. It yields the converse at odd good ordinary primes under residual irreducibility and an auxiliary ramification hypothesis [71]. In rank one, the problem becomes the non-torsion of a Heegner point. Skinner uses anticyclotomic Iwasawa theory to deduce this from a one-dimensional Selmer group with nonzero localization at p , under ordinary,

residual, and ramification hypotheses [70, Theorem B]. Wei Zhang proves nonvanishing of derived Heegner classes by level-raising and rank-lowering arguments. Under his ordinary and residual hypotheses, this proves the corank-one converse without the nonzero-localization assumption [80, Theorem 1.4]. These two approaches have different hypotheses; their common task is to recover analytic nonvanishing from arithmetic information that does not itself exhibit a non-torsion Heegner point.

Later work has removed many of the restrictions in these arguments. For good ordinary $p > 3$ with irreducible $A_0[p]$, Burungale–Castella–Skinner remove the earlier auxiliary residual-ramification conditions [16]. Reducible residual representations require a different analysis: Castella–Grossi–Lee–Skinner prove the converse at odd good Eisenstein primes with a local-character restriction, and Keller–Yin remove that restriction and treat multiplicative and potentially good ordinary cases, including rational p -torsion and some additive reduction [24, 48, 49]. At good supersingular primes greater than three, Castella–Wan prove the corank-one converse for semistable curves [22]. Castella treats multiplicative reduction under local and residual hypotheses [21, Theorems 1.1 and 1.3]; the zeta-element constructions of Burungale–Skinner–Tian–Wan yield further supersingular rank-zero and ordinary applications [19, Theorems 1.6 and 1.10].

Complex multiplication also permits small-prime results. Burungale–Castella–Skinner–Tian prove the corank-one converse at good ordinary primes, including 2 and 3, when the associated Hecke character has conductor exactly divisible by the different of the CM field [15, Theorem A]. Burungale–Tian prove the rank-zero converse for every CM elliptic curve at every prime [18]. Kriz’s preprint states a low-corank converse at the ramified prime for curves with CM by the full ring of integers, together with its cube-sum application [51, Theorems 1.1 and 1.3].

The present proof uses two distinct results of OpenAI [62]: the unrestricted converse at 2 settles that prime, while the analytic twist-density theorem supplies the auxiliary fields for the odd-prime argument. Their precise statements are recorded in Section 2. The remaining construction keeps the specified odd prime arbitrary. In particular, it must control local conditions even when the elliptic curve is not ordinary and the residual representation is reducible.

1.2 Proof strategy: two incompatible lengths

Fix an odd prime p and suppose $s_p(A_0) \in \{0, 1\}$. The first goal is to produce a non-torsion Heegner point. The density input selects an imaginary quadratic field K and a biquadratic CM field L containing K . The rational Selmer spaces of A_0 and an auxiliary quadratic twist over K are both lines; the latter is generated by a known non-torsion point. Section 2 shows that non-torsion of the conductor-one Heegner trace y implies the desired conclusion by Gross–Zagier and Kolyvagin. Assume, for a contradiction, that y is torsion. We will obtain more Galois extensions than the Selmer hypothesis allows.

The cohomological upper bound. Let $V = T_p A_0 \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$. At a sequence of split auxiliary primes q_i , cyclic p -quotients of growing ring class groups give a tame character ψ with coefficients in

$$R_b = k[t]/(t^b), \quad b \geq 1,$$

where $k/\mathbb{Q}_p$ is finite and $\psi \equiv 1 \pmod{t}$. We form ψ by taking the p -adic ultralimit of the finite-stage characters on sequences of Galois elements over K , and restrict it to L below. Its cohomology is represented by bounded finite-stage cochains whose identities hold to uniformly increasing precision; Section 3 calls these cochains *admissible*. Choose a place v of K above p and the places of L above v . The one-sided Greenberg group $\mathcal{H}_b = H_{\text{Gr}}^1(L, V \otimes \psi)$ allows unrestricted local classes at the chosen

places and requires zero classes at their conjugates, with relaxed conditions at the remaining fixed places.

If the Selmer line of A_0/K has nonzero localization at p , then $\mathcal{H}_b = 0$. Otherwise that line is everywhere locally zero; we call this the strict case. There a Heisenberg commutator modifies the tame direction so that no nonzero base Greenberg class lifts to first order. The exact coefficient sequence then confines every admissible class to the last t -layer and gives $\text{length}_{R_b} \mathcal{H}_b \leq 2$, independently of b . Consequently, with

$$b_* = 1 \quad \text{outside the strict case,} \quad b_* = 3 \quad \text{in the strict case,}$$

the cohomological upper bound is strictly smaller than b_* .

A theta congruence retaining the tame parameter. The weighted Heegner logarithm is a truncated t -series formed from unnormalized sums of local logarithms of ring-class Heegner points, weighted by the finite-stage tame characters. The torsion assumption on y makes its constant coefficient tend to zero. In the strict case, conjugation and the zero localizations give the same conclusion for the next two coefficients. This logarithmic vanishing is distinct from the first-order obstruction used for the upper bound. It is the input to the automorphic construction.

Sections 4–6 construct a holomorphic theta family on a unitary group of signature $(3, 1)$ at both real places of the maximal real subfield of L . The construction uses Weil’s oscillator representation and the theta and period framework of Kudla and Rallis [52, 63, 74]. An algebraic square identity relates its constant term to toric periods, one of which specializes to the Heegner logarithm. The passage from raised CM values through an inverse differential operator to that logarithm follows the method of Bertolini–Darmon–Prasanna [10]; the argument here includes the additional bad-reduction comparisons. Uniform integral moment estimates make the constant terms small, while a separate positive Fourier–Jacobi coefficient has bounded divisibility and prevents the entire theta series from disappearing.

The geometric input to cusp lifting is the ordinary locus of the target PEL variety, not ordinary reduction of A_0 . Using the compactification and ordinary deformation frameworks of Lan and Katz [47, 54, 55], Section 7 lifts the theta series to genuine cusp forms modulo increasing powers of p . Its integral coefficient lattice and normalized Hecke traces preserve the whole truncated tame ring. Evaluating the congruence only at individual characters would discard the nilpotent parameter that measures the required length.

The extension lower bound. The passage from an automorphic congruence to a Galois extension goes back to Ribet and Wiles [64, 75]. Skinner–Urban use congruences on larger unitary groups [71]; the semiordinary $\text{GU}(3, 1)$ construction of Castella–Liu–Wan is a close antecedent allowing a nonordinary two-dimensional source [23, 25]. Here Section 8 constructs four-dimensional characteristic-zero Galois representations from the cusp forms and proves the local identities needed to extract one-sided Greenberg classes. Two normalized parabolic operators are units, although the elliptic representation remains unrestricted at p .

The limiting trace separates into a two-dimensional twist of V and two characters; this is not a decomposition assumed for each cuspidal Galois representation. Section 9 uses the off-diagonal modules of a finite limiting matrix algebra to record extensions between these constituents, in the spirit of the generalized matrix algebras of Bellaïche–Chenevier [7]. The main tasks are to eliminate an unwanted cyclotomic extension quotient between the two characters and to impose the strict local condition on extensions of one character by the two-dimensional constituent. Uniformly many algebra generators reduce each limiting functional to finitely many matrix values, so the resulting

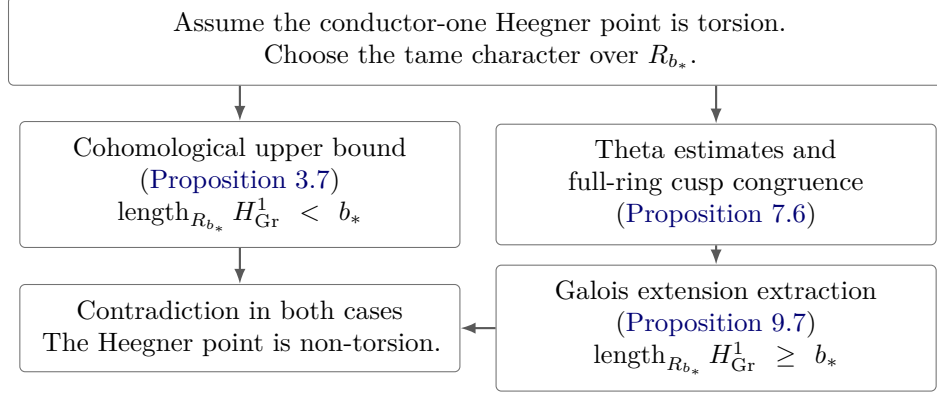


Figure 1: The two bounds concern the same admissible Greenberg module over R_{b_*} . Nonzero localization gives upper bound 0 and $b_* = 1$; the strict case gives upper bound at most 2 and $b_* = 3$.

cochains remain admissible. The local identities hold over the entire Artin ring, and a Fitting ideal calculation gives

$$\text{length}_{R_{b_*}} \mathcal{H}_{b_*} \geq b_*.$$

This contradicts the upper bound and proves that y is non-torsion. Figure 1 summarizes the comparison.

The proof therefore separates into the arithmetic reduction and upper bound in Sections 2–3, the theta and cusp congruence in Sections 4–7, and the Galois construction and extraction in Sections 8–9. Section 10 gives the cube-sum application. The simultaneous transfer in Section 8.1 uses Labesse’s base-change theorem, with its sign property verified for the holomorphic discrete series at both real places. Appendix A records a separate normalized global measure comparison and a change-of-characteristic principle for nonstandard weighted orbital integrals.

2 Arithmetic reduction and conventions

The converse at 2 settles the even prime. For an odd prime, we choose auxiliary quadratic fields so that the problem reduces to proving that one conductor-one Heegner trace is non-torsion. The resulting compact Selmer spaces will also supply the initial data for the tame deformation.

2.1 Selmer groups and established inputs

For a number field M , put $T_p A_0 = \varprojlim_n A_0[p^n]$ and $V_p A_0 = T_p A_0 \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$. For a finite place v of M , the finite local condition is the Kummer image

$$H_f^1(M_v, V_p A_0) = \text{im}(A_0(M_v)_p^\wedge \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \longrightarrow H^1(M_v, V_p A_0)).$$

Here $A_0(M_v)_p^\wedge = \varprojlim_n A_0(M_v)/p^n A_0(M_v)$. The rational compact Selmer group is

$$H_f^1(M, V_p A_0) = \ker \left(H^1(M, V_p A_0) \longrightarrow \prod_v H^1(M_v, V_p A_0) / H_f^1(M_v, V_p A_0) \right).$$

Continuous cohomology and the usual archimedean local conditions are understood. The Kummer exact sequences and local/global duality give

$$\dim_{\mathbb{Q}_p} H_f^1(M, V_p A_0) = \text{corank}_{\mathbb{Z}_p} \text{Sel}_{p^\infty}(A_0/M). \quad (2.1)$$

We use rational coefficients throughout the cohomological argument; integral lattices will be retained whenever a uniform valuation bound is required.

We use the following two established results.

Theorem 2.1 (Established converse at 2). *If $E/\mathbb{Q}$ is any elliptic curve and $\text{corank}_{\mathbb{Z}_2} \text{Sel}_{2^\infty}(E/\mathbb{Q}) = r \in \{0, 1\}$, then*

$$\text{ord}_{s=1} L(E, s) = \text{rank } E(\mathbb{Q}) = r, \quad \text{Sha}(E/\mathbb{Q}) \text{ is finite.}$$

Theorem 2.2 (Established analytic twist-density theorem). *For any elliptic curve $E/\mathbb{Q}$, among nonzero squarefree integers d ordered by $|d|$, the quadratic twists $E^{(d)}$ of analytic rank zero and one each have density $1/2$. In particular the exceptional set of twists of analytic rank at least two has density zero.*

These are OpenAI [62, Theorems 1.1 and 1.2]. Theorem 2.1 treats the prime 2. For auxiliary-field selection we use the density-zero exceptional-set consequence of Theorem 2.2 in fixed positive-density congruence classes. The remainder of the paper supplies the argument at odd primes.

Fix A_0 and $r = s_p(A_0) \in \{0, 1\}$. By Theorem 2.1, we may assume $p > 2$. Write N for the conductor and $V = V_p A_0$. The Weil pairing gives $V \simeq V^*(1)$ with alternating pairing. The p -parity theorem [33, Theorem 1.4] gives the global root number

$$w(A_0) = (-1)^r. \quad (2.2)$$

Modularity and the Gross–Zagier–Kolyvagin theorems will be used in their rank-zero and rank-one forms [13, 37, 50].

2.2 Two auxiliary discriminants

Lemma 2.3. *There are coprime negative odd fundamental discriminants D, D' such that every prime dividing $2Np$ splits in both quadratic fields,*

$$\left(\frac{D}{|D'|} \right) = 1,$$

and each of $A_0^D, A_0^{D'}, A_0^{DD'}$ has the smallest analytic rank permitted by its sign. They may also be chosen so that

$$K = \mathbb{Q}(\sqrt{D}), \quad F = \mathbb{Q}(\sqrt{DD'}), \quad L = \mathbb{Q}(\sqrt{D}, \sqrt{D'})$$

satisfy the following properties: L/F is CM and unramified at every finite prime, L has only the roots of unity $\{\pm 1\}$, and L does not contain the CM field of A_0 if A_0 has CM.

Proof. Negative odd fundamental discriminants in any admissible fixed congruence class have positive density. The splitting conditions at $2Np$ are finitely many such conditions. We can exclude the finitely many unwanted quadratic fields and then apply Theorem 2.2 to choose D for which A_0^D has analytic rank at most one. Its rank is then minimal for its sign.

Keep D fixed. The conditions on D' of being coprime to D , splitting $2Np$, and satisfying the displayed Jacobi-symbol condition again contain admissible congruence classes of positive density. For example, impose $D' \equiv -1 \pmod{|D|}$ and $D' \equiv 1$ at a sufficiently divisible modulus supported on $2Np$. The moduli are coprime, and $|D'| \equiv 1 \pmod{|D|}$ gives the required Jacobi symbol. Exclude the finitely many remaining forbidden fields. Apply the density theorem to both A_0 and A_0^D . The union of their two exceptional sets has density zero, so some D' in the prescribed classes makes both $A_0^{D'}$ and $(A_0^D)^{D'} = A_0^{DD'}$ have analytic rank at most one.

This use of density is unchanged if parameters are described by fundamental discriminants: for squarefree d the corresponding discriminant is d or $4d$, so a zero-density exceptional set remains zero-density. In the odd classes used here the parameter itself is the discriminant.

The three quadratic subfields have discriminants D, D', DD' . Consequently the biquadratic discriminant formula gives

$$|\operatorname{disc}(L)| = D^2(D')^2 = \operatorname{disc}(F)^2.$$

The discriminant formula for L/F therefore has relative discriminant of norm one, proving finite unramifiedness. The two imaginary subfields give the CM involution over the real field F . The exclusions made above remove any extra roots of unity and the possible CM field of A_0 . $\square$

Proposition 2.4. *With these choices,*

$$\dim H_f^1(K, V) = 1, \quad \dim H_f^1(K, V^{DD'}) = 1.$$

The second line localizes nontrivially at both places of K over p . Let $y \in A_0(K)$ be the norm of the conductor-one Heegner point under a modular parametrization taking the cusp ∞ to zero. If y is non-torsion, Theorem 1.1 follows.

Proof. For a fundamental discriminant prime to N , the quadratic-twist root-number formula is $w(A_0^d) = w(A_0)\chi_d(-N)$. The splitting conditions and the signs of D, D' therefore give

$$w(A_0^D) = w(A_0^{D'}) = -(-1)^r, \quad w(A_0^{DD'}) = (-1)^r.$$

By Lemma 2.3, their analytic ranks are $1 - r, 1 - r, r$, respectively. Gross–Zagier–Kolyvagin gives the same Mordell–Weil and p -power Selmer ranks and finite Sha for these three twists. Restriction and corestriction for the quadratic extension $K/\mathbb{Q}$ give

$$H_f^1(K, V) = H_f^1(\mathbb{Q}, V) \oplus H_f^1(\mathbb{Q}, V^D),$$

and the analogous decomposition for $V^{DD'}$. Thus the two dimensions are $r + (1 - r) = 1$ and $r + (1 - r) = 1$.

For the second representation both summands come from twists with known analytic rank at most one. Its Selmer line is therefore spanned by a non-torsion global point on the corresponding twist over K . Such a point has nonzero image under the local logarithm at either place above p . Indeed, on a fixed p -adic completion, a finite-index open subgroup is a formal group on which the logarithm is injective after shrinking; a point with zero logarithm consequently has a torsion multiple in that subgroup, and is torsion. This argument uses no reduction hypothesis.

Every prime dividing N splits in K , so the classical Heegner hypothesis holds. The Gross–Zagier formula identifies the nonzero height of y with simple vanishing of

$$L(A_0/K, s) = L(A_0, s)L(A_0^D, s)$$

at $s = 1$. Since the second factor has order $1 - r$, a non-torsion y implies $a(A_0) = r$. The analytic rank-zero/one theorems then give $\operatorname{rank} A_0(\mathbb{Q}) = r$ and finiteness of the whole $\operatorname{Sha}(A_0/\mathbb{Q})$. $\square$

For the rest of the proof we assume

$$y = 0 \quad \text{in } A_0(K) \otimes \mathbb{Q}. \tag{2.3}$$

We will derive a contradiction. In particular, every subsequent choice is made after A_0, p, D, D' have been fixed. The first Selmer line in Proposition 2.4 may have zero localization, whereas the second cannot. Section 3 distinguishes these two possibilities before deforming the representations.

2.3 Coefficient, character, and measure conventions

Fix embeddings into $\mathbb{C}$ and $\overline{\mathbb{Q}}_p$. Let Σ be the CM type on L induced from one embedding of K , and let Σ_p be the corresponding half of the places of L above p . Complex conjugation is denoted by c . All the fields introduced above are split at p , and the places of K above p will be distinguished according to this choice.

Smooth characters and normalized Satake parameters are evaluated on uniformizers using the reciprocity convention corresponding to geometric Frobenius. Write ϵ for the cyclotomic character. A de Rham weight of ϵ^{-1} is called 1. When a local cohomology computation instead uses arithmetic Frobenius, this will be stated; inversion changes neither the vanishing assertions nor the invertibility requirements used below.

Characters of $U(1)_{L/F} = \ker N_{L/F}$ are pulled back to characters of $L^\times \backslash \mathbb{A}_L^\times$ by $z \mapsto z/z^c$; the same convention applies to $K/\mathbb{Q}$. Choose a character $\mathcal{C}$ of $U(1)_{K/\mathbb{Q}}$ with infinity type $s \mapsto s$ in the chosen embedding and with finite conductor supported on split primes away from $pDD'N$. These primes may be required to split completely in L . The ray-class construction gives such a character: a sufficiently deep split conductor removes the finite unit obstruction, and the prescribed algebraic character on principal ideles then extends across the finite ray-class quotient. We use the same symbol for its norm pullback to $U(1)_{L/F}$ and for the corresponding algebraic p -adic character.

At the distinguished split component above p , its finite smooth part has uniformizer value of valuation -1 with these conventions. This records the algebraic infinity contribution; it is not a claim that the smooth character is unitary in the p -adic norm. All archimedean weights $m \rightarrow +\infty$ below also satisfy $m \rightarrow 0$ p -adically on a fixed residue branch, including the finite torsion orders of the characters raised to the m th power.

Let k be a sufficiently large fixed finite extension of $\mathbb{Q}_p$, with ring of integers $\mathcal{O}$. We enlarge it finitely when fixed algebraic normalization factors require this. Valuation bounds refer to fixed bases and integral lattices. A *uniform* bound may depend on the fixed data and on a fixed truncation degree, but not on the moving prime, its growing p -power quotient, or the growing weights. When coefficient fields for classical eigensystems vary, we retain their embeddings in $\overline{\mathbb{Q}}_p$ and use valuation bounds; no compactness of their union is assumed.

For an algebraic group H over a number field M , write $[H] = H(M) \backslash H(\mathbb{A}_M)$.

Measures on compact source groups and archimedean integration tori have algebraic normalization, and maximal compact subgroups have volume one at almost every finite place. Passing between these measures and invariant norm formulas costs a fixed global scalar. All toric sums called *raw* below are sums without division by the varying ring-class order. This distinction is essential for the estimates at the moving primes.

3 A tame deformation and its Selmer length

We retain the fields and the contradiction assumption of [Section 2](#). In particular, the conductor-one Heegner trace $y \in A_0(K)$ is torsion. Put $T = T_p A_0$ and $V = T \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$. Initially all cohomology in this section has $\mathbb{Q}_p$ coefficients; extension to the fixed finite coefficient field k is made by tensor product. Enlarge the finite, conjugation-stable set S to contain the places over p , the bad places of A_0 , and all the fixed conductors used below. It may be enlarged by a further *fixed* finite set at any stage. In particular, the auxiliary finite characters and tests of [Section 6](#) are chosen independently of the moving primes, and their conductors are included before the final prime sequence is realized. All the constructions below allow this fixed enlargement. Write $G_{K,S}$ for the Galois group unramified outside S .

We need two conclusions for the same tame deformation. The one-sided local condition, full at one p -adic place and zero at its conjugate, will give a cohomology module of length zero or at most two. Under the torsion assumption on y , the first one or three coefficients of the corresponding weighted Heegner logarithms will tend to zero, respectively. These are [Propositions 3.7](#) and [3.9](#). The logarithm estimate supplies the constant-term estimate of [Proposition 5.1](#); the length bound will contradict the extensions extracted in [Section 9](#).

3.1 The undeformed Selmer spaces at the two places above p

The two places v, v^c of K above p have completion $\mathbb{Q}_p$. The alternating Weil pairing

$$e : V \times V \longrightarrow \mathbb{Q}_p(1)$$

and the local invariant give a symmetric perfect pairing on $H^1(K_w, V)$: the sign from interchanging two degree-one cochains cancels the sign of e . The local Kummer line $H_f^1(K_w, V)$ is its own orthogonal complement. Thus, for $w \mid p$, this is a hyperbolic plane. These statements include arbitrary reduction at p : local torsion is finite, the rational Kummer space has dimension one, and the local Euler characteristic gives $\dim H^1(K_w, V) = 2$. At a finite place $w \nmid p$ we instead have

$$H^0(K_w, V) = H^2(K_w, V) = H^1(K_w, V) = 0. \quad (3.1)$$

Indeed $H^0 = 0$ by finiteness of local torsion, $H^2 = 0$ by local duality, and the Euler characteristic away from p is zero. We use the local and global duality sequences in [\[57, Chapter I, Sections 2–4\]](#) and [\[59, Section 8.6\]](#).

For the moment let W denote either V or its twist $V^{DD'}$ over K . Its compact Selmer space has dimension one by [Proposition 2.4](#). Let $H_{\text{rel}}^1(K, W)$ have full conditions at v, v^c , and the usual unramified conditions outside S ; conditions at the other places of S make no difference by [Equation \(3.1\)](#). Let $H_{\text{Gr}}^1(K, W)$ have full condition at v and zero condition at v^c , and let $H_{\text{Gr}}^1(K, W)$ have the opposite conditions. These are the two one-sided Greenberg conditions.

Lemma 3.1. *If the compact Selmer line of W/K has nonzero localization, then $H_{\text{Gr}}^1(K, W) = H_{\text{Gr}}^1(K, W) = 0$. The only remaining possibility is the following strict case in the V summand: $r = 1$, and the compact Selmer space is generated by a class a such that $a^c = a$ and every localization of a is zero. In that case*

$$H_{\text{Gr}}^1(K, V) = \langle a, h \rangle, \quad H_{\text{Gr}}^1(K, V) = \langle a, h^c \rangle, \quad (3.2)$$

where h has nonzero localization at v and zero localization at v^c . The $V^{DD'}$ summand always has zero Greenberg space.

Proof. Let $F_w = H_f^1(K_w, W)$ and let $Q_w = H^1(K_w, W)/F_w$. Poitou–Tate duality identifies the image of

$$H_{\text{rel}}^1(K, W) \longrightarrow Q_v \oplus Q_{v^c}$$

with the annihilator of the localization of the compact Selmer line in $F_v \oplus F_{v^c}$. If that localization is nonzero, both of its coordinates are nonzero: the one-dimensional global space is stable under c , which interchanges the two places. Its annihilator is therefore a line whose two coordinates are both nonzero for every nonzero vector. A class satisfying a zero condition at one place has zero singular image, hence belongs to the compact Selmer line; the zero condition then kills that line as well.

Suppose the compact Selmer line is everywhere locally zero. The same duality sequence makes the singular image all of $Q_v \oplus Q_{v^c}$. The kernel of localization is precisely the compact Selmer line.

Consequently the relaxed local image has dimension two and maps isomorphically to the singular plane. In each local hyperbolic plane choose the isotropic complement E_w of F_w , choosing the two complements compatibly with c . The relaxed local image is the graph of a map

$$E_v \oplus E_{v^c} \longrightarrow F_v \oplus F_{v^c}.$$

Global reciprocity makes the graph isotropic. In mutually dual bases its matrix is therefore skew-symmetric, hence of the form $\begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}$. Conjugation interchanges both pairs of basis vectors and preserves the local pairings. Invariance of the graph therefore changes u to $-u$, so $u = 0$. Lifting the first coordinate line gives h ; conjugating gives h^c and proves [Equation \(3.2\)](#).

The auxiliary summand has a nontorsion global point and nonzero localization, by [Proposition 2.4](#). If $r = 0$, the unique Selmer line of V/K likewise comes from a nontorsion point on the quadratic twist $A_0^D/\mathbb{Q}$, so it cannot be strict. When $r = 1$, the twist by D has zero Selmer space and the line comes from $H_f^1(\mathbb{Q}, V)$, giving $a^c = a$. $\square$

3.2 Primes with a controlled tame logarithm

In the prime construction and cohomological calculations of this section, Galois Frobenius elements are arithmetic Frobenius elements. The following construction gives a tame direction that is trivial at the fixed places and has uniform local bounds. In the strict case we will refine its limiting Frobenius element after determining the obstruction to first-order lifting.

Proposition 3.2. *There are rational primes $q_i \notin S$, split completely in L , distinguished places $\mathfrak{q}_i$ of K above them, cyclic p -quotients Δ_i of the ring class group of conductor q_i , and surjections*

$$\ell_i : G_K \longrightarrow \Delta_i \simeq \mathbb{Z}/p^{n_i}\mathbb{Z}, \quad n_i \longrightarrow \infty,$$

with the following properties.

- (i) ℓ_i is unramified outside the pair above q_i , its restriction to inertia at $\mathfrak{q}_i$ is surjective, $\ell_i^c = -\ell_i$, and $\ell_i|_{G_{K_s}} = 0$ for every $s \in S$.
- (ii) $v_p(q_i - 1) - n_i$ is bounded. At either place above p , the full ring class field of conductor q_i is unramified and its local degree has bounded p -part.
- (iii) Frobenius at $\mathfrak{q}_i$, on all fixed data unramified outside S , tends to an element $\gamma \in G_L$ such that

$$\epsilon(\gamma) = 1, \quad \det(V(\gamma) - 1) \neq 0, \quad \mathcal{C}(\gamma) \text{ is not a root of unity.}$$

Moreover γ fixes all p -power radicals of a fixed, conjugation-stable finite list in $K^\times$ spanning $\mathcal{O}_{K,S}^\times \otimes \mathbb{Q}_p$ under the Kummer map.

Proof. Choose representatives $\mathfrak{a}_j$ for generators of the ideal class group, prime to S . For their orders, their multiplication relations, and the expressions of the primes in S in these generators, choose principal generators. Adjoin generators of the finite-rank S -unit group and their conjugates to this finite list, denoted $\alpha_1, \dots, \alpha_d$. Fixed class-group orders and roots of unity will cost only a fixed index.

For a split prime q , the kernel of the ring class group map to the ordinary ideal class group is the quotient of $(\mathcal{O}_K/q\mathcal{O}_K)^\times$ by diagonal residue units and global units. Its p -part is cyclic, of order $p^{v_p(q-1)}$ up to a fixed factor. Require the α_j to have zero residue logarithm modulo p^{n_i} at the pair over q_i . This is ensured by splitting in their p^{n_i} -radical extensions, together with the corresponding roots of unity. The ring class presentation now allows us to set the logarithms of all $\mathfrak{a}_j$ equal to zero

and to take the surjective logarithm on the residue quotient. Each relation is respected, and the expressions chosen for the primes of S give $\ell_i|_{G_{K_s}} = 0$. Conjugation inverts the residue quotient and the ring class action, giving $\ell_i^c = -\ell_i$.

We verify that imposing the full radical condition leaves room for γ . Over the cyclotomic tower the Kummer Galois group is a closed subgroup of $\mathbb{Z}_p(1)^d$. The joint image of V and $\mathcal{C}$ is a p -adic Lie group. A common Lie quotient with this translation group is abelian, so it kills the derived Lie algebra of the image of V . On the remaining torus Lie algebra, conjugation by an open cyclotomic subgroup is trivial; on the Kummer translations it is multiplication by the cyclotomic character. Any equivariant common quotient is consequently zero. This argument can be made after a fixed finite extension, which does not change Lie dimensions. Thus killing the radicals removes no Lie dimension from the joint image over the cyclotomic tower.

For a non-CM elliptic curve the determinant-one image has Lie algebra $\mathfrak{sl}_2$, by the open image theorem [68]; for a CM curve its identity component contains the norm-one torus, by the CM description of the Tate-module character [67]. In either case the locus $\det(V(g) - 1) \neq 0$ is nonempty and open in every sufficiently small identity neighborhood. The infinity type of $\mathcal{C}$ makes its restriction over the cyclotomic tower nonconstant; its torsion locus is a proper closed subset in a sufficiently small identity neighborhood. The two complements intersect. We may also impose $g \in G_L$ and any fixed open-kernel conditions. This gives γ fixing the full radicals and the cyclotomic tower with the required properties.

It remains to control $v_p(q_i - 1)$, rather than merely make it large. The cyclotomic-radical image is a closed subgroup of

$$\mathbb{Z}_p(1)^d \rtimes \mathbb{Z}_p^\times$$

with open cyclotomic projection. Choose an element in a pro- p subgroup whose cyclotomic projection is nontorsion. Its p^m -th powers tend to the identity; their translation coordinates are divisible by p^{m-C_0} , and the valuation of their cyclotomic coordinate minus one is $m + C_1$, for fixed C_0, C_1 . Multiplying γ by such a power with $m = n_i + C_2$ therefore leaves all radicals trivial modulo the requested precision and makes the cyclotomic valuation exactly $n_i + C$, for a fixed C . Chebotarev applied to a growing sequence of finite quotients realizes these conditions and increasingly accurate approximations on all fixed data. Choosing a place above each prime selects the indicated Frobenius representative. The primes can be taken outside any fixed finite set.

Finally the p -part of the order of the full ring class group is $p^{v_p(q_i-1)+O(1)}$. The kernel of its map to Δ_i therefore has bounded p -part. The Frobenius of every $s \in S$ belongs to this kernel. At p the extension is unramified, so the p -part of the local degree, which is the p -part of the Frobenius order, is bounded. This proves the proposition. $\square$

3.3 Admissible cochains and division by the parameter

Fix a nonprincipal ultrafilter $\mathcal{U}$ on the prime indices. A statement “eventually” in the construction below means on a set in $\mathcal{U}$. For every fixed $b \geq 1$, put

$$R_b = k[t]/(t^b).$$

At finite stages the substitution of $(1+t)^{\ell_i}$ is defined only to finite p -adic precision. There is a constant c_b such that

$$u \equiv u' \pmod{p^n} \implies \binom{u}{j} \equiv \binom{u'}{j} \pmod{p^{n-c_b}} \quad (0 \leq j < b). \quad (3.3)$$

This follows by writing each binomial coefficient as a polynomial with denominator $j!$. Thus truncated substitution of $(1+t)^{\ell_i}$ defines an action modulo $p^{n_i-c_b}$ and sends integral group-ring elements of Δ_i to coefficients with a fixed denominator. No division by $|\Delta_i|$ is involved.

For a sequence (g_i) , let $\ell((g_i))$ be the p -adic ultralimit of $\ell_i(g_i) \in \mathbb{Z}/p^{n_i}\mathbb{Z}$: each fixed residue modulo p^m is well-defined eventually and has a unique ultralimit. This is an additive character on the group of sequences of elements of $G_{K,S_{q_i}}$, modulo equality on a set in $\mathcal{U}$. We can therefore define

$$\psi = (1+t)^\ell, \quad M_b^\eta = V \otimes_{\mathbb{Q}_p} R_b(\psi^\eta), \quad \eta \in \{1, -1\}.$$

Constant representations have their usual compact p -adic ultralimit.

Definition 3.3. An *admissible cochain* with coefficients in M_b^η is a function on sequences of Galois elements which is the coefficientwise ultralimit of continuous finite-stage cochains, in the fixed basis of $V \otimes R_b$, with one denominator p^C independent of i . Finite-stage cochain identities are required modulo precisions $A_i \rightarrow \infty$ along $\mathcal{U}$, with $A_i \leq n_i - c_b$. Equivalently, the limiting identity holds on *every* sequence of arguments. Cohomology is the cohomology of this cochain complex, so coboundaries are also required to be admissible. Local complexes are defined by the chosen embeddings of decomposition groups. The same definition applies over L by restriction and inverse image of the finite-stage Galois groups.

The equivalence in the definition uses boundedness: if an identity failed uniformly modulo p^m on a set in $\mathcal{U}$, choosing one violating tuple at every such index would violate the identity on that sequence. A diagonal choice of precisions then gives $A_i \rightarrow \infty$. This observation also applies when replacing two cochain systems with the same limit. In particular, pointwise agreement only on fixed Galois elements would not suffice.

Lemma 3.4. For constant coefficients V , admissible H^1 is precisely $H^1(G_{K,S}, V)$, and likewise over L . For $\mathbb{Q}_p(1)$ the same assertion holds when unramifiedness at the moving primes is imposed. The fixed local conditions specialize to the usual rational local conditions. Coefficient reduction and multiplication by t give short exact sequences of admissible cochain complexes

$$0 \longrightarrow C^\bullet(M_{b-1}^\eta) \xrightarrow{t} C^\bullet(M_b^\eta) \longrightarrow C^\bullet(V \otimes k) \longrightarrow 0. \quad (3.4)$$

Zero local conditions at p and local Kummer conditions are preserved by the resulting division of a class with zero specialization by t .

Proof. Choose a stable lattice in V . A bounded finite-stage cocycle modulo p^{A_i} has trivial wild inertia at q_i . On tame p -inertia the Frobenius relation implies

$$(V(\text{Fr}_i) - q_i)z_i(\tau) = 0 \pmod{p^{A_i-C}}.$$

The inverses of $V(\text{Fr}_i) - q_i$ have a fixed denominator: these matrices converge to the invertible matrix $V(\gamma) - 1$. The same is true at the conjugate prime and over L . After one fixed loss of precision, the cocycle kills the normal subgroup generated by moving inertia, hence factors through $G_{K,S}$. For each fixed finite coefficient module, there are only finitely many continuous cocycles on $G_{K,S}$: cohomological finiteness follows from the finiteness of extensions with fixed ramification and bounded degree, and the group of coboundaries is finite. Taking the compatible ultralimits of these finite sets yields an actual continuous integral cocycle; then undo the fixed scaling. For cyclotomic coefficients the proof is the same once moving inertia has been required to vanish. Equality of classes is compatible with this procedure because bounded zero-cochains have compact ultralimits.

It also shows that a representative of a given ordinary class can be replaced by a chosen continuous representative, up to a bounded coboundary and uniformly vanishing errors.

Exactness of Equation (3.4) is coefficientwise: lift the finitely many coordinates continuously, and divide a zero constant coefficient by t by shifting coordinates. These operations do not enlarge a uniform denominator except for the fixed truncation loss in Equation (3.3). The character is exactly trivial at the fixed places in S , so there

$$H^1(K_s, M_b^\eta) = H^1(K_s, V \otimes k) \otimes_k R_b, \quad H_f^1(K_s, M_b^\eta) = H_f^1(K_s, V \otimes k) \otimes_k R_b.$$

These are fixed linear subspaces. A strictly zero local class can be made a zero local cocycle with bounded corrections: since $H^0(K_s, V) = 0$, finitely many elements $g_1, \dots, g_u$ make

$$V \longrightarrow V^u, \quad A \longmapsto ((V(g_j) - 1)A)_j$$

injective. A fixed left inverse bounds the denominator of every such correction. Therefore division preserves the strict and Kummer conditions, both in rational cohomology and in their finite-precision representatives. $\square$

Over L we impose full local conditions at the places of Σ_p , zero conditions at their conjugates, and relaxed conditions at the other places of S . This is the notation $H_{\text{Gr}}^1(L, M_b^\eta)$ used below, always with admissible cohomology. Restriction, the two idempotents for $\text{Gal}(L/K)$, and corestriction decompose it into the corresponding spaces for V and $V^{DD'}$ over K . Their denominators are fixed (in fact 2 is a unit). Thus all length computations may be made over K .

3.4 The first-order obstruction

We now work in the strict case of Lemma 3.1. Choose fixed continuous cocycles for a, h, h^c , taking the representatives equivariantly under c by averaging. Cup products below contract with e , and δ denotes the cochain differential.

Let x be a fixed cocycle representing a class in $H_{\text{Gr}}^1(K, V)$. A first-order lift has the form $x + tx'_i$. Comparing the coefficients of t in its cocycle equation gives

$$\delta x'_i = -\eta \ell_i \cup x \tag{3.5}$$

modulo precision tending to infinity. To obstruct a lift with the Greenberg local conditions, we pair this equation with a class having the opposite local conditions. The proof below justifies the bounded corrections needed to use the fixed representative x .

For $d \in H_{\text{Gr}}^1(K, V)$, the product $x \cup d$ vanishes at every place: at p one of the two classes is zero, and away from p use Equation (3.1). The injection of global degree-two cyclotomic cohomology into its localizations (Brauer reciprocity) gives a continuous cochain u on $G_{K,S}$ satisfying

$$\delta u = x \cup d. \tag{3.6}$$

Choose primitives for the finitely many pairs in the two fixed bases, and thereafter use bilinear extension. Each has compact image, hence all these choices have one fixed denominator.

For an element γ as in Proposition 3.2, define

$$z_\gamma(x, d; u) = u(\gamma) - e((V(\gamma) - 1)^{-1}x(\gamma), d(\gamma)), \tag{3.7}$$

$$H_\gamma(x, d) = z_\gamma(x, d; u) - z_\gamma(x^c, d^c; u^c). \tag{3.8}$$

The chosen trivialization of $\mathbb{Q}_p(1)$ makes these scalar expressions; changing it multiplies them by one nonzero scalar.

Lemma 3.5. *The expression H_γ is independent of the primitive and bilinear. If a nonzero class $x \in H_{\text{Gr}}^1(K, V \otimes k)$ is the specialization of an admissible class in $H_{\text{Gr}}^1(K, M_2^\eta)$, then*

$$H_\gamma(x, d) = 0 \quad \text{for every } d \in H_{\text{Gr}}^1(K, V \otimes k). \quad (3.9)$$

The same necessary condition holds for either η .

Proof. We give the finite-precision argument, so that no continuity assertion about the varying Galois groups is implicit. For $\eta = 1$, write a first-order lift as $x + tx'_i$ satisfying Equation (3.5). Indeed, Lemma 3.4 permits subtraction of one bounded global coboundary and replacement of its constant coefficient by the chosen fixed cocycle x . Equality on all sequences makes the error uniformly small. Its linear coefficient remains bounded. The cochain

$$x'_i \cup d - \ell_i \cup u \quad (3.10)$$

is consequently a two-cocycle modulo the reduced precision. Its local invariants at S vanish to increasing precision. At p this follows from opposite strict conditions and $\ell_i = 0$; away from p it follows from the vanishing of rational H^1 and the bounded exponents of the fixed integral local cohomology groups. At places outside Sq_i it is unramified and has zero invariant.

At q_i , the fixed cocycles x, d are unramified and are coboundaries, say $x = \delta A_i$, $d = \delta B_i$, where

$$A_i = (V(\text{Fr}_i) - 1)^{-1}x(\text{Fr}_i), \quad B_i = (V(\text{Fr}_i) - 1)^{-1}d(\text{Fr}_i).$$

These vectors have bounded denominators, because the inverse matrices converge to $(V(\gamma) - 1)^{-1}$. By Equation (3.5), $x'_i - \ell_i \cup A_i$ is a cocycle. Its cup product with d is a coboundary. The invariant of Equation (3.10) therefore equals minus the invariant of

$$\ell_i \cup (u - A_i \cup d). \quad (3.11)$$

The second factor is an unramified cyclotomic one-cocycle locally, since its differential is $x \cup d - (\delta A_i) \cup d = 0$. The tame local pairing evaluates this unramified factor at Frobenius and pairs it with the inertia logarithm. With compatible local invariant conventions, the multiplier at stage i is a unit ν_i modulo the working precision, because the logarithm on tame inertia is surjective. At the conjugate prime the same calculation uses x^c, d^c, u^c and $\ell_i^c = -\ell_i$; functoriality of the local invariant gives the same multiplier ν_i . Choose unit lifts to $\mathbb{Z}_p$ and let $\nu \in \mathbb{Z}_p^\times$ be their compact ultralimit. The global sum of invariants then gives $\nu H_\gamma(x, d) = 0$, hence Equation (3.9). This calculation never divides by $q_i - 1$. We use arithmetic Frobenius in this calculation; replacing it throughout by geometric Frobenius changes the conventions compatibly and not the zero condition. For $\eta = -1$, replace ℓ_i by $-\ell_i$.

Here is the precise bound on all reductions of precision. Let c_{fix} bound the fixed cocycles and their finitely many primitives, and let c_{lift} bound the particular admissible lift. Let c_γ clear both inverses $V(\text{Fr}_i) - 1$ and $V(\text{Fr}_i) - q_i$ eventually. Let c_S bound the fixed left inverses detecting local $H^0 = 0$ and the exponents of integral H^1, H^2 at the fixed places away from p . Finally use the binomial constant c_b of Equation (3.3). All displayed corrections use finitely many sums and products of these objects. A constant $C = C(c_{\text{fix}}, c_{\text{lift}}, c_\gamma, c_S, c_b)$ therefore bounds every loss. If the original cochain identities hold to precision A_i and their replacement accuracies are B_i , make the calculation to any precision

$$N_i \leq \min\{n_i, A_i, B_i\} - C, \quad N_i \longrightarrow \infty. \quad (3.12)$$

The tame pairing above is defined at this precision since $N_i \leq n_i$. The constants may depend on a fixed lift and on b , but never on i, q_i, n_i . Only finitely many fixed primitives were chosen; no bounded contracting homotopy on all cochains is being asserted.

For independence of u , the difference of two primitives is in

$$H^1(G_{K,S}, \mathbb{Q}_p(1)) = \mathcal{O}_{K,S}^\times \otimes_{\mathbb{Z}} \mathbb{Q}_p.$$

The equality is the Kummer sequence and finiteness of the S -class group. The radical list in [Proposition 3.2](#) spans this space, including its conjugate, up to a fixed index. Every such Kummer class evaluates to zero at γ ; a coboundary also evaluates to zero because $\epsilon(\gamma) = 1$. Changing u therefore changes neither term of [Equation \(3.8\)](#). Bilinearity follows from the fixed bilinear choices of primitives. $\square$

Lemma 3.6. *The element γ in [Proposition 3.2](#) can be chosen so that, in the bases of [Equation \(3.2\)](#), the obstruction matrix is*

$$\begin{pmatrix} 0 & \kappa \\ -\kappa & 0 \end{pmatrix}, \quad \kappa \in \mathbb{Q}_p^\times. \quad (3.13)$$

Thus no nonzero base Greenberg class lifts to order two.

Proof. First $H_\gamma(x, d) = H_\gamma(d, x)$ whenever the pairs under consideration have locally zero cup product. To check the sign explicitly, put $q(g) = e(x(g), d(g))$. The cocycle identities give

$$\delta q = d \cup x - x \cup d,$$

so $u + q$ is a primitive for the switched cup product. At a place where $x = \delta A$ and $d = \delta B$, the difference between the two corrected evaluations is $(\epsilon(\gamma) - 1)e(A, B) = 0$. This proves the symmetry for the expression in [Equation \(3.7\)](#), and hence for H . Its definition also gives $H_\gamma(x^c, d^c) = -H_\gamma(x, d)$. Since $a^c = a$ and $p > 2$, these identities give both diagonal zeros in [Equation \(3.13\)](#) and the opposite signs of its off-diagonal entries.

It remains to make $\kappa = H_\gamma(a, h^c)$ nonzero. Set $d^- = h^c - h$. It is anti-invariant, and its cup product with a is everywhere locally zero. Average a primitive to be anti-invariant as well. Then

$$H_\gamma(a, d^-) = 2z_\gamma(a, d^-; u), \quad H_\gamma(a, d^-) = 2H_\gamma(a, h^c).$$

It is enough to make $z_\gamma(a, d^-; u)$ nonzero.

Let $G = G_{K,S}$ and $N_V = \ker(G \rightarrow \mathrm{GL}(V))$. The classes a, d^- are linearly independent: the former is nonzero and invariant under c , whereas the latter is anti-invariant with nonzero local image. The image of G on V contains a central scalar different from one. The usual central-scalar argument gives $H^1(\mathrm{im}(G \rightarrow \mathrm{GL}(V)), V) = 0$: for a cocycle f and a central scalar $z \neq 1$, the cocycle identity is $(z - 1)f(g) = (g - 1)f(z)$, so f is a coboundary. Inflation–restriction thus makes restriction to N_V injective on $H^1(G, V)$. On N_V , both cocycles are additive homomorphisms. Their joint image spans $V \oplus V$: its span is a G -submodule, and absolute irreducibility with $\mathrm{End}_G(V) = \mathbb{Q}_p$ says that a proper submodule projecting onto each factor would impose a scalar linear relation between the two restrictions. That contradicts the preceding injectivity. The absolute irreducibility here follows from open image in the non-CM case; in the CM case it follows because neither K nor L contains the CM field. Passing to the finite-index subgroup over L leaves the span unchanged.

The Weil pairing implies $\epsilon = 1$ on N_V . On this group, [Equation \(3.6\)](#) reads

$$u(gg') = u(g) + u(g') - e(a(g), d^-(g')).$$

Consequently

$$u([g, g']) = -e(a(g), d^-(g')) + e(a(g'), d^-(g)). \quad (3.14)$$

The alternating form on $V \oplus V$ on the right is nonzero and nondegenerate. Since the joint translations span $V \oplus V$, some $g, g' \in N_V \cap G_L$ have a nonzero right-hand side. Write $\sigma = [g, g']$. Both translations vanish on σ , but $u(\sigma) \neq 0$.

Every Kummer character in our radical list is an additive homomorphism on N_V , because $\epsilon = 1$ there. It therefore vanishes on σ , so σ fixes all the radicals. A one-dimensional character such as $\mathcal{C}$ also kills σ . Replacing γ by $\gamma\sigma^m$, for an integer m , thus preserves its V , $\mathcal{C}$, cyclotomic, and radical values. The two translations entering the correction in Equation (3.7) do not change, whereas its primitive value changes by $mu(\sigma)$. Some m makes the result nonzero. If further fixed open conditions are required, first replace σ by a suitable nonzero power; its nonzero primitive value survives. This constructs the required γ . In particular the inverse matrix $V(\gamma) - 1$, and hence its denominator bound, is unchanged by the modification. Applying the prime construction to this element proves Equation (3.13) for both signs of the tame twist. $\square$

In the strict case we henceforth use this refined γ in Proposition 3.2, and q_i, ℓ_i denote the sequence realized from it. With the same ultrafilter $\mathcal{U}$, we henceforth form ℓ, ψ, M_b^n , and the admissible global and local cochain complexes from this reselected sequence. The preserved properties of the prime construction ensure that the specialization, division, and obstruction arguments above still apply. As stipulated at the start of the section, the final realization is made after all fixed conductors and tests have been chosen. Any additional fixed open-kernel conditions can be imposed in the construction and preserved by the commutator modification.

3.5 The exact Artin length bound

Proposition 3.7. *For the primes of Proposition 3.2, chosen in the strict case as in Lemma 3.6, every fixed $b \geq 1$, and either $\eta = 1$ or $\eta = -1$, one has*

$$\text{length}_{R_b} H_{\text{Gr}}^1(L, V \otimes \psi^\eta) \begin{cases} = 0, & \text{outside the strict case,} \\ \leq 2, & \text{in the strict case.} \end{cases} \quad (3.15)$$

In the strict case this module is killed by t and is the image of its two-dimensional base space under multiplication by t^{b-1} . Put

$$b_* = 1 \quad \text{outside the strict case,} \quad b_* = 3 \quad \text{in the strict case.} \quad (3.16)$$

Thus in both cases the length in Equation (3.15) is strictly smaller than b_ .*

Proof. It suffices to work in each of the two summands over K . Let $\mathcal{S}_b^\bullet$ be the cone, shifted by -1 , of the map from the admissible global cochain complex to the local complex at the strict p -place. Its degree-one cohomology is exactly the indicated Greenberg group. Indeed global and local H^0 vanish for V and, by the t -filtration, for every M_b . The coefficient sequence in Equation (3.4) gives a short exact sequence of these cone complexes. The resulting cohomology sequence therefore identifies

$$\ker(H^1(\mathcal{S}_b^\bullet) \longrightarrow H^1(\mathcal{S}_1^\bullet)) = \text{im}(t : H^1(\mathcal{S}_{b-1}^\bullet) \longrightarrow H^1(\mathcal{S}_b^\bullet)). \quad (3.17)$$

There is no flatness assumption on these cohomology groups.

If the base group is zero, induction in Equation (3.17) gives zero at every order. This deals with the auxiliary summand and with both summands outside the strict case. In the strict summand,

a class at any order $b \geq 2$ reduces to an order-two class, whose base specialization is zero by Lemmas 3.5 and 3.6. Equation (3.17), iterated, gives

$$H^1(\mathcal{S}_b^\bullet) = \text{im}(t^{b-1} : H^1(\mathcal{S}_1^\bullet) \longrightarrow H^1(\mathcal{S}_b^\bullet)).$$

This image is killed by t and has k -dimension at most two. Its R_b -length is therefore at most two, as asserted. $\square$

3.6 A uniform local logarithm estimate

The cohomological length bound is now proved. Under the torsion assumption on y , we next establish the weighted Heegner-logarithm vanishing needed for the theta congruence. The argument first gives vanishing of local Kummer classes. The following estimate converts this into a valuation statement, uniformly in the unramified fields of definition that occur at p .

Lemma 3.8. *Fix an integer $d_0 \geq 0$. There is a constant c such that, for every finite unramified extension $E/\mathbb{Q}_p$ with $v_p([E : \mathbb{Q}_p]) \leq d_0$,*

$$\log_{A_0,p}(A_0(E)) \subseteq p^{-c}\mathcal{O}_E. \quad (3.18)$$

If $P \in A_0(E)$ and the Kummer class of $p^C P$ vanishes in $H^1(E, A_0[p^n])$, where $C, n \geq 0$ are integers, then

$$v_p(\log_{A_0,p} P) \geq n - C - c.$$

Proof. On a fixed sufficiently small formal subgroup, the formal logarithm and its inverse are integral, uniformly under unramified base extension. The intervening formal-group quotient has bounded p -exponent. In the Néron special fiber the geometric component group has fixed order, the additive part has bounded p -exponent, and a torus in characteristic p has no p -torsion. For the étale p -power torsion in the abelian part, let α be a unit Frobenius eigenvalue. It is not a root of unity, by its complex absolute value. If $[E : \mathbb{Q}_p] = d$, the exponent is controlled by $v_p(\alpha^d - 1)$. Writing α as its finite torsion part times a principal unit shows

$$v_p(\alpha^d - 1) \leq C_\alpha + v_p(d)$$

whenever the left side is positive. It is therefore bounded for these extensions. A fixed p -power, followed if necessary by an integer prime to p , moves any point into the formal subgroup. Dividing its logarithm by that multiplier proves Equation (3.18), including at bad reduction.

For the precision assertion, the Kummer injection

$$A_0(E)/p^n A_0(E) \hookrightarrow H^1(E, A_0[p^n])$$

gives $p^C P = p^n Q$ for some $Q \in A_0(E)$. Consequently

$$v_p(\log_{A_0,p} P) \geq n - C - c$$

by Equation (3.18). Thus the comparison loses only a fixed precision, without division by the local extension degree. $\square$

3.7 Weighted Heegner traces and their local logarithms

Let H_i/K be the full ring class field of conductor q_i , and let $y_i \in A_0(H_i)$ be the corresponding Heegner point. At the split prime q_i it is the conductor- q_i translate obtained by $n(1/q_i)$ in torus-adapted coordinates, with the opposite sign if the embedding convention requires it. Write $\mathcal{G}_i = \text{Gal}(H_i/K)$, and view ℓ_i also as its quotient character. The sums in the following statement are raw sums over $\mathcal{G}_i$.

Proposition 3.9. *At the distinguished place over p , and for either sign,*

$$\sum_{g \in \mathcal{G}_i} (1+t)^{\pm \ell_i(g)} \log_{A_0, p}(y_i^g) \longrightarrow 0 \pmod{t^{b_*}}. \quad (3.19)$$

More precisely, every coefficient of degree less than b_ has valuation tending to $+\infty$ along $\mathcal{U}$. The coefficients through any other fixed degree have a common lower valuation bound. These assertions hold in a common p -adic algebraic closure, with its fixed extension of the valuation.*

Proof. Choose integral Kummer cocycles for y_i and corestrict with the tautological group-ring coefficients. Corestriction is a sum: its terms have integral coefficients regardless of the size of $\mathcal{G}_i$. Truncate by the substitution in Equation (3.3). We obtain bounded admissible classes with coefficients $M_b^\pm$, for every fixed b , satisfying Kummer conditions at both p -places. The split Heegner norm relation gives augmentation equal to the Kummer class of

$$(a_{q_i} - 2)y.$$

Indeed the $q_i + 1$ cyclic subgroups in the Hecke correspondence comprise two horizontal CM isogenies, corresponding to the two prime ideals over q_i , and the $q_i - 1$ descending isogenies whose target order has conductor q_i . The latter are the orbit over the conductor-one ring class field. Thus the relative trace is the Hecke action minus the two horizontal translates. Tracing further to K turns each horizontal translate into y , and the Hecke eigenvalue on A_0 is a_{q_i} . This proves the displayed relation; see also the correspondence and CM action conventions in [37, Chapter I, Sections 2–3]. Since y is a fixed torsion point, one fixed scalar kills all these augmentations in integral Kummer cohomology. Their rational augmentation is zero.

For $b_* = 1$ this proves the cohomological assertion. In the strict case divide the limiting Kummer trace once by t , as in Lemma 3.4; its first specialization, say z_1 , is an ordinary compact Selmer class of V/K . The cohomology sequence of Equation (3.4) and $H^0(K, V \otimes k) = 0$ make multiplication by t injective on $H^1(M_{b-1}^\pm) \rightarrow H^1(M_b^\pm)$, so this divided class is unique. All local Kummer conditions survive this division because the tame character is trivial at p . The operation merely shifts bounded coordinates after subtraction of a fixed bounded coboundary; no denominator depending on $[H_i : K]$ is introduced.

We verify the conjugation identity with its effect on this coefficient, using the CM coordinates of [37, Chapter II, Section 1]. Write a CM point in fractional-ideal coordinates as

$$x(\mathfrak{a}, \mathfrak{n}) = (\mathbb{C}/\mathfrak{a} \longrightarrow \mathbb{C}/\mathfrak{n}^{-1}\mathfrak{a}), \quad \mathfrak{n}\bar{\mathfrak{n}} = (N).$$

The dual cyclic isogeny gives $w_N x(\mathfrak{a}, \mathfrak{n}) = x(\mathfrak{n}^{-1}\mathfrak{a}, \bar{\mathfrak{n}})$. Consequently

$$x(\mathfrak{a}, \mathfrak{n})^c = w_N x(\mathfrak{n}\bar{\mathfrak{a}}, \mathfrak{n}).$$

In the ring class group, $[\bar{\mathfrak{a}}] = [\mathfrak{a}]^{-1}$, so this is inversion followed by a fixed translation. The level ideal is invertible in every order in question because $(q_i, N) = 1$. If φ is the modular parametrization, the

differentials of $\varphi \circ w_N$ and $-\text{sign}(A_0)\varphi$ agree. Their difference is therefore a constant Q , the image of a cusp. The Manin–Drinfeld theorem [34, 56] makes Q of fixed torsion order. In the strict case $r = 1$, so the Fricke scalar $-\text{sign}(A_0)$ is $+1$. Conjugation also inverts the anticyclotomic weight, namely

$$t \mapsto (1+t)^{-1} - 1 = -t + t^2 - \dots \quad (3.20)$$

The translation is multiplication by a group-ring unit of augmentation one. Because the constant Kummer coefficient has already vanished, that unit has no effect on the leading linear coefficient. Equation (3.20) thus gives $z_1^c = -z_1$. But the compact Selmer line is generated by the invariant class a . Therefore $z_1 = 0$.

Divide once more by t . Its specialization z_2 again belongs to the compact Selmer line, by the same local Kummer and bounded division argument. That line is everywhere locally zero in the strict case. Thus the original trace has zero local Kummer class through degree two, which is the assertion required for $b_* = 3$.

It remains to convert the local Kummer vanishing into Equation (3.19). At p , Proposition 3.2 makes H_i/K unramified with local degree of bounded p -part, so Lemma 3.8 applies with a common constant c . Restricting the local twisted Kummer trace to the unramified field of definition makes the coefficient action trivial. It becomes exactly the Kummer class of the raw weighted sum, modulo the precision allowed by Equation (3.3). The vanishing through degree $b_* - 1$ holds to precision tending to infinity as in Equation (3.12). After one fixed factor p^C clears the cochain denominators, the local comparison gives the lower bound $n - C - c$ at Kummer precision n . Thus the corresponding logarithm coefficients have valuations tending to infinity. For any fixed higher truncation, the Kummer cocycles, logarithms, and binomial coefficients all have a fixed denominator, so those coefficients remain bounded as well. This proves the proposition. $\square$

When a subsequent construction squares a series, we form the preceding traces first at any fixed order $b \geq 2b_*$. Boundedness of all the retained coefficients and Equation (3.19) then permit the required product estimates without an interchange of an unbounded truncation and a limit.

4 The holomorphic theta family

We construct forms on a unitary group of signature $(3, 1)$ at each real place of F , with unit eigenvalues for two normalized positive parabolic operators at p and coefficients of uniformly bounded denominator. These are the forms whose constant and positive Fourier–Jacobi coefficients will be compared in the next two sections. The operator calculation allows arbitrary reduction of A_0 at p ; the denominator bound must hold as both the weight and the tame level grow. Throughout this section a finite-order character at the moving prime means the universal character with values in a group algebra. All integrality assertions are made before specializing that algebra at its characters.

4.1 Hermitian spaces and the compact automorphic input

Let $\delta = \sqrt{D}$, with $\text{Im } \sigma(\delta) > 0$ for $\sigma \in \Sigma$. We use the Hermitian space and the skew-Hermitian space

$$A = (L^2, 1_2), \quad W = B \oplus \mathbb{H}, \quad B = (L^2, b_0 \delta 1_2), \quad (4.1)$$

where $\mathbb{H} = Le \oplus Lf$ is a hyperbolic plane and $b_0 \in F^{\times, +}$ is a unit at the primes above p . Choose b_0 so that

$$v(b_0 \delta \mathfrak{d}_F) \equiv 0 \pmod{2} \quad \text{at every nonsplit finite place of } L/F. \quad (4.2)$$

Here $\mathfrak{d}_F$ is the different and we use the trace additive character. To check the existence of this choice, prescribe the local norm classes of b_0 needed to cancel the parity of the additive conductor on each paired line. The exact norm-invariant sequence for the quadratic extension L/F says that these prescriptions come from a global element if and only if their product is one. The possible odd contributions are at the primes of D' that are inert in the relevant quadratic extension; the condition $(D/|D'|) = 1$ in Lemma 2.3 makes their number even. The discriminants are odd and coprime, and the places over 2 are split. The real prescriptions are positive. The product condition therefore holds. Multiplication by a global norm, using approximation at the split p -places, makes b_0 a p -unit without changing the prescribed classes.

At a nonsplit finite place the extension is unramified. Scaling a line coefficient by a norm changes its valuation by an even integer, so (4.2) gives a self-dual Schrödinger lattice for each paired line, compatible with the trace character. Choose compatible hyperbolic bases. These choices give the unramified models and spherical tests used below. At p all coordinates are unit self-dual coordinates. We pair the sign of the alternating form with the additive character so that a positive unary theta series has exponential $\exp(2\pi i a x^2 z)$ with $a > 0$.

Write π for the representation on the compact group $U(A)$ obtained from the weight-two form of A_0 by real quadratic base change and Jacquet–Langlands transfer, followed by pullback to $U(A)$; see [4, 41]. More explicitly, $PU(A)$ is the projective multiplicative group of the quaternion algebra ramified at the two real places of F and at no finite place. The real component of π is trivial, as is its adelic central character. The successive base changes remain cuspidal: the only possible quadratic inducing field in the dihedral case was excluded in the choice of L . Its standard base change to L , denoted π_L , is thus the cuspidal base change of the representation of A_0 .

At a nonsplit finite place take hyperspecials with the B -lattice homothetic to the A -lattice. The hyperspecial maps onto the hyperspecial of the adjoint quotient because the unramified kernel is connected. On dual groups the quotient map is the inclusion $SL_2 \hookrightarrow GL_2$ with trivial pinned Galois action on the former. Consequently the spherical standard parameters are the restrictions of those of the quaternionic transfer. We use algebraic-valued vectors in the compact automorphic model of π ; a rational tensor vector can be taken real algebraic. They are spherical at nonsplit finite places. No local condition is imposed on π_p . The precise multiplicity and pullback assertion needed for the binary lift is proved in Section 5.

For the moving prime q_i , let ν_i denote the universal character of Δ_i , pulled back to the norm-one torus of L/F via the norm $N_{L/K}$. Put

$$\eta = \mathcal{C}^m \nu_i, \quad k_0 = m - 1. \quad (4.3)$$

The integer $m > 0$ will grow in the ordinary sense and tend to zero p -adically on the fixed torsion branch. The finite smooth conductor of the factor $\mathcal{C}^m$ is fixed, or is killed by that branch. The tame character ν_i is trivial at every fixed place where this is required by Proposition 3.2. We often suppress i in η .

4.2 The Fock vector and its algebraic weight

Take the oscillator representation on $A \otimes_L W$ with both even unitary splitting characters trivial, in the conventions of [52]. For a bare compact vector $z \in \pi^\vee$, the theta integral has the form

$$\Theta_{i,m}(g; \Phi, z) = \int_{[U(A)]} \theta_\Phi(g, h) z(h) \eta^{-1}(\det h) dh. \quad (4.4)$$

The quotient of integration is compact, so this is an ordinary convergent theta integral.

At each real place use Fock polynomials of degree k_0 in the 2×2 minors of the 2×3 positive-coordinate matrix. Their space is $\text{Sym}^{k_0}(\wedge^2 \text{Std}_3)$. The vacuum contributes $\det$ on the source $U(2)$, while each minor contributes another determinant. Thus the source type is $\det^m$, as required by (4.4). On the target the vacuum has weight $(1, 1, 1; -1)$, and the polynomial contributes $(m-1, m-1, 0; 0)$. The resulting weight is

$$\lambda = (m, m, 1; -1). \quad (4.5)$$

The first three entries refer to the distinguished holomorphic Lie quotient; the last refers to the remaining distinguished direction, whose conjugate is holomorphic. The antiholomorphic lowering operators are contractions involving the negative coordinates, which do not occur in these polynomials. They therefore annihilate the vector. The mixed expansion below proves holomorphy also at the cusps.

We specify the integral coefficient lattice, including its frame convention. Let $\mathcal{H} = \omega^+$ be the rank-three positive Hodge differential bundle and put $d = m - 1$. The positive creation directions belong to the holomorphic Lie quotient, dual to $\mathcal{H}$. Write frames as row tuples. If a differential frame changes from e to eg , its dual creation frame changes by g^{-t} , and the minor frame changes by $\wedge^2 g^{-t}$. This passive change of differential frame is distinct from the active compact-group action on the Fock polynomials used to compute (4.5): the Lie quotient has active type Std_3 , while its differential dual has type $\text{Std}_3^\vee$.

Theta evaluation is linear in its Fock test polynomial. Equivalently, the covariance identity $\Theta(gk; P) = \Theta(g; \omega(k)P)$ says that its coefficient array transforms dually to the input polynomial lattice. In the preceding differential frames this gives

$$\mathcal{P}_d(\mathcal{H}) = \text{Sym}^d((\wedge^2 \mathcal{H})^\vee), \quad \mathcal{D}_d(\mathcal{H}) = \mathcal{P}_d(\mathcal{H})^\vee = \Gamma^d(\wedge^2 \mathcal{H}). \quad (4.6)$$

Here Γ^d denotes the divided-power module, defined by the displayed duality for finite locally free modules. We use the literal dual basis to the Fock monomials throughout. In particular, extracting a displayed monomial coefficient does not multiply it by a binomial coefficient or a factorial. Including the vacuum, the positive coefficient bundle is $\det(\mathcal{H}) \otimes \mathcal{D}_d(\mathcal{H})$, and the opposite differential line occurs to exponent 1. Section 7 uses this same integral bundle. All constructions are tensored over the two embeddings of F .

We use the PEL model with common rational similitude to give these coefficient bundles their algebraic meaning, and restrict to the isometry components. Analytically these are the Hermitian symmetric domain times finite isometry-group cosets modulo the rational isometry group. Fixing the multiplier class of the finite frame and absorbing positive principal multipliers into the polarization realizes them as components or finite covers of the PEL model, with compatible cyclotomic constants. The positive rational units preserving a fixed compact multiplier level are trivial. Thus the isometry Hecke correspondences and the expansion principles below are compatible with this model. No extension of the theta kernel to a chosen similitude theta representation is required.

4.3 Split finite tests and the two parabolic operators

At a split finite place identify the isometry groups with their actions on the distinguished factors. In the linear-pair polarization the Schrödinger variables are dual covectors $Y \in M_{2,4}$, and the action is

$$\omega(h, g)\Phi(Y) = |\det h|^2 |\det g| \Phi({}^t h Y g). \quad (4.7)$$

For the binary target the exponent on $|\det h|$ is 1 instead of 2. There is no further determinant character: both even splitting characters are trivial. At nonsplit unramified places the matched self-dual lattice test is spherical. At every other fixed finite place we use integral tests up to a

fixed denominator and at fixed level. Finite additive Fourier transforms away from p introduce only powers of the residue characteristic, roots of unity, and square roots of moduli. These are p -units; in particular no inverse order of a multiplicative residue group is introduced.

Order the distinguished columns at p as (e, B_1, B_2, f) . The e direction is multiplicative. Our tests have the form

$$\Phi_p(Y) = \phi_p(Y_{[1,2]}) \mathbf{1}_{\mathbb{Z}_p^2}(Y_3) \mathbf{1}_{\mathbb{Z}_p^2}(Y_4), \quad \text{supp } \phi_p \subset \text{GL}_2(\mathbb{Z}_p), \quad (4.8)$$

where ϕ_p is locally constant at a fixed depth s . They are fixed by a subgroup $J \subset \text{GL}_4(\mathbb{Z}_p)$ with upper block pattern $(2, 1, 1)$: its upper block unipotent is full, its first Levi is a principal congruence subgroup of depth s , its two remaining unit factors are full, and its lower block entries have depth s . Increasing the fixed s if needed makes the same description work for all the tests. On the canonical component the rank-three distinguished flag member modulo p^s is the multiplicative subgroup. This level records only flags and frames in that subgroup; it does not choose a splitting of the étale quotient. Its integral geometric description will be given in Section 7.

At either distinguished p -place put

$$t_j = \text{diag}(1_j, p^{-1} \mathbf{1}_{4-j}), \quad U_j = [J t_j J], \quad j = 2, 3, \quad (4.9)$$

where U_j is the unnormalized right-coset sum.

Lemma 4.1. *Every test (4.8) satisfies $U_2 \Phi_p = p^2 \Phi_p$ and $U_3 \Phi_p = p^2 \Phi_p$. Thus both operators $p^{-2} U_j$ have value 1 on the theta form of weight (4.5).*

Proof. The upper shifts in the double coset are indexed by $M_{j,4-j}(\mathbb{F}_p)$, of cardinality $p^{j(4-j)}$. The expanding columns must become divisible by p before the multiplication by p^{-1} . Since the first two columns have rank two modulo p , these equations have $p^{(j-2)(4-j)}$ solutions whenever the original test is supported, and none otherwise. The equations do not change the chosen head function. The determinant factor in (4.7) is p^{4-j} . Hence the surviving factor is

$$p^{4-j} p^{(j-2)(4-j)} = p^{(j-1)(4-j)} = p^2$$

for both $j = 2$ and $j = 3$. The calculation is independent of the source representation and of its reduction at p . $\square$

At each of the two places of F over $q = q_i$ take the corresponding pivot test

$$\Phi_q(Y) = \mathbf{1}_{\text{GL}_2(\mathcal{O}_q)}(Y_{[1,2]}) \eta_q(\det Y_{[1,2]}) \mathbf{1}_{M_{2,2}(\mathcal{O}_q)}(Y_{[3,4]}). \quad (4.10)$$

Only the tame determinant factor is relevant on the displayed compact support. Its transformation under $\text{GL}_2(\mathcal{O}_q)$ cancels the η_q^{-1} in source integration. Consequently this integration uses a compact subgroup of fixed volume, rather than the kernel of a character of growing order. The coefficients in (4.10) are group-like units in the tame group algebra. The corresponding two-column test is the reference test in the binary kernel. Section 6 will impose the other finite local controls without changing either (4.8) or (4.10).

Lemma 4.2. *At a split spherical place the standard parameter of the theta form, in unitary normalization, is*

$$(\pi \otimes \eta) \oplus |\cdot|^{-1/2} \oplus |\cdot|^{1/2}. \quad (4.11)$$

Here $\pi \otimes \eta$ means tensoring the standard two-dimensional parameter by the determinant-twist character.

Proof. This is the unramified theta correspondence for the matched lattices. At split places it can also be read directly from the open rank-two matrix orbit: its coinvariants are the normalized parabolic induction of the determinant-twisted source with the trivial binary representation. The latter has standard parameters $|\cdot|^{-1/2}, |\cdot|^{1/2}$. The open-orbit calculation, including the exclusion of the lower ranks near the comparison point, is given in Lemma 6.4. $\square$

The spaces, coefficient lattice, and local tests are now fixed. We record the resulting family before proving the uniform denominator bound, which occupies the rest of this section.

Proposition 4.3. *For the characters (4.3) and every sufficiently large positive m on the prescribed branch, the theta construction gives holomorphic algebraic forms $\Theta_{i,m}$ of weight (4.5). In the canonical ordinary monomial-dual lattice their coefficients have a common bounded denominator, coefficientwise in the tame group algebra and in every fixed truncated substitution R_b . They have spherical parameters (4.11), and at both distinguished p -places they satisfy*

$$p^{-2}U_2\Theta_{i,m} = \Theta_{i,m}, \quad p^{-2}U_3\Theta_{i,m} = \Theta_{i,m}.$$

These assertions allow arbitrary fixed split tests of the stipulated bounded levels. They remain valid for the particular tests with all the finite local controls in Proposition 6.5.

4.4 Frames, periods, and unary moments

The remaining bound reduces, through the mixed Fourier–Jacobi expansion, to moments of unary theta series. We first fix their periods and differential frames. The essential estimate will bound every polynomial combination of moments by the supremum norm of that polynomial on the position lattice; bounds for individual degrees alone would not give the required interpolation.

Use the PEL Hodge bundles, with logarithmic differentials in toric directions. In complex linear coordinates analytic-to-de Rham comparison uses $2\pi i du$ for each holomorphic differential, including those of the conjugate Lie quotient. Normalize each Fock creation generator so that its leading Schrödinger position coordinate is linear, rather than multiplied by $\pi^{1/2}$. Diagonal CM and hyperbolic frames can be chosen with algebraic p -unit scales, allowing square roots. The base heights, quadratic coefficients, and comparison bases can likewise be chosen to be units at p .

On one positive paired line, restriction to the symplectic SL_2/F gives a Hilbert unary theta function at the CM point of type Σ . Write $d = e_0 + 2j$ with $e_0 \in \{0, 1\}$, independently at each embedding when necessary. The d -th Fock vector is the j -fold normalized Maass–Shimura raise of the base theta of weight $\frac{1}{2} + e_0$. Indeed the compact-circle raising operator multiplies by the square of the Fock variable, while the leading term of the normalized derivative of the Gaussian exponential is its quadratic form. Comparison of leading terms identifies the two vectors up to a fixed algebraic unit raised to the degree. There is no factor $j!$.

If Ω_σ compares the linear differential with the chosen integral algebraic CM differential, normalization multiplies by

$$\left(\frac{2\pi i}{\Omega_\sigma}\right)^{1/2+d}. \quad (4.12)$$

Use compatible square roots in products. Since the type is induced from K , rational tensor decompositions of homology over K , with bases localized at p , compare these periods with the elliptic CM period by algebraic p -units. Equivalently use the Serre tensor construction and prime-to- p isogenies; p is completely split. Passing to canonical Igusa differentials gives the analogous comparison with p -adic CM periods, again with unit factors.

Convention 4.4. The position-moment functional is defined first in fixed Igusa differential frames. Fixed powers of algebraic unit scales arising from conversion to another CM frame are restored afterwards. An equivalent formulation puts the polynomial norm on the actual scaled compact support. We do not identify that support with $\mathbb{Z}_p$ merely because the scale is an algebraic p -unit: a unit in a coefficient extension need not belong to $\mathbb{Z}_p$. The restored unit powers are analytic on a sufficiently narrow weight branch and have valuation zero.

Lemma 4.5 (Uniform unary moments). *Fix the integral position supports, the locally constant tests at p , and their levels. At a fixed ordinary CM test object, the period-normalized unary jets, in the frames of Convention 4.4, are the moments of a bounded measure on the product of their integral position lattices. Its norm is bounded in terms of the coefficients of the fixed tests and their p -levels. The bound is independent of the moment degrees and of the moving tame level q_i .*

Cutting a position measure to units gives at classical degrees the theta jet with the corresponding unit cut on the finite position test. The corresponding moments converge to these unit moments when every degree tends to infinity and converges on its fixed p -adic branch. On a sufficiently narrow branch the unit moments, including the fixed CM unit-scale powers, are analytic functions of the degrees.

Proof. For a scalar position test ϕ , write $J_{\mathbf{d}}$ for the normalized jet of multi-degree $\mathbf{d}$ at the fixed CM object. For a polynomial $P(X) = \sum_{\mathbf{d}} a_{\mathbf{d}} X^{\mathbf{d}}$ on the product position lattice Λ , put

$$L_{\phi}(P) = \sum_{\mathbf{d}} a_{\mathbf{d}} J_{\mathbf{d}}, \quad \|P\|_{\Lambda} = \sup_{x \in \Lambda} |P(x)|_p.$$

We will prove $|L_{\phi}(P)|_p \leq C \|P\|_{\Lambda}$, where C depends only on the stated test bounds and p -levels. The proof constructs the square of this combination as an algebraic Igusa function, bounds its expansions, and removes its initial denominator.

The square argument in this proof is scalar. For a test with values in a tame group algebra, first expand it in the group basis and apply the argument separately to each scalar coefficient test. These tests have the same coefficient bound and the same position supports and levels at p . Only after obtaining the scalar estimates do we reassemble the group-algebra-valued moments. In particular, no coefficientwise bound is inferred by taking a square root inside a growing group algebra.

Algebraic squares in a common Igusa ring. For a fixed tame Schwartz level N_1 , which may contain q_i , let $\mathcal{I}_{N_1}$ be the complete ordinary Hilbert Igusa tower with its canonical multiplicative trivialization. In its canonical logarithmic differential frames all integral Hodge weights are functions. Let $\mathcal{A}_{N_1}$ be the p -adically complete ring of these regular functions. Finite sums of different weights therefore lie in one ring; a finite Igusa level may be chosen separately at each reduction precision. We use Katz's expansion principle [45, Section 5.2] and its Hilbert formulation [3, Theorem 6.10 and Sections 11.1–11.7]. The coefficient-multiplier formula for differentiation is Andreatta and Goren [3, Sections 15.21–15.24], which identifies these operators with those of Katz [46, Corollary 2.6.25]. For the CM comparison we use the equality of the ordinary unit-root and CM Hodge projections in Hida and Tilouine [40, Section 1.4, Theorem 1.2], attributed there to Katz (2.6.7). The prime p splits completely in our CM field, and the chosen multiplicative direction defines the required ordinary CM type. After the fixed tame-level enlargement, the level and polarization hypotheses in that comparison hold. We apply it to the algebraic integral-weight products constructed below; their algebraization and descent are part of this proof.

We first show that the square of a finite combination of jets belongs to $\mathcal{A}_{N_1}[1/p]$. Enlarge the fixed prime-to- p level, including its level at 2, to fix the metaplectic multipliers. Every product

of two base half-weight thetas is then an ordinary integral-weight algebraic form, by its Fourier expansion. Products of their normalized nearly holomorphic derivatives are rational expressions in the jets of these base products: apply Leibniz and Gauss–Manin, initially allowing powers of base squares in the denominators. There are no horizontal poles. On a smooth characteristic-zero Hodge-frame chart the square of a base theta has even horizontal divisor, as is seen in its holomorphic half-weight realization over $\mathbb{C}$. Locally write this square as h^2u , with u invertible. Adjoining $\sqrt{u}$ is étale, and the half-weight coefficient $h\sqrt{u}$ is regular. On the torsor of complements to the Hodge filtration, projected Gauss–Manin differentiation has a regular connection matrix; its half-weight action multiplies the scalar weight by $1/2$. Differentiating $h\sqrt{u}$ therefore introduces powers of 2 and u in denominators, but no inverse of h . All fixed iterates are regular. Products descend because the base cross-products fix the signs of the local square roots. In logarithmic cusp charts, choose fixed widths clearing the half-integral exponents; the operators $q_\sigma \partial / \partial q_\sigma$ preserve nonnegative orders. The same argument applies there, and specializing the splitting variable gives the required jets. Omit an identically zero base theta.

Projection by the ordinary unit-root splitting consequently produces a function $F_P \in \mathcal{A}_{N_1}[1/p]$ from the square of the combination defining $L_\phi(P)$. Its initial vertical denominator may depend on the orders and on N_1 . At a CM object the unit-root splitting is the CM splitting, so these products agree with the normalized analytic jets, including products of different orders and weights. In particular, $F_P(\text{CM}) = L_\phi(P)^2$. Compatible square roots also give algebraicity of the individual normalized values. The common Igusa ring is what makes this construction possible when P combines different weights.

Compatible cusps and the position norm. We next choose cusp frames in which the bound by $\|P\|_\Lambda$ is preserved. On each paired unary line, choose the position vector in the distinguished L -factor selected by Σ_p , the same factor used for the linear-pair coordinates Y . This is the CM multiplicative eigenspace. The unit local pairing makes this an integral symplectic frame. Specify the finite position tests in this frame from the outset. At a place over p , choose s such that all these tests are supported on $\mathbb{Z}_p$ and constant modulo $p^s\mathbb{Z}_p$. Their additive Fourier transforms are supported on $p^{-s}\mathbb{Z}_p$. With the unramified additive character and unit quadratic coefficient, the lower unipotent with parameter $c \in p^{2s}\mathbb{Z}_p$ fixes each test: conjugating by Fourier transform makes its action multiplication by the trivial phase $\psi(cQ(x))$ on that support. Upper unipotents with integral parameter preserve integral position support, and diagonal units rescale it by units. Enlarge a fixed depth M to cover all tests and fixed character conductors. Their congruence stabilizer then contains the subgroup with $a, d \equiv 1$, $c \equiv 0 \pmod{p^M}$ and $b \in \mathbb{Z}_p$.

Use the canonical $\Gamma_0(p^M)$ section over the tame ordinary moduli space, together with the multiplicative Igusa marking. At a Tate cusp its connected subgroup is μ_{p^M} , the same position direction used at the CM object. Choose cusp representatives integral at p with unit diagonal entries, by choosing their cusp ideals prime to p and applying approximation to the bases. Their effective frame changes lie in $B(\mathbb{Z}_p)K(p^M)$, where B is the upper triangular subgroup and $K(p^M)$ the principal congruence subgroup. The congruence part fixes the finite test; the triangular part preserves its support and coefficient norm. Thus this choice involves no initial Fourier interchange of the position and momentum directions.

These cusps test every component. Indeed the finite Igusa cover over each connected tame ordinary component is a Galois torsor under the units of the multiplicative character group [3, Sections 7.2 and 11.1]. Its deck group acts transitively on the connected components of the cover. The canonical unramified Tate-cusp chart of Andreatta and Goren [3, Sections 6.5 and 11.7] gives one such cusp; translating its marking by the deck units gives a cusp on each component. These

changes are diagonal units and preserve the position norm. We use these canonical charts and their unit-deck translates as the compatible cusps. The expansion principle applies componentwise, so one such cusp per component will suffice to detect vertical divisibility. All of this concerns the position frames before restoring the separate unit scales of Convention 4.4.

The uniform bound and removal of vertical denominators. By scaling P , it suffices to prove the estimate when $\|P\|_\Lambda \leq 1$. At a compatible cusp the base expansion is $\sum_x \phi(x)q^{Q(x)}$. The normalized jet of degree $e_0 + 2j$ has expansion

$$\sum_x \phi(x)x^{e_0}Q(x)^j q^{Q(x)}.$$

Katz differentiation multiplies by the quadratic index. The quadratic coefficient is a p -unit; normalize the position coordinate first, or record its power as a separate fixed unit-scale factor. A polynomial P of supremum norm at most 1 on the integral position lattice therefore gives coefficients

$$\sum_{Q(x)=\beta} \phi(x)P(x), \tag{4.13}$$

each bounded by the fixed bound for ϕ . Every sum is finite. The square has bounded convolution coefficients, with twice the valuation loss. In several variables the same argument applies to multiorders. Parity pieces are separated by the sign projectors $2^{-r} \sum_{\varepsilon \in \{\pm 1\}^r} \varepsilon^e[\varepsilon]$, which are integral because $p > 2$; the base cross-products give their common descent data.

At primes other than p , changes of polarization preserve this bound, since their normalizations are residue-characteristic powers and roots of unity. At q_i the scalar coefficient tests obtained from the determinant character have integral coefficients. Additive Fourier transforms there likewise have p -unit normalization. Thus F_P has the same expansion bound at every compatible cusp, independently of the degree and q_i . Clear that fixed bound and call the resulting function $G \in \mathcal{A}_{N_1}[1/p]$. Its expansions are integral. Choose the least $r \geq 0$ for which $\varpi^r G \in \mathcal{A}_{N_1}$. If $r > 0$, all expansions of $\varpi^r G$ vanish modulo ϖ . At a finite level containing this reduction, the expansion principle on each ordinary component implies that its reduction is zero: every component meets one of the compatible cusps just constructed, and the finite étale Igusa cover introduces no denominator. Thus $\varpi^r G$ is divisible by ϖ , contradicting minimality.

This saturation argument removes the entire initial vertical denominator, regardless of its dependence on N_1 and the degrees. Evaluation at the integral ordinary CM object now bounds $F_P(\text{CM}) = L_\phi(P)^2$. Taking a scalar square root proves $|L_\phi(P)|_p \leq C$ and hence the required polynomial-norm estimate.

The measure and its unit restriction. We have obtained a bounded linear functional on polynomials on the compact integral position lattice. Polynomial density extends it to continuous functions, hence to a bounded measure. Approximate the characteristic function of the units by polynomials. Applying the same square and cross-product argument to the difference proves that cutting this measure corresponds exactly to cutting the finite test. On the complement of the product unit locus, the monomials tend uniformly to zero when every coordinate degree tends to infinity. On units, after fixing the torsion component, the formula $x^s = \omega(x)^s \exp(s \log \langle x \rangle)$ proves analyticity on a sufficiently narrow branch. The finitely many fixed coefficient-field unit scales have the same property after narrowing the branch once. $\square$

4.5 Mixed Fourier–Jacobi expansions and their integrality

At the rank-one boundary use $x \in A$ for the hyperbolic variables and the Schrödinger realization of $A \otimes B$ for the remaining variables. The Fourier–Jacobi indices are the positive norms (x, x) , together with zero. At zero Jacobi argument a power of the positive hyperbolic Fock column inserts the corresponding power of the coordinates of x . There are no lower Hermite terms: the used creations are the isotropic half of the variables for the contractions, namely holomorphic linear coordinates on the positive trace form of A_∞ . The positive and conjugate multiplicative frames in this direction are toric logarithmic frames, so these insertions incur no CM periods. The B -columns have precisely the unary normalization of Lemma 4.5.

At a rational Jacobi torsion point, use Heisenberg translation of the finite tests and its translated Fourier trivialization. A rational unipotent moves this point to zero; for prime-to- p torsions this is a prime-to- p isogeny move and supplies a splitting of the semiabelian differentials. Consequently the same assertion holds monomial by monomial according to column degree. No estimate in an unrelated analytic trivialization at a nonzero Jacobi argument is required. Restricting to $B \oplus \mathbb{H}$ gives another description: after removing $\exp(2\pi i(x, x)\tau)$, the hyperbolic coefficient is the harmonic position polynomial. In particular, at index zero only the minor using the two B -columns remains; it is the binary theta lift of parallel determinant weight m .

Lemma 4.6. *The mixed Fourier–Jacobi expansions of (4.4), in algebraic ordinary differential frames and in the monomial-dual lattice, have a common bounded denominator. The bound is independent of m , i , and the tame character, and is coefficientwise in the tame group ring. The same assertion holds after substitution into any fixed R_b .*

Proof. Choose source integration representatives integral at p , by weak approximation in the rational unitary group, or by changing the rational Hermitian basis. The source quotient then gives finite sums at a level of fixed volume denominator. This remains true at q_i because the determinant equivariance of (4.10) cancels the varying source character. The archimedean transformation factors cancel in the integral, and character evaluations on the remaining representatives are algebraic p -units.

A boundary representative may be chosen integral at p by Iwasawa decomposition, sending the specified multiplicative member to $e + B$ modulo the fixed level. The translated B -tests remain integrally supported. The test in the column Fourier-dualized to produce x is invariant under integral translations, so x too is integral at p . Rational Jacobi torsions can use unipotent parameters divisible by the fixed p -level; these torsions are still dense and preserve this structure. Representatives and unipotents can simultaneously be chosen integral at the moving and fixed exceptional places, using approximation in the cusp parabolic and the rational Cayley chart on its binary unitary factor.

After the source-equivariance cancellation just made, expand each remaining finite test in the tame group basis and decompose each scalar coefficient test into simple unary tensors, retaining integral support and the original coefficient bound. Apply the scalar unary estimate before reassembling the group-algebra coefficients. Additive Fourier transforms away from p cost only p -units; at p the levels are fixed. Each index coefficient at a prime-to- p Jacobi torsion is consequently a finite sum of the bounded moments in Lemma 4.5, multiplied by integral powers of the coordinates of x and by the unit scales already separated in Convention 4.4. The nonarchimedean bound is unaffected by the number of summands.

At each fixed weight and Fourier–Jacobi index, these algebraic evaluations first give an algebraic coefficient section. Indeed the coefficient bundle on the abelian chart is algebraic. By proper algebraic–analytic comparison, the holomorphic coefficient lies in the complexification of its finite-dimensional algebraic space of sections. Choose finitely many algebraic prime-to- p torsion evaluations

that separate this space, in the algebraic split differential frames just used. In an algebraic basis their evaluation matrix has full rank. Algebraicity of the values therefore gives algebraicity of the section, with an initial denominator that may depend on the weight, index, and tame level.

The full family of these torsion evaluations detects vertical integrality. The prime-to- p torsions, and those satisfying the fixed congruences just imposed, are dense on the special fiber. Their split differential frames are obtained by prime-to- p isogenies. After clearing an individual denominator, a section whose normalized torsion evaluations vanish modulo the uniformizer is therefore zero modulo it. The same least-denominator argument as above proves the uniform bound for the coefficient section. These algebraic coefficients first give algebraicity of the form by the characteristic-zero expansion principle. The ordinary Fourier–Jacobi expansion principle then transfers their uniform bound to the form itself. The canonical cusp charts with their finite étale level are those of Lemma 7.1. Their construction uses only ordinary PEL geometry and the flag and frame data of J , so it is independent of the theta estimates and the cusp-lifting conclusion.

Finally the substitution defining R_b uses only the finitely many binomial coefficients through degree $b - 1$. Its factorial loss is a fixed constant depending on b . This proves the last assertion. $\square$

Proof of Proposition 4.3. The Fock calculation gives the weight and holomorphy, the mixed expansion gives its extension at cusps and its algebraic integral interpretation, Lemma 4.1 gives the exact operator values, and Lemma 4.2 gives the spherical parameters. Lemma 4.6 gives the uniform bounds. $\square$

5 The constant terms and a square-root period comparison

We retain the theta data of Proposition 4.3 and the integer b_* of Section 3. At each of the two places of F above $q = q_i$, the upper theta test, in the column order (e, B_1, B_2, f) , is supported on

$$Y_{[e, B_1]} \in \mathrm{GL}_2(\mathcal{O}_{F_v}), \quad Y_{B_2}, Y_f \in \mathcal{O}_{F_v}^2,$$

and has factor $\eta_v(\det Y_{[e, B_1]})$ on the invertible block. Its transformation under the full source maximal compact cancels the determinant twist in the source integral. The corresponding two-column test is the reference test for the binary theta lift. All unspecified exceptional tests have fixed support and level and uniformly bounded coefficients.

Proposition 5.1. *Assume that the conductor-one Heegner point y is torsion. The positive weights m_i in Proposition 4.3 can be chosen with $m_i \rightarrow +\infty$ and $m_i \rightarrow 0$ on the prescribed p -adic branch so that every constant Fourier–Jacobi coefficient of the upper theta form tends coefficientwise to zero modulo t^{b_*} . The assertion holds in bounded algebraic ordinary differential bases, uniformly over all boundary representatives and all polynomial components. The weights may be required to satisfy any additional lower bounds and any further p -adic approximation conditions tending to zero.*

Here and below, a coefficientwise limit uses the full tame Artin deformation of Section 3. In particular, it is stronger than a statement about separate specializations at finite-order characters.

At index zero, the hyperbolic variable in the mixed expansion is zero. Every minor involving its column vanishes, and the remaining Fock polynomial is the determinant power on B . A constant coefficient is therefore a value of the binary theta lift. We first identify the single global scalar in that lift. Its square will be expressed by two raw toric periods, one of which approaches the Heegner logarithm from Proposition 3.9.

5.1 The binary scalar and its multiplicity

Since multiplying a Hermitian form by the scalar $b_0\delta$ does not change its isometry group, $U(B) = U(A)$. The binary theta lift of the source data belongs to the compact automorphic representation $\pi\eta(\det)$. We record the multiplicity assertion needed here. At split places equal-rank local theta correspondence gives the prescribed representation; at nonsplit finite places we use only its spherical line, and at infinity the binary polynomial gives the determinant character. Removing $\eta(\det)$ gives trivial adelic central action.

There is a direct description of the resulting automorphic space that avoids a multiplicity formula for general inner-form packets. Let D_{quat}/F be the quaternion algebra ramified at the two real places and split at all finite places, so that $PU(A) = P(D_{\text{quat}}^\times)$. Write $\omega_{L/F}$ for the quadratic idele-class character. The image of the central quotient of $U(A)$ in the projective quaternion automorphic quotient is the part on which

$$\omega_{L/F}(\text{Nrd}(g)) = 1.$$

Locally the obstruction to lifting is the reduced norm modulo norms from L . The Hasse norm theorem supplies rational lifts when the local obstructions vanish, and rational quaternion norms are precisely the totally positive elements. Quadratic norm reciprocity then identifies the adelic image modulo rational translation with the displayed part.

Extend a function on this part by zero to the projective quaternion quotient. This respects all split-place actions and the trivial archimedean weight. After Jacquet–Langlands transfer, the split good Hecke data determine the prescribed cuspidal Hilbert representation up to the twist $\omega_{L/F}$. To see the last assertion, use the Galois representations of the parallel-weight-two Hilbert transfers [60, Theorem 1.1 and Remark 1.2] and Chebotarev at the places splitting in L . Their restrictions to G_L agree and are absolutely irreducible, so their extensions to G_F differ by at most the quadratic character of L/F . Strong multiplicity one gives the assertion for the automorphic representations. The one-dimensional reduced-norm representations do not have these cuspidal Hecke data. The two possible quadratic twists restrict identically to the indicated part. Moreover restriction of the prescribed representation is injective: a nonzero function supported only on the complementary part would belong also to its quadratic twist, contradicting their distinctness. Finally each nonsplit hyperspecial factor contributes a single spherical line. This proves the required multiplicity one with exactly the local levels being used. The base-change and Jacquet–Langlands inputs are the standard cuspidal ones [4, 41]; the avoidance of the inducing CM field ensures the required cuspidality.

At a split finite place the binary local intertwiner is

$$J_v(\Phi, z) = \int_{\text{GL}_2(F_v)} \Phi({}^th) |\det h|_v \pi_v^\vee(h) \eta_v^{-1}(\det h) z d^\times h, \quad \text{vol}(\text{GL}_2(\mathcal{O}_{F_v})) = 1. \quad (5.1)$$

Here z belongs to the bare $\pi_v^\vee$ space; the determinant twist has been written in the integrand. The output is read by inverse transpose, which is field conjugation in the chosen split bases. Equation 5.1 is the Godement–Jacquet integral at the unitary center, interpreted by rational continuation where necessary. Equivariance, local theta uniqueness, and a nonzero open test identify it with the theta intertwiner. We normalize by this map at the exceptional split places, by spherical-to-spherical maps elsewhere, and by the determinant highest vector at infinity.

The global binary map is consequently a single scalar $S_i(m)$ times these fixed intertwiners. Include in $S_i(m)$ the conversion to CM evaluation at a unit-framed object, namely

$$\prod_{\sigma: F \hookrightarrow \mathbb{R}} (2\pi i / \Omega_\sigma)^{2m},$$

with the fixed algebraic unit powers prescribed in Section 4. This scalar has bounded tame group-ring coefficients. Indeed, choose integral open tests at q and fixed-level open tests at p that map a fixed

compact vector to itself under Equation 5.1; the volume loss at p is fixed, whereas the full source maximal compact at q has volume one. The binary moment estimate bounds the reference lift. Its values are $S_i(m)$ times values of one fixed nonzero algebraic compact automorphic function, times determinant-character units. Evaluate at a fixed nonzero value, choosing finite representatives integral at p . Division by that value costs a fixed denominator. The cancellation of determinant twists at finite compact levels prevents the source summation level from growing with q .

Applying these intertwiners to the projected test at any boundary component expresses its constant coefficient as $S_i(m)$ times a compact automorphic evaluation and local operators. Thus two estimates remain: $S_i(m)$ must tend to zero through tame degree $b_* - 1$, and the projected local operators must have uniform bounds. The latter will use the same local factors as the period comparison.

5.2 Raw toric periods and the comparison identity

Let f_0 and f_1 be the weight-two newforms attached to A_0 and $A_0^{D'}$, respectively, and let π_0, π_1 denote their unitary automorphic representations. For $m \geq 1$, form toric periods over K of the $(m-1)$ st normalized Maass–Shimura raises of these forms, against $\eta_K^{-1} = (\mathcal{C}^m \nu)^{-1}$. Normalize CM evaluations by integral differentials. We denote the resulting periods by $P_{0,i}(m)$ and $P_{1,i}(m)$, suppressing the tautological tame character from the notation. The following local choices fix their normalization.

1. At q , translate the spherical Whittaker vector by $n(1/q)$ in torus coordinates whose first direction is the distinguished multiplicative direction. Multiply the fixed-measure torus integral by $q-1$. Thus $P_{j,i}$ is a raw sum at conductor q , with no averaging denominator depending on q .
2. At p , use the Kirillov function $W(\text{diag}(x, 1)) = \mathbf{1}_{\mathbb{Z}_p^\times}(x)$. This is the canonical orientation of the p -depletion of the newvector. The function belongs to the Kirillov model of every generic local representation of GL_2 , including ramified principal series, special representations, and supercuspidal representations.
3. At all other places use fixed tests with nonzero local toric pairing at weight zero. For $P_{0,i}$ these can be the oriented Heegner newvectors. At a split level prime their diagonal Mellin integrals are the nonzero local central L -factors. For $P_{1,i}$, a prime dividing D' that is nonsplit in K presents no obstruction: the local representation is a tempered principal series, and the field torus acts transitively on the projective line. In the induced model this gives the toric functional directly, since the central character is trivial. A fixed deeper level is allowed at these finitely many places.

The coefficients of both $P_{j,i}$ have a common lower valuation bound. Indeed, all CM objects are ordinary with canonical multiplicative structure, the p -tests are integral of fixed level, and Katz differentiation gives bounded moments in integral differential frames. The expansion and coefficient-multiplier results are Katz [45, Section 5.2] and Andreatta and Goren [3, Theorem 6.10, Sections 11.1–11.7 and 15.21–15.24]. The ordinary CM comparison is Hida and Tilouine [40, Section 1.4, Theorem 1.2]; its application to the algebraized theta moments is Lemma 4.5. Fixed CM-conductor and ideal-class denominators can be cleared once. Changes to Igusa differential frames contribute fixed unit powers, and the raw conductor- q sum introduces no further denominator. All these assertions hold before projecting the tame group algebra to characters.

The relation between these periods and the binary scalar is the following cleared identity. Its augmentation condition is exactly what will allow division after a fixed Artin truncation.

Lemma 5.2 (Cleared period comparison). *There are group-ring polynomial factors $A_i(m), B_i(m)$ with uniformly bounded coefficients such that*

$$A_i(m)S_i(m)^2 = B_i(m)P_{0,i}(m)^2P_{1,i}(m)^2. \quad (5.2)$$

For each i , after taking m sufficiently close to zero on its branch, $A_i(m)$ has nonzero augmentation whose valuation is bounded above by a constant independent of i . The required neighborhood of zero may depend on i . The identity holds in the full characteristic-zero tame group algebra, including at zeros of the global central L -values.

We prove this comparison below by matching the two bilinear period formulas and their local factors. First we show why $P_{0,i}$ is small. Its square will then be small through twice the required tame order; the bounded coefficients of S_i will recover the required order for S_i itself.

5.3 The Heegner logarithm limit

We claim that

$$P_{0,i}(m) \longrightarrow 0 \pmod{t^{b_*}}, \quad (5.3)$$

where for each fixed i one first takes m sufficiently close to zero on its branch, and then lets i increase.

The passage from raised CM values to a Heegner logarithm follows the p -adic period method of Bertolini et al. [10, Introduction and Proposition 3.24]. Their construction is made under $p \nmid N$. We give the primitive and logarithm comparison for the reduction types occurring here.

Write $u = q_{\text{Tate}}$ for a modular cusp parameter, to distinguish it from the auxiliary prime q_i , and put

$$f_0^{[p]}(u) = \sum_{p \nmid n} a_n u^n, \quad F_0(u) = \theta^{-1} f_0^{[p]}(u) = \sum_{p \nmid n} \frac{a_n}{n} u^n.$$

On the ordinary Igusa tower, the normalized derivatives $\theta^{m-1} f_0^{[p]}$ converge as $m \rightarrow 0$ on the chosen branch, because $n^{m-1} \rightarrow n^{-1}$ uniformly on the unit indices. Their limiting Igusa weight is zero. The ordinary expansion principle therefore constructs a global analytic function F_0 on the completed ordinary locus of the good tame modular curve; weight-zero invariance descends it from the Igusa tower. Its differential and its cusp constants are

$$dF_0 = f_0^{[p]}(u) d \log u, \quad F_0|_{\text{each compatible cusp}} = 0. \quad (5.4)$$

Each component in use has such a compatible cusp. The canonical $\Gamma_0(p^{\text{ord}_p N})$ section and the canonical-subgroup Frobenius maps allow the p -depleted form to be read on this tame ordinary locus, even when $p \mid N$.

To make the logarithm comparison explicit, choose a modular parametrization φ with $\varphi^* \omega_0 = \kappa f_0(u) d \log u$, where ω_0 is a fixed invariant differential of A_0 and $\kappa \neq 0$ is fixed. If the depletion polynomial is $E_p(X) = \sum_r e_r X^r$, and Fr is canonical-subgroup Frobenius, then

$$dF_0 = \kappa^{-1} \sum_r e_r p^{-r} (\varphi \circ \text{Fr}^r)^* \omega_0. \quad (5.5)$$

The factors p^{-r} arise because substitution $u \mapsto u^{p^r}$ multiplies $d \log u$ by p^r . There are only finitely many such maps, all determined by the fixed newform. They extend over a wide-open thickening of the ordinary affinoid.

The primitive itself extends to a slightly smaller such thickening. For completeness, clear one denominator and write its expansion near a missing special-fiber point, with local parameter z , in the completed localization of $\mathcal{O}[[z]][1/z]$:

$$F_0 = \sum_{n \in \mathbb{Z}} a_n z^n.$$

Overconvergence of dF_0 implies exponential decay of the negative coefficients na_n . Division by n loses only the subexponential factor $|1/n|_p \leq |n|$, so a_n still decays exponentially on any slightly smaller collar. Exactness rules out a logarithmic term. The positive part is bounded and converges for $|z| < 1$. These expansions retain the coefficients of the original F_0 , hence introduce no independent constants. One can glue them on a model with collar $zw = \varpi'$ and $|\varpi'| < 1$ sufficiently close to one: modulo each uniformizer power the usual localization–completion sequence

$$A \longrightarrow A[1/z] \oplus \hat{A} \longrightarrow \hat{A}[1/z]$$

is injective at the first term and exact at the middle term. This follows from flatness of completion and invariance of local cohomology with support. Thus the continued primitive is an analytic function on a connected wide open in every component under consideration.

We use Berkovich–Coleman integration on this neighborhood, with the functoriality and fundamental theorem of [44, Definition 3.5]. After a finite extension, A_0 has either good or split multiplicative reduction. In the good reduction case integration of its invariant differential equals its abelian logarithm [44, Corollary 3.18]. In the multiplicative case write $A_0^{\text{an}} = \mathbb{G}_m^{\text{an}}/Q_0^{\mathbb{Z}}$ and normalize the pullback differential to dz/z . Starting with any branch Log , choose

$$\text{Log}_{Q_0}(z) = \text{Log}(z) - \frac{v_p(z)}{v_p(Q_0)} \text{Log}(Q_0).$$

This branch kills Q_0 , and integration along every lifted path is the abelian logarithm on the quotient; see [44, Example 3.11]. All maps in Equation 5.5 have the same elliptic target, so this single branch suffices.

For a path from a compatible cusp c to x , functoriality and Equations 5.4–5.5 now give

$$F_0(x) = \kappa^{-1} \sum_r e_r p^{-r} \log_{A_0, p}(\varphi(\text{Fr}^r x)). \quad (5.6)$$

Indeed the cusp constant is zero, and every image of a cusp is torsion by the Manin–Drinfeld theorem [34, 56]. This is an equality obtained by path integration on the connected wide open; equality of derivatives alone would not suffice to remove constants on different residue discs.

At the CM points entering $P_{0,i}$, the canonical Frobenius quotients are the horizontal translations at p of the conductor- q_i Heegner points, including when a nontrivial canonical p -level is present. Taking their raw tame weighted sums in Equation 5.6 therefore gives a fixed finite linear combination of the logarithm sums in Proposition 3.9. Both signs of the tame character are covered by Equation 3.19. The CM differential and character unit powers tend to one as $m \rightarrow 0$. This proves Equation 5.3, with bounded higher coefficients in every fixed truncation. No ordinary hypothesis on A_0 at p has entered this argument.

5.4 Period identities and the two local functional equations

Proof of Lemma 5.2. The equal-rank Rallis inner product formula and the Waldspurger toric period formula give

$$S_i(m)^2 = C_i(m) P_{0,i}(m)^2 P_{1,i}(m)^2. \quad (5.7)$$

We use these formulas with bilinear dual data, so the squares are algebraic squares, rather than absolute squares. For the theta identity, the precise Siegel–Weil input is the anisotropic boundary case of [76, Theorems 2.1(i) and 2.2(i)], also stated in [35, Theorem 7.1]. In the notation of those results, $m = n = 2$, $r = 0$, and $s_0 = (m - n)/2 = 0$. On the doubled group $U(2, 2)$ the Siegel Eisenstein series is holomorphic at s_0 and $E(0, \Phi) = 2I(\Phi)$, where I is the ordinary convergent theta integral over the compact binary group with quotient volume one. The conversion to our global measure convention contributes only the fixed scalar permitted in Section 2.3. Restrict this identity to the two copies of $U(A)$ and integrate against the source cusp forms. The theta seesaw gives their bilinear theta pairing on one side; unfolding the Eisenstein series gives the doubling zeta integral on the other. Its Euler factorization is the central standard L -value times the normalized local integrals used below. This proves the equal-rank Rallis identity needed here directly from that boundary Siegel–Weil identity. The doubling and local factorization conventions are those of [35, Sections 11.3, 11.5–11.6]. The source groups are compact, the standard base-change parameter is cuspidal and tempered, and its zero-space theta lift vanishes, since that lift could only be one-dimensional. The Waldspurger identities [73, Section II.7 and Section III.3, Proposition 7] apply to cuspidal transfers with compatible central character and opposite torus characters in the two slots. Thus the identities involve central values, not residues or derivatives.

For the theta pairing, pull back the same data by $(h, g) \mapsto (h^c, g^c)$. Field conjugation is anti-symplectic on the tensor product and identifies the oscillator with its dual by Poisson summation, without a global conductor factor. Pair fixed compact vectors nontrivially; if necessary two different vectors can be used, since the scalar is the same. On the toric side, translate the vector by a rational torus normalizer inducing conjugation. This turns the second period into the first, up to a fixed character value from a base-point translation. At p and q the normalizer exchanges the distinguished coordinates with unit scales.

The global central L -values in the formulas match by quadratic base change over K :

$$L(\tfrac{1}{2}, \pi_L \otimes \eta_L) = L(\tfrac{1}{2}, \pi_{0,K} \otimes \eta_K) L(\tfrac{1}{2}, \pi_{1,K} \otimes \eta_K). \quad (5.8)$$

Here η_L is induced by norm from η_K , and simultaneous inversion of the character gives the same comparison. One may retain raw local integrals at every removed place and use partial L -values. Fixed adjoint, norm, and global zeta factors then contribute a single fixed scalar. We compare the remaining local factors explicitly.

The matrix functional equation. Let E be a split local field, write $\chi = \eta_v^{-1}$ in the distinguished multiplicative coordinate, and let ψ_{add} be the trace additive character. Write π for the bare elliptic local representation, so $\pi \simeq \pi^\vee$ and its central character is trivial. For a matrix coefficient c and $F(X) = \Phi({}^t X)$, put

$$Z_G(F, c, \chi, s) = \int_{\text{GL}_2(E)} F(g) c(g) \chi(\det g) |\det g|^{s+1/2} d^\times g.$$

At $s = 1/2$ this is a scalar coefficient of Equation 5.1. The Fourier transform uses the pairing $\psi_{\text{add}}(\text{tr}(XY))$. Field conjugation exchanges the two dual Lagrangians and acts on the source by inverse transpose, so the second integral uses the full matrix Fourier transform and $c^\vee(g) = c(g^{-1})$. With these identifications the functional equation is

$$Z_G(\widehat{F}, c^\vee, \chi^{-1}, 1-s) = \gamma(s, \pi \otimes \chi, \psi_{\text{add}}) Z_G(F, c, \chi, s). \quad (5.9)$$

Apply the matrix zeta identity [41, Theorem 13.1(iii),(v)] to $\sigma = \pi \otimes (\chi \circ \det)$: its determinant exponent is $s + 1/2$ and its trace Fourier transform uses the self-dual additive measure. The L -

and epsilon factors are the same as in the Whittaker functional equation of Section 2.18 of that reference. In particular, normalizing the first map to be one on the chosen vector gives the factor

$$\Gamma_v(\pi \otimes \chi) = \varepsilon\left(\frac{1}{2}, \pi \otimes \chi, \psi_{\text{add}}\right) \frac{L\left(\frac{1}{2}, \pi^\vee \otimes \chi^{-1}\right)}{L\left(\frac{1}{2}, \pi \otimes \chi\right)}. \quad (5.10)$$

There is only one application of the functional equation in this bilinear pairing.

To relate this calculation to the Rallis local integral, integrate first over invertible matrices in $M_2(E)$. Additive measure becomes a fixed multiple of $|\det h|^2 d^\times h$, so the integral factors as the pairing of the two maps in Equation 5.1. For self-dual additive measure the spherical factor is $(1 - Q^{-1})(1 - Q^{-2})$, where Q is the residue cardinality. It cancels exactly when the local integral is normalized by its spherical value: the spherical intertwiner itself gives $L(1/2, \pi^\vee \otimes \chi)$ with no such factor. In particular this normalization never leaves a denominator $1 - Q^{-1}$ at the moving prime.

The toric functional equation. Put $a(x) = \text{diag}(x, 1)$, let $w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and write

$$M(W, \chi, s) = \int_{E^\times} W(a(x)) \chi(x) |x|^{s-1/2} d^\times x, \quad W^w(g) = W(gw).$$

The corresponding Whittaker functional equation is

$$M(W^w, \chi^{-1}, 1-s) = \gamma(s, \pi \otimes \chi, \psi_{\text{add}}) M(W, \chi, s). \quad (5.11)$$

For the twist σ , use $W_\sigma(g) = \chi(\det g)W(g)$ and its central character χ^2 in Jacquet and Langlands [41, Theorem 2.18(ii)–(iv)]. Since $\det w = 1$, the dual integral has character $\chi(x)\chi(x)^{-2} = \chi(x)^{-1}$. The identity $wa(x^{-1}) = x^{-1}a(x)w$ then gives the displayed Weyl translate, with no extra $\chi(-1)$ factor. Both first integrals, in Equations 5.9 and 5.11, use the same χ in the same distinguished coordinate. The invariant Kirillov pairing factors the toric local matrix integral into these Mellin integrals; its proportionality constant also cancels under spherical normalization. Consequently its factor is

$$\Gamma_v(\pi \otimes \chi) M(W, \chi, 1/2)^2,$$

with exactly the same gamma orientation as the matrix factor.

At p , the character χ is unramified and the chosen Kirillov test gives $M(W, \chi, 1/2) = 1$. Thus all gamma, Euler, and epsilon factors cancel between the two period formulas. No bound for $\Gamma_v(\pi \otimes \chi)$ itself is asserted or needed; indeed $v_p(\chi(p)) = m$ varies with the weight. The functional equations are identities of rational zeta functions and apply by continuation at these parameters. Their validity includes special representations [41, Proposition 3.6 and Corollary 3.7]; a fixed compact open test normalizes the matrix map even when π_p is ramified. Additive-character signs and changes of the chosen unit frames introduce only character values on units and fixed unit powers. Since p splits completely in L , there are two matching local comparisons; the quadratic twist by D' does not change either local elliptic representation.

The raw factor at q . Let L_{fake} be the degree-two unramified central Euler factor formed using $\chi(q)$ while ignoring its tame inertia. The spherical Whittaker expansion after translation by $n(1/q)$ gives

$$F_q(\chi) := (q-1)M(W, \chi, 1/2) = \sum_{a \in \mathbb{F}_q^\times} \chi(a)(\psi_{\text{add}}(a/q) + L_{\text{fake}} - 1). \quad (5.12)$$

To obtain the formula, separate valuation zero from positive valuations. On units the translated Whittaker function has the additive factor $\psi_{\text{add}}(a/q)$; on positive valuations its sum is $L_{\text{fake}} - 1$.

Multiplication by $q - 1$ converts the unit integrals to the displayed raw sum. For nontrivial tame χ the second sum vanishes, leaving the Gauss sum. At augmentation one instead has

$$F_q(1) = -1 + (q - 1)(L_{\text{fake}} - 1).$$

In the group ring put $D_q = L_{\text{fake}}^{-1}$. Clearing this Euler denominator gives the polynomial factor

$$H_q = D_q F_q = D_q \sum_{a \in \mathbb{F}_q^\times} \chi(a) \psi_{\text{add}}(a/q) + (1 - D_q) \sum_{a \in \mathbb{F}_q^\times} \chi(a), \quad H_q(1) = (q - 1) - q D_q(1). \quad (5.13)$$

Both D_q and H_q have bounded group-ring coefficients. The no-eigenvalue-one condition on $V(\gamma)$ implies that the limit of $D_{q_i}(1)$ at weight zero is nonzero; the conventions of Section 2.3 identify this with the corresponding Frobenius Euler determinant, up to factors tending to one. Thus its valuation is bounded above. Since $q_i \rightarrow 1$, the same is true of $H_{q_i}(1)$, once m is sufficiently close to zero for each i . The two raw toric periods contribute F_q^2 in their respective comparisons. Therefore the ratios in Equation 5.7 divide by these full factors, and never by an isolated $(q - 1)^2$, by a conductor projector, or by individual Gauss sums. The identical gamma factors cancel just as at p .

The archimedean factors. Write $k_0 = m - 1$. In the 2×2 Fock model the Gaussian norm ratio of $\det(Z)^{k_0}$ to the vacuum is

$$k_0!(k_0 + 1)!. \quad (5.14)$$

For example, expand the determinant by the binomial theorem. The different monomials are orthogonal, and each of the $k_0 + 1$ summands contributes $(k_0!)^2$ after including its squared binomial coefficient. Their sum is Equation 5.14. The definite quaternion real polarization has positive quadratic norm equal to this determinant. Its scalar symplectic SL_2 generates from the vacuum the holomorphic discrete series of lowest weight two, and raising is multiplication by the determinant in Fock coordinates. The norm ratio of its k_0 th raised vector is the same $k_0!(k_0 + 1)!$. This is also the ratio in the archimedean toric pairing with the Weyl-dual raised weight-two vector.

The factorials therefore cancel. The remaining comparison with normalized Maass–Shimura raising is a fixed algebraic p -unit to the k_0 th power: this follows by comparing the leading linear Schrödinger coordinates, as in Lemma 4.5, with quaternion frames chosen as unit bases at p and CM heights chosen as algebraic p -units. The anti-symplectic duality contributes only phases of the same kind. There is one comparison at each real embedding of F , and the compatible CM normalization on both sides has weight $2m$ per embedding, with the factors $(2\pi i/\Omega_\sigma)^{2m}$ already included in $S_i(m)$. No factorial denominator remains.

Fixed factors and zeros. At fixed exceptional finite places away from p , the local factors are rational functions of unramified smooth-character parameters, with fixed inertia. On our branch those parameters tend to their weight-zero values. At the tempered center the zeta denominators have no pole, and the toric functionals were chosen nonzero there. After shrinking the branch, the denominators have nonzero specialization of bounded valuation, so these ratios cost a fixed denominator. The same statement applies to the fixed rational local intertwiners. The remaining global measure and norm constant is independent of i, m, ν . It is algebraic by evaluating the comparison at any parameter for which both sides are nonzero, since all normalized periods there are algebraic. If there is no such parameter, the vanishing theta comparison already gives the required zero scalar.

The factors just computed, notably D_q and H_q , supply the polynomial clearing factors in Equation 5.2, with bounded coefficients and with nonzero augmentation of $A_i(m)$ of bounded

valuation. Fixed algebraic constants and unit powers are absorbed into A_i and B_i . The period formulas prove this identity first at every finite-order character. Clearing denominators gives an identity in the full characteristic-zero tame group algebra, because those characters separate that algebra. There is no division by a global central L -value: use Equation 5.8 in the two identities and cross-multiply the local factors. Hence the cleared identity also holds at their global zeros. We do not assert that the inverse of A_i is a bounded measure on the entire finite group. $\square$

5.5 Projected tests at every cusp

The period comparison now controls the scalar shared by all binary theta values. To apply it to every constant coefficient, we must bound the projected local operators in the factorization at the start of this section, uniformly in the boundary representative. The moving prime will use the factor D_q just computed.

At p , canonical projection leaves the binary test supported on integral matrices. Expanding Equation 5.1 by the determinant valuation gives the factor

$$p^{(m-1)v_p(\det h)}.$$

The right-coset volumes have fixed level, and translates of the fixed compact automorphic function have bounded sup norm. For large m the integral-matrix expansion therefore converges p -adically with a uniform bound. Its determinant truncations are the usual Godement–Jacquet series, so by continuation it computes the same rational zeta operator. This argument applies at both places over p , without imposing any condition on the local elliptic representation.

At q , the projected test is supported integrally, depends only on reduction modulo q , and retains determinant equivariance under the source $\mathrm{GL}_2(\mathcal{O}_{F_v})$. On the full-rank stratum this gives a bound directly with maximal-compact volume one. On a deficient-rank stratum the determinant of the stabilizer surjects onto Δ_i . Therefore its group-ring coefficient is invariant under multiplication by every element of Δ_i and is a bounded multiple of the norm element

$$N_{\Delta_i} = \sum_{\delta \in \Delta_i} [\delta].$$

This is an integral invariance statement: all its group coefficients are equal. It involves no division by $|\Delta_i|$.

The remaining integral, now untwisted on units, is a sum over integral image lattices $h\mathcal{O}_{F_v}^2$ with weight $|\det h|\chi(\det h)$, and a coefficient constant on each prescribed reduced image subspace. For a line or the zero subspace, express the condition of having that exact reduced image by subtraction of the conditions that the image be contained in it. Each containment sum is a translate and a determinant power times the spherical full-lattice series L_{fake} . All translation determinants and volume factors are powers of q , hence p -units. Thus the already controlled factor D_q clears the operator denominator. This rank-stratum argument is valid coefficientwise in the full group ring.

At all other places the operators are spherical or among fixed rational operators with the no-pole property proved above. Boundary Iwasawa representatives can be chosen integral at p , compatible with the canonical multiplicative flag. Evaluation of the compact model at the conjugate representatives then introduces only unit factors. Source sums have a fixed volume denominator, and all nonsplit spherical lines and fixed-level coefficients have common bounds. Although the number of boundary representatives may grow at q_i , their projected tests have precisely the uniformly bounded forms just described. This proves uniformity over every cusp, rather than only over a chosen finite list of evaluations.

5.6 Truncation, square roots, and choice of weights

We finish by spelling out the algebra behind the coefficientwise estimate. Fix $h \geq 2b_*$. In a coefficient extension containing the finite test values, substitute

$$[g] \longmapsto (1+t)^{\ell_i(g)} \pmod{t^h}$$

in the tame group algebra. At finite precision this is a ring map modulo $p^{n_i-c_h}$ for a constant c_h depending only on h : the relation $[g]^{p^{n_i}} = 1$ maps to $(1+t)^{p^{n_i}} - 1$, whose coefficients below degree h have valuations at least $n_i - c_h$. Consequently products and identities may be truncated with an error tending to zero. Bounded group-ring coefficients give bounded coefficients in this fixed Artin truncation. All estimates below are unaffected by these errors.

If $D(t) = d_0 + \cdots + d_{h-1}t^{h-1}$ has bounded coefficients and $v_p(d_0)$ is bounded above, its inverse in the truncated rational ring is bounded by a constant depending only on h and those bounds:

$$D(t)^{-1} = d_0^{-1} \sum_{r=0}^{h-1} (-d_0^{-1}(D(t) - d_0))^r \pmod{t^h}. \quad (5.15)$$

Thus the clearing factors must be cleared in the group algebra *before* truncation, after which their inverses cost a fixed denominator. No assertion about bounded character idempotents or bounded inverses on the full group ring is required.

By Equation 5.3, the coefficients of $P_{0,i}$ below degree b_* tend to zero, and all coefficients below degree h of both toric periods are bounded. Every coefficient of $P_{0,i}^2$ below degree $2b_*$ contains a factor of degree below b_* . Hence

$$P_{0,i}^2 P_{1,i}^2 \longrightarrow 0 \pmod{t^{2b_*}}.$$

Apply Equation 5.15 to Equation 5.2 to obtain $S_i(m)^2 \rightarrow 0 \pmod{t^{2b_*}}$.

The passage to the square root uses only valuations, not compactness of the fields of coefficients. Write $S_i = \sum_{j < h} s_{ij} t^j$, with all s_{ij} bounded. First $s_{i0}^2 \rightarrow 0$, hence $s_{i0} \rightarrow 0$. Inductively, for $0 < j < b_*$,

$$[t^{2j}]S_i^2 = s_{ij}^2 + 2 \sum_{a < j} s_{ia} s_{i,2j-a}.$$

The sum tends to zero by the preceding steps and boundedness, so $s_{ij}^2 \rightarrow 0$ and then $s_{ij} \rightarrow 0$. Therefore $S_i(m) \rightarrow 0 \pmod{t^{b_*}}$. The same fixed-truncation inversion handles the projected-test factors at every cusp. Their uniform bounds prove the asserted vanishing of all constant coefficients.

Finally, for each i , let m approach zero in the prescribed branch until the interpolation of $P_{0,i}$ has the precision supplied by Equation 3.19, and until all fixed and moving clearing factors satisfy the bounds above. Arbitrarily large positive integers exist in every sufficiently small such congruence class. Choose one that also satisfies any imposed archimedean lower bound and additional p -adic approximation requirements. Taking the precision to infinity with i gives the required diagonal choice of weights, and proves Proposition 5.1.

6 A positive Fourier–Jacobi coefficient

The constant-term estimate would be useless if the entire theta form had content tending to zero. We now exhibit a positive Fourier–Jacobi coefficient whose valuation is bounded above already at the trivial tame character. We then impose the additional finite local controls needed for the Galois argument, while retaining that coefficient bound. No average over a group whose order grows with q_i will occur in this section.

6.1 The norm-one orbit

Use the Fourier–Jacobi index represented by $x = e_1 \in A$, and let $T = U(Le_2) \subset U(A)$ be its stabilizer. At nonsplit finite places choose the unramified hyperbolic test so that its mixed x -lattice is $A(\mathcal{O}_L)$; keep this Lagrangian lattice and adjust its complement dually. This is compatible with the scaled binary lattice of (4.1). Primitive norm-one vectors form one orbit of the compact unitary group at such a place.

Choose a fixed tensor vector $\varphi \in \pi^\vee$ and a finite-order character d_0 of $[T]$, unramified at nonsplit finite places, for which

$$I_T(\varphi, d_0) := \int_{[T]} \varphi(s) d_0(s) ds \neq 0. \quad (6.1)$$

Here is an existence argument with the local conditions included. A nonzero compact automorphic vector, spherical at nonsplit places, can be translated at split places to be nonzero somewhere on T . Determinants at nonsplit places are units realized by the maximal compact, and strong approximation on $SU(A)$ away from a split place removes the remaining nonsplit components. Its restriction to the compact torus quotient is therefore nonzero. Since its infinity type is trivial, Fourier expansion on that quotient supplies a finite-order character d_0 with (6.1). Sphericity forces d_0 to be unramified at the nonsplit places. A fixed diagonal translation at p makes φ_p invariant under the integral lower unipotent. This translation multiplies the torus period by a nonzero scalar, so it does not destroy (6.1).

For $d_1, d_2 \geq 0$ satisfying $d_1 + d_2 = m - 1$, take at each real embedding the polynomial coefficient

$$[e, B_1]^{d_1} [e, B_2]^{d_2}. \quad (6.2)$$

The brackets denote the corresponding 2×2 minors. At $x = e_1$ these minors select the second row of the respective binary columns.

Lemma 6.1. *There are finite tests satisfying the p - and q_i -requirements of Section 4 for which the coefficient (6.2), at index (e_1, e_1) , unfolds into the product of a fixed row-one Heisenberg theta function and*

$$\int_{[T]} \varphi(s) \eta^{-1}(s) \vartheta_{1,d_1}(sb_1) \vartheta_{2,d_2}(sb_2) ds. \quad (6.3)$$

Here $b_1, b_2 \in U(1)(\mathbb{A}_F)$ translate the binary lines, and known splitting-character factors in b_j have been removed. At q_i the only additional row-two condition is a unit cut in the first binary column. The unfolding introduces a fixed nonzero volume factor and no factor $q_i - 1$.

Proof. At a fixed split place away from p , use small support transverse to the orbit of e_1 in the mixed polarization, leaving arbitrary desired unary tests on the remaining coordinates. Unfolding the rational norm-one orbit then cuts the finite integration to its stabilizer T . At a nonsplit place the single compact orbit of primitive norm-one vectors gives the same cut, with all remaining functions spherical.

At $q = q_i$, integral vector–covector pairs pairing to 1 are one $\mathrm{GL}_2(\mathcal{O}_q)$ -orbit. The pivot test (4.10) therefore effects the same cut. At augmentation its determinant character is trivial. Setting the position vector equal to e_1 makes invertibility of the head precisely the condition that the second entry of Y_{B_1} is a unit; the other remaining entries are integral. The orbit volume is the unmodified ratio $\mathrm{vol}(K_{A,q}) / \mathrm{vol}(T(\mathcal{O}_q))$, so it does not introduce the residue unit-group order.

At p , take Y_e in a small fixed neighborhood of e_1 , Y_{B_1} integral with any prescribed locally constant test on its second entry supported on units, and Y_{B_2}, Y_f integral. The dual coordinate of x is integral and pairs to 1 with Y_e . Such pairs form one orbit under a sufficiently deep source

congruence subgroup enlarged by the integral lower unipotent. The latter fixes φ_p by its choice above, and its transpose on the covector test leaves the relevant second entry unchanged. Thus the row-two B_1 test can have arbitrary fixed locally constant unit level while the row-two B_2 test is the integer test. The depth of Y_e and all other p -levels can be fixed once and for all. These tests still have the form (4.8).

At infinity, rotating x introduces exactly the determinant multiplier already present in the source type. The minor coefficient therefore becomes the specified second-row powers. The Heisenberg translation acts only on the first row; the second row gives (6.3). On a line, the two torus factors act by their product, so translating the binary argument by b_j is the same as $s \mapsto sb_j$, up to its explicitly removable splitting character. This proves the factorization. $\square$

The row-one function in this lemma is nonzero at some prime-to- p Jacobi torsion for every one of the finitely many torus representatives that will be needed. Indeed, the Heisenberg realization with nontrivial central character is injective by its Fourier expansion (or by the irreducibility of that representation). More concretely, with finite components fixed, finite-adelic approximation finds a rational position where the translated finite Schwartz function is nonzero. Its real Gaussian is nonzero, and modulation in the real Heisenberg variable separates the distinct rational positions. Prime-to- p torsion arguments, with compatible finite Heisenberg shifts if necessary, are dense; they therefore detect this nonzero function. Move each chosen torsion to zero by a rational unipotent and use the translated split differential frames. At p the parameters can be integral and preserve the fixed deep level. These moves affect only row one. Their normalized values form a fixed finite list of nonzero constants, costing only a fixed denominator and valuation.

6.2 The two unary characters and their separate root numbers

Split the action of T in (6.3) into the two line oscillators. Choose inverse splitting characters for the binary columns, each restricting to $\omega_{L/F}$ on $\mathbb{A}_F^\times$, with opposite infinity exponents $\pm\frac{1}{2}$ in unitary base-change normalization, that is, $z \mapsto (z/|z|)^{\pm 1}$. They may be taken unramified at p and at every nonsplit finite place, with all additional conductor supported at split places. To construct them, prescribe the restriction, infinity types, and unit conditions in the ray class construction. A possible principal relation has ratio to its conjugate a unit root of unity. A sufficiently deep split conductor forces that ratio to be 1, reducing the compatibility condition to the prescribed one on $F^\times$. Extend over the remaining finite ray class quotient and take the inverse for the other column.

Use compatible tensor splittings. Their product is trivial, as is required by the binary action of Section 4. Alternatively, a discrepancy between such splittings is a fixed finite-order character: its infinity type is trivial by the vacuum formula, and its nonsplit finite components are spherical by the matched lattices. It can be absorbed into the finite character product below, in its first slot at p . The hyperbolic factor uses the trivial splitting and contributes no stabilizer character.

The two T_∞ -weights are consequently parallel $d_j + c_j$, where c_j are fixed integers and $c_1 + c_2 = 1$. Choose characters ξ_j of these weights with

$$\xi_1 \xi_2 = \mathcal{C}^m d_0, \quad (6.4)$$

unramified at nonsplit places, and with ξ_2 unramified at p . They are powers of $\mathcal{C}$ times fixed characters; the first slot carries any required fixed p -character. The same split-conductor construction makes these choices possible.

Averaging the translates in (6.3) against $\xi_j(b_j)^{-1}$ separates the variables. At the trivial tame character the result is

$$I_T(\varphi, d_0) T_1(d_1) T_2(d_2), \quad T_j(d_j) = \int_{[U(1)]} \vartheta_{j,d_j}(s) \xi_j(s)^{-1} ds, \quad (6.5)$$

with the nonzero row-one factor understood. For the moment omit the extra unit cut at q_i .

The two degrees must eventually satisfy a joint condition: $m_i = d_{1i} + d_{2i} + 1$ tends to zero p -adically. Their p -adic limits will therefore lie on the line $d_1 + d_2 = -1$, although the classical approximating degrees are positive and tend to infinity. We will obtain a nonzero analytic period for each line oscillator and then choose a point on this compatible pair of degree discs avoiding their zeros.

The discs can be fixed before choosing the auxiliary finite-order characters in the next lemma. Represent the torus classes integrally at p , translate only their finite tests, and evaluate at fixed CM frames. The factors varying with degree are powers of the fixed $\mathcal{C}$, the fixed unit period and coordinate factors, and the unit moments of Lemma 4.5. The values of $\mathcal{C}$ on these representatives lie in the units of a fixed finite extension. A new prime-to- p finite test changes its fixed coefficients, but not the width required for those variable powers. Thus, for every fixed choice of the finite characters and tests, sufficiently narrow fixed residue discs give convergent analytic functions $\mathcal{T}_j(s)$ for the unit-cut periods. The lemma supplies a nonzero classical value for each function and makes the two discs compatible with $d_1 + d_2 = -1$.

Lemma 6.2. *The fixed finite-order parts of ξ_1, ξ_2 can be chosen, preserving (6.4), so that suitable permitted local tests give $\mathcal{T}_j(d_{j0}) \neq 0$ at positive classical cut weights d_{10}, d_{20} . The weights can be chosen so that $d_{10} + d_{20} + 1$ is divisible by every prescribed torsion order and a prescribed sufficiently large fixed power of p .*

Proof. Begin with large weights satisfying the indicated congruences. The local theta Hom for each individual line is nonzero: at a nonsplit finite place this is the unramified correspondence for the matched self-dual lattice, at infinity it is the chosen Fock degree, and at a split place there is no local obstruction.

We check the global root number separately for each line. If α is its $U(1)$ -character and χ_W the corresponding splitting character, the standard unitary Hecke character entering its central L -value is

$$\chi_W^{-1} \alpha_L, \quad \alpha_L(z) = \alpha(z/z^c).$$

Since α_L is trivial on $\mathbb{A}_F^\times$ and $\chi_W|_{\mathbb{A}_F^\times} = \omega_{L/F}$, this character restricts to $\omega_{L/F}$. The dimension-one epsilon dichotomy identifies its local epsilon factor, for the chosen trace-zero element and additive character, with the product of the two local line invariants [12, Theorem 3.5]. The product formula for these invariants is 1 for each coherent global pair of lines. Thus each of the two global root numbers is +1; it is not only their product that is known to be positive.

At infinity the pullback of ξ_j has exponent $2(d_j + c_j)$, modified by ± 1 by the splitting. After the arithmetic norm-half shift, and if necessary conjugating the CM orientation using self-duality, its critical algebraic character has type

$$(a_j + 1)\Sigma - a_j\Sigma^c, \quad a_j = d_j + O(1) \geq 0, \quad (6.6)$$

and restriction $\omega_{L/F}| \cdot |_{\mathbb{A}_F}$. Its conductor has only split prime factors.

For the simultaneous nonvanishing, use the characteristic-zero theorem of Burungale–He–Tian–Ye [17, Theorem 1.1 and Corollary 4.20]. Write $u_j = \chi_{W,j}^{-1}(\xi_j)_L$ for the unitary standard character of the j th line and put

$$\lambda_j = u_j| \cdot |_{\mathbb{A}_L}^{-1/2}.$$

This is the opposite norm-half shift from the arithmetic character in (6.6). Since the idelic norm on L restricts to the square of that on F ,

$$\lambda_j|_{\mathbb{A}_F^\times} = \omega_{L/F}| \cdot |_{\mathbb{A}_F}^{-1}, \quad L(1, \lambda_j\nu) = L(1/2, u_j\nu).$$

Thus λ_j has the self-duality convention of that theorem. For the large weights already chosen, its infinity type has the required form after choosing the corresponding CM orientation; the theorem allows an arbitrary CM type, including an induced one.

Choose a degree-one prime $\mathfrak{l}$ of F above an odd rational prime ℓ , split in L and outside all fixed conductors and p . Let $\Gamma_{\mathfrak{l}}$ be the Galois group of the maximal $\mathbb{Z}_{\ell}$ -free anticyclotomic extension of L unramified outside $\mathfrak{l}$; it is isomorphic to $\mathbb{Z}_{\ell}$. The two base signs are separately $+1$ by the preceding local calculation. At this split prime they remain $+1$ for all sufficiently ramified finite-order characters of $\Gamma_{\mathfrak{l}}$ [17, Lemma 4.12 and Section 4.4.3]. The nonvanishing theorem therefore gives finite exceptional sets for each of $L(1/2, u_1\nu)$ and $L(1/2, u_2\nu^{-1})$. Choose ν outside their union.

Because ℓ is odd, ν has a unique square root among the ℓ -power-order characters. Restrict $\nu^{1/2}$ to $U(1)(\mathbb{A}_F)$ and call the resulting character τ . Anticyclotomicity gives

$$\tau(z/z^c) = \nu^{1/2}(z/z^c) = \nu(z).$$

Replacing ξ_1, ξ_2 by $\xi_1\tau, \xi_2\tau^{-1}$ preserves Equation 6.4 and gives the two nonzero central values just selected. This adds conductor only at the split prime $\mathfrak{l}$, so all prescribed local conditions are preserved. Only characteristic-zero nonvanishing is used here.

The equal-rank line theta norm formula, equivalently the global line-lift criterion [12, Theorem 4.2 and Proposition 4.3], now supplies nonzero unary periods with suitable local tests. At p use the appropriate unit-character test for row-two B_1 and the unit cut of the unramified test for row-two B_2 ; their split local Mellin integrals are nonzero. Keep the spherical tests at nonsplit places. Compact line groups and the nontrivial infinity powers give the cuspidal situation with no occurrence from dimension zero. The nonzero output on the other line is a character, so it is already nonzero at identity: translating that argument factors through the input torus, up to the known splitting scalar. This proves the assertion at the chosen cut weights. Fix this auxiliary finite-order choice permanently. Its conductor and every additional local denominator are now fixed; enlarge the finite set in Proposition 3.2 before the final moving-prime sequence is chosen. $\square$

6.3 Interpolation and the moving-prime unit cut

Lemma 6.3. *Before imposing the additional fixed-place controls, there are positive d_{1i}, d_{2i} , with $m_i = d_{1i} + d_{2i} + 1$ tending to zero p -adically and to infinity archimedeanly, for which the theta forms have a nonconstant Fourier–Jacobi coefficient of valuation bounded above at augmentation. The choices can simultaneously satisfy the constant-term precision of Proposition 5.1.*

Proof. Apply Lemma 4.5 to the two unary periods. For the actual B_2 integer test, positive degrees tending to infinity pick out its unit cut. Thus the analytic limiting function specializes at d_{20} to the nonzero cut value obtained in Lemma 6.2; the same is true for B_1 . Hence neither $\mathcal{T}_j$ is identically zero on its chosen disc. This is the reason for testing nonvanishing with unit cuts, while retaining the integer test in the actual third column of (4.8).

By the congruence imposed on $d_{10} + d_{20} + 1$, the discs can be arranged so that s in the first disc implies $-1 - s$ in the second. The analytic functions $\mathcal{T}_1(s)$ and $\mathcal{T}_2(-1 - s)$ are nonzero functions. Choose a point a avoiding their zero sets. We may take positive integer sequences $d_{1i} \rightarrow a$ and $d_{2i} \rightarrow -1 - a$, both tending to infinity, on the required torsion branches. Their sum plus one then tends to zero. The two period values stay bounded away from zero.

Restore the unit cut above q_i . Subtracting the uniformizer translate of the relevant unary Schwartz function from its integer test and using torus equivariance expresses its effect by finitely many factors

$$1 - \alpha_{iv}\mathcal{C}(\mathrm{Fr}_{iv})^{\pm d_1}. \tag{6.7}$$

Here α_{iv} consists of residual-cardinality powers, fixed finite characters, splitting characters, and the fixed weight shifts. Its limit is a nonzero p -unit; the dependence through $d_2 = m - 1 - d_1$ is incorporated in the sign and these shifts. Pass to sublimits for the bounded-degree roots occurring in these normalizations. Proposition 3.2 gives $\mathcal{C}(\text{Fr}_{iv}) \rightarrow \mathcal{C}(\gamma)$, which is a p -unit because $\mathcal{C}$ is a continuous p -adic Galois character, and is not a root of unity. On a fixed weight branch its unit logarithm is consequently nonzero. None of the limiting analytic factors (6.7) is identically zero. Refine the choice of a to avoid their finitely many zero sets as well. The product of the two unary periods, all these local factors, the fixed period (6.1), and the chosen row-one evaluations is then bounded away from zero. The remaining m -dependent normalization factors in (6.2) are only the unit powers already treated in the mixed expansion.

There is no growing averaging denominator in turning this product back into a coefficient. The auxiliary finite-order choices and the torus levels have been fixed; the q_i unit tests are invariant under local torus units. Thus the averages over the b_j in (6.5) use a fixed finite set of representatives, chosen away from the varying q_i . If every one of the corresponding coefficient evaluations had valuation tending to infinity, so would their fixed finite weighted sum, contrary to the product just obtained. Hence one of finitely many evaluations of the positive-index coefficient has valuation bounded above. Passing to the filter chooses one option if needed.

Finally choose the positive integers d_{1i}, d_{2i} sufficiently close to their limits at each stage, while keeping them arbitrarily large. Their sum plus one, m_i , can be as close to zero as required by the fixed- i constant-term estimate. Simultaneous congruences and lower bounds on the positive integers cause no conflict. This proves the assertion with the required constant-term precision. $\square$

6.4 Fixed-place replacements with bounded denominators

We now prescribe the remaining levels at split places outside p . The comparison point is $\eta = 1$, where (4.11) consists of the elliptic block and the final pair $|\cdot|^{-1/2}, |\cdot|^{1/2}$. The separation used here excludes a ratio Q_v^n , $n \in \mathbb{Z}$, between a Frobenius root of the elliptic block and one of the final pair; Q_v is the residue cardinality. This is the separation applied in Lemma 8.7. At a place of potentially good reduction, or whenever the two blocks of (4.11) have no norm-power linkage at the comparison point, use an upper (2, 2) level: deep first Levi and opposite radical, full last Levi and upper radical. This includes a ramified quadratic Steinberg twist when the Frobenius lift is chosen with its quadratic minus sign. Use open-pivot tests with $Y_{[1,2]} \in \text{GL}_2(\mathcal{O})$, nontrivially paired with the local source vector, and the remaining columns integral. At an unramified Steinberg twist where linkage may occur, use a maximal (3, 1) parahoric and make no additional Hecke refinement there.

Lemma 6.4. *At each such fixed split place, the tests just prescribed and their translates span the tests used in Lemma 6.3 in theta coinvariants, after inverting a Laurent polynomial nonzero at weight zero. For weights tending to zero, this spanning uses finitely many translates and a common bounded denominator. All replacements can be made independently while leaving the p - and q_i -tests unchanged.*

Proof. Work in the split local theta coinvariants over the Laurent ring of the unramified determinant deformation, localized near its central value. Filter locally constant Schwartz functions on $M_{2,4}$ by rank. The rank-zero contribution would require a character quotient of the generic source representation, which is impossible. For a rank-one image fix its first covector. The stabilizer Levi $\text{diag}(1, d)$ would require exponent $|d|^{-1}$ on the raw lower Jacquet module of $\pi^\vee$: the Weil factor is $|d|^2$ and the vector-orbit Jacobian is $|d|$. In normalized Jacquet conventions the required exponent is $|d|^{-3/2}$, which is excluded by temperedness of the elliptic source representation. For a supercuspidal

source the relevant Jacquet module is zero; for a tempered principal or special source its exponents have the usual tempered bounds and do not equal $-3/2$.

These are calculations over the deformation ring, not merely at isolated characters. Use restriction of locally constant functions for the rank filtration and integration along the vector orbit, or smooth compact-induction coinvariants, on compact open charts of the row/image parameters. The difference between the required Levi character and each of the finitely many actual Jacquet characters is invertible after localization at the comparison point. Thus the lower-rank coinvariants vanish in that localization, and the open rank-two part surjects onto all coinvariants.

On the open rank-two orbit, direct orbit induction identifies the coinvariants with normalized parabolic induction of the determinant-twisted source and the trivial binary representation. At weight zero this induction is irreducible by irreducibility of unitary parabolic products for general linear groups [72]. The pivot test is nonzero there, since its head can be paired with a nonzero local matrix coefficient. Its translates consequently generate. In the Steinberg case there are nonzero $(3, 1)$ -parahoric invariants: in the induced model place the two Steinberg positions on different sides of the single-step flag. Their finite-Weyl sign condition then imposes no forbidden merging of those positions. Project an open test nontrivially to these invariants; its translates also generate the central specialization.

Only finitely many original test vectors at one fixed level must be spanned. In that finite-dimensional induced model, select a finite collection of the permitted translates whose spanning matrix has a minor f with $f(1) \neq 0$. Cramer's rule expresses the needed vectors after inverting f . The coefficients are rational functions of the unramified parameter. Since $\eta_m \rightarrow 1$ and ν_i is locally trivial at this fixed place, $f(\eta_m) \rightarrow f(1) \neq 0$. All these coefficients therefore have bounded denominator. There are only finitely many fixed places, so tensoring the finite expressions gives one bound. No inverse involving the varying place q_i has entered this argument. $\square$

If all the resulting forms with the prescribed controls had content tending to zero, then their finitely many fixed prime-to- p translates would too. Such translations preserve ordinary integrality: they are prime-to- p isogeny correspondences with unit differential comparisons, and the finitely many fixed averaging factors cost at most a fixed constant. The bounded expression in Lemma 6.4 would force the original form of Lemma 6.3 to have content tending to zero, a contradiction. At least one controlled choice therefore retains the bounded-content coefficient. A filter subsequence fixes the choice if necessary. If a finer neat level is needed, pass to a fixed-index neat cover at one fixed auxiliary place and descend by its fixed averaging factor; this changes only the uniform constant.

6.5 Positive Hecke controls on the final binary block

At each upper $(2, 2)$ level, including the moving q_i level, retain the strictly expanding center operator U_2 and the degree-one spherical operator of the last GL_2 in the normalized Jacquet module. They act on our test with the final allocation in (4.11), namely the two characters $|\cdot|^{-1/2}, |\cdot|^{1/2}$ alone. In particular the center expansion value is invertible.

For completeness this assertion follows at the level of the tests. Represent double cosets of the expanding monoid in the last block and lift them with the upper shifts. The invertible head determines exactly one surviving upper shift for each last-block coset. For both $\varpi^{-1}1_2$ and $\mathrm{diag}(\varpi^{-1}, 1)$, the determinant factor in (4.7) is exactly the factor that undoes the Jacquet half-modulus. After normalized Jacquet passage, the two integral trailing columns therefore carry the spherical action of the trivial binary representation. Its Satake pair is $(Q^{-1/2}, Q^{1/2})$, where Q is the residue cardinality. The first block may have the principal congruence level needed at q_i ; it does not change this count.

Iwahori decomposition for the block parabolic makes these positive operators commute. Their normalizations use only powers of the local residue characteristic. Inverting the strictly contracting operator identifies them with the stated Jacquet operators by Jacquet lifting [20]. At the fixed places the replacements in Lemma 6.4 can be made with these tests from the outset. The p - and q_i -requirements remain unchanged.

Proposition 6.5 (The combined theta output). *Fix the truncation b_* selected in Section 3. There exist positive integers $m_i \rightarrow \infty$, tending p -adically to zero on the prescribed torsion branch, and theta forms Θ_i of weight $(m_i, m_i, 1; -1)$ with the following simultaneous properties.*

- (i) *Their ordinary monomial-dual lattices have a uniform denominator bound, coefficientwise in the full tame group algebra and after every fixed truncated substitution used in the argument. One fixed scaling clears this bound.*
- (ii) *All constant Fourier–Jacobi coefficients tend to zero through degree $b_* - 1$ of the tame variable, in compatible ordinary differential bases, uniformly at the boundary representatives. This is a full-ring estimate in R_{b_*} .*
- (iii) *At augmentation, a nonconstant Fourier–Jacobi coefficient has an integral-frame evaluation of valuation bounded above by a constant independent of i . Thus the forms do not acquire unbounded common content.*
- (iv) *At both distinguished places above p , the fixed canonical $(2, 1, 1)$ level is J and the operators $p^{-2}U_2, p^{-2}U_3$ have value 1. At every controlled upper $(2, 2)$ place, including q_i , the expanding center is invertible and the last binary spherical data are those of the final pair in (4.11). At the specified unramified Steinberg places the level is $(3, 1)$ parahoric, without an additional refinement. All nonsplit finite places are hyperspecial.*
- (v) *At split good spherical places the Hecke parameters are (4.11) with $\eta = \mathcal{C}^{m_i}\nu_i$.*

All constants may depend on the fixed original data, the fixed local tests and b_ , but not on i, q_i, n_i or the growing weights.*

Proof. Lemma 6.3 chooses the degrees and obtains the content bound simultaneously with Proposition 5.1. The bounded fixed-place spanning and the preceding content argument impose all additional controls without losing that bound. Every selected test still satisfies the hypotheses of Proposition 4.3 and the constant-term proof, which allowed arbitrary bounded fixed split tests. The explicit positive-operator calculation gives the last binary allocations. If finitely many choices remain, make the constant-term approximation sufficiently accurate for all of them and then choose the one with bounded content. This gives all the properties at once. $\square$

7 Cusp lifting on the ordinary locus

We pass from the theta sections of Propositions 4.3, 5.1 and 6.5 to the Hecke algebra acting on classical cusp forms. Throughout this section, ordinary refers to the integral PEL variety for the target unitary group. Its integral datum is hyperspecial and totally split at p ; the argument places no reduction condition on A_0 .

There are two steps. We first lift the theta sections, whose constant terms vanish modulo increasing precision, to genuine characteristic-zero cusp forms in a higher weight. We then use their bounded nonconstant content to show that every relation among the cusp-form Hecke operators vanishes at the theta comparison values. The second step produces a homomorphism from the entire operator order to the truncated tame ring. Both steps use the same integral coefficient lattice.

Put $b = b_*$ and $\mathcal{O}_b = \mathcal{O}[t]/(t^b)$. We use the following combined output of the preceding sections. After one fixed clearing of denominators, the theta sections have bounded coefficients in the dual-Fock lattice of Equation (4.6), expressed in ordinary differential frames, and their constant terms tend to zero coefficientwise in $\mathcal{O}_b$. At augmentation, one nonconstant Fourier–Jacobi coefficient, evaluated in a unit frame, has valuation bounded above by a constant independent of i . Their weight at each real place is $(m, m, 1; -1)$, and U_2 and U_3 have eigenvalue p^2 at each distinguished p -adic place. All the fixed finite local conditions of Proposition 6.5 are retained. In particular, these include the binary controls at the indicated $(2, 2)$ levels and the $(3, 1)$ parahoric condition at the indicated Steinberg places. The coefficient fields of individual sections and evaluations may vary with i ; all valuation bounds below are uniform under these finite coefficient extensions. We normalize $v_p(p) = 1$.

7.1 The canonical branch of the actual level

Choose a neat tame base level. The depth s of the p -level is fixed. Enlarge k once to contain the required p^s -th roots of unity and choose compatible cyclotomic frames. All models and finite normalizations in this section are taken after this fixed base extension. Write X^{tor} and X^{min} for compatible hyperspecial integral toroidal and minimal compactifications. We use the canonical differential bundles, the formal toroidal charts over abelian extension-data schemes, and the ample total Hodge line bundle $\mathcal{L}$ on X^{min} . These are the good-prime PEL constructions of Lan [54]; the formal-chart statement used here is Theorem 6.4.1.1, and the Fourier–Jacobi descriptions are in Section 7.1.2, of the author revision specified in that reference. The additional p -level will be imposed by finite normalization, rather than by an assumption that this finer level is hyperspecial.

For the corresponding ordinary-level constructions, see Lan [55, Definition 3.2.2.9 and Propositions 5.2.3.3, 5.2.3.18]. We identify the particular branch and prove the base-change statement needed here below.

Let $\pi : Y^{\text{tor}} \rightarrow X^{\text{tor}}$ be the finite normalization in the generic-fiber level J . At each distinguished p -adic factor the ordinary p -divisible group has multiplicative height three and etale height one. The level J has the upper $(2, 1, 1)$ block structure: the first rank-two Levi is congruent to the identity at the chosen fixed depth, the last two unit factors and the upper radical are full, and the lower blocks have the prescribed depths. Equivalently, it specifies a rank-two flag with the indicated frame inside a rank-three flag. It does not mark lifts of the etale quotient.

Lemma 7.1 (Canonical ordinary level). *Over the ordinary formal completion, the locus on which the rank-three flag equals the multiplicative subgroup is an open and closed branch of Y^{tor} . Its residual flag and frame data are finite etale data on the Cartier-dual character group. On a toroidal cusp chart this branch adds finite etale covers of the abelian extension-data scheme and does not ramify the center Fourier variable.*

Proof. The ordinary group has a canonical multiplicative subgroup at every fixed level p^s [47, proof of Theorem 2.1, pp. 149–150]. Once the rank-three flag is this subgroup, its rank-two subflag and frame are equivalent by Cartier duality to quotients and markings of its finite etale character group. The full upper radical of J is essential: specifying a splitting of the etale quotient would impose different data. The canonical subgroup separates the corresponding generic level choices. Their idempotents extend in the finite normalization over the ordinary completion; equivalently, one may normalize after completion, since the base is excellent. This proves the asserted branch decomposition.

At a rank-one boundary component, the Raynaud semiabelian group is an extension of the binary abelian group by a rank-one torus. The preimage of the binary multiplicative torsion is

an extension of multiplicative-type finite flat groups, hence is of multiplicative type: its Cartier dual is an extension of finite étale groups. This is the entire distinguished rank-three multiplicative subgroup. Its construction precedes quotienting by the Raynaud period lattice and does not involve the center Fourier parameter. Its character-group level therefore produces only the asserted finite étale chart covers.

After passage to a completed strict boundary base and choice of an origin, a connected finite étale cover of an abelian chart is an abelian scheme and the covering map is an isogeny. On a geometric fiber this follows by lifting addition to the pointed cover: the subgroup defining the cover in the abelian fundamental group is preserved by addition, and uniqueness of pointed lifts gives the group laws. The same construction extends over the strict base by the equivalence for finite étale covers under henselian thickening. Thus the new charts have the same abelian geometry needed for Fourier–Jacobi coefficients; compare the ordinary chart description in Lan [55, Proposition 4.2.1.34 and Corollary 4.2.1.36]. $\square$

We will use ordinary expansions to test vertical integrality. Every component at the neat hyperspecial base level has an ordinary cusp: the boundary datum is split ordinary CM, and component comparison for the smooth good-prime compactification identifies these components with those in characteristic zero. The ordinary locus is dense. Each canonical finite étale component also meets the corresponding completed cusp charts, since a connected finite étale cover maps onto its base component. Consequently vanishing of all reduced expansions detects a vertical factor of a section; successive reduction gives the same assertion modulo every power of a uniformizer.

The expansion representatives can be chosen integral at p , at q_i , and at all the fixed exceptional places simultaneously. Indeed, Iwasawa decomposition reduces this to weak approximation in the cusp parabolic; the binary unitary factor has a rational Cayley chart. The same approximation permits the prime-to- p Jacobi torsions used for bounded content, with denominators prime to the other specified places. At the remaining places spherical Iwasawa decomposition applies. The unipotent changes do not alter a constant term, and the Levi changes translate its binary argument or multiply its frame by a unit at p . Thus the bounds from the preceding sections apply in all these charts.

7.2 Positive Fourier–Jacobi coefficients and base change

Let D be the reduced toroidal boundary of X^{tor} . For the weights in use, put $d = m - 1$ and retain the coefficient functor from Equation (4.6):

$$\mathcal{D}_d(\mathcal{H}) = (\text{Sym}^d((\wedge^2 \mathcal{H})^\vee))^\vee.$$

Here $\mathcal{H} = \omega^+$ is the rank-three positive Hodge bundle. The inverse Fock-frame convention fixed there identifies this functor with the literal dual of the Fock monomial lattice. Define $\mathcal{E}_\lambda$ on Y^{tor} using, at each distinguished embedding, the positive factor

$$(\det \omega^+)^{\lambda_3} \otimes \mathcal{D}_d(\omega^+)$$

and the opposite differential line to exponent $-\lambda_4$. In particular this is the same integral coefficient bundle as in the theta expansions; a Hasse shift changes only its two line-bundle factors. Define the coherent sheaf

$$\mathcal{M}_\lambda = (Y^{\text{tor}} \longrightarrow X^{\text{min}})_*(\mathcal{E}_\lambda(-\pi^* D)).$$

The pulled-back boundary is used in this definition. By Lemma 7.1, its restriction to the canonical ordinary cusp charts has the usual integral Fourier indices.

Lemma 7.2 (Integral operations on the coefficient lattice). *The functor $\mathcal{D}_d$ is finite locally free and commutes with base change. Integral changes of Hodge frame preserve its lattice. If a map on $\mathcal{H}$ is diagonal with entries c_1, c_2, c_3 , its action on the basis dual to the minor monomials has entries*

$$(c_1 c_2)^{a_{12}} (c_1 c_3)^{a_{13}} (c_2 c_3)^{a_{23}}, \quad a_{12} + a_{13} + a_{23} = d.$$

A filtration of $\mathcal{H}$ by subbundles with constant locally free graded pieces induces such a filtration on $\mathcal{D}_d(\mathcal{H})$.

Proof. For a finite locally free module $\mathcal{W}$, its divided-power module is

$$\Gamma^d(\mathcal{W}) = (\mathrm{Sym}^d(\mathcal{W}^\vee))^\vee.$$

Use this integral definition with $\mathcal{W} = \bigwedge^2 \mathcal{H}$. A minor monomial and its dual basis vector have pairing one, without a factorial. All the modules in this definition are finite locally free, so it commutes with base change. Functoriality for linear maps proves preservation under integral frame changes and gives the displayed diagonal entries, including when some c_j is not a unit.

To check the filtration assertion, split the filtration locally. Exterior and symmetric powers have their degree filtrations, with graded pieces given by tensor products of the corresponding powers of the original graded pieces. Dualizing the symmetric-power filtration gives the divided-power filtration. Block-triangular changes between local splittings preserve these filtrations, so they descend. Their graded pieces are constant whenever those of $\mathcal{H}$ are constant. No rational splitting or rational identification with ordinary symmetric powers is used. $\square$

Lemma 7.3 (Cusp base change on the canonical branch). *An ordinary section modulo p^M on the canonical branch whose constant terms vanish, extended by zero on the other branches, is a section of $\mathcal{M}_\lambda/p^M$ over the ordinary open. The assertion remains true with coefficients in $\mathcal{O}[t]/(p^M, t^b)$ and after finite extension of coefficients.*

Proof. The boundary has rank one over L , and the residual binary Shimura datum is zero dimensional. In its formal charts, a cusp tuple is indexed by strictly positive integral $\alpha \in F$. Here positivity means positivity at both real embeddings. There are no nonzero rational indices on the boundary of the positive cone: an element of F with one embedding zero is zero. Vanishing of the constant term therefore imposes positive order on every relevant cone ray.

For $\alpha > 0$, the coefficient is a section on the abelian extension-data chart C of

$$\mathcal{L}_\alpha \otimes \mathcal{E}_\lambda|_C,$$

where the Jacobi line bundle $\mathcal{L}_\alpha$ is ample. This description persists on the finite etale abelian covers in Lemma 7.1. The semiabelian differential sequence has constant locally free graded pieces over the fixed boundary base. Apply Lemma 7.2, and tensor with the determinant and opposite differential line factors. This gives $\mathcal{E}_\lambda|_C$ a finite filtration with constant graded pieces, in precisely the integral lattice used by the theta section. This construction needs no rational splitting of the semiabelian differential sequence.

We recall why the required vanishing is valid in every residue characteristic. If L is ample on an abelian variety, choose multiplication by an integer n prime to the characteristic. Its trace splits $\mathcal{O}$ inside $[n]_* \mathcal{O}$, so $H^r(L)$ is a summand of $H^r([n]^* L)$. For sufficiently large such n , the latter vanishes for $r > 0$: $[n]^* L$ has arbitrarily large ample degree, and Serre vanishing is uniform under the algebraically trivial twists occurring in the theorem of the square, by properness of the dual abelian variety. The filtration just described gives the same vanishing after tensoring with $\mathcal{E}_\lambda|_C$.

Cohomology and base change, or induction through the exact sequences for successive uniformizer powers, then lifts every positive coefficient modulo p^M integrally.

It remains to lift the arithmetic identifications between coefficients. Modulo translations, the stabilizer acts on positive indices by unit norms. Fixing a positive index forces this norm to be one; the resulting CM norm-one unit group is finite. The remaining binary arithmetic stabilizer is finite by definiteness. Neatness removes these finite stabilizers. We may therefore choose one lift in each free index orbit and transport it around the orbit, without averaging or dividing by an orbit order.

The resulting tuples converge on every formal cone. In fact, an interior rational ray gives a linear functional strictly positive on the nonnegative index cone. A bound on its value bounds both real coordinates of a positive index, and hence leaves only finitely many points of the index lattice. This is exactly the formal convergence condition on the completed monoid algebra. The formal-chart description and formal functions now identify these tuples with the completed pushforward sections; see Lan [54, Section 7.1.2] and Hartshorne [39, Chapter III, Theorem 11.1]. Thus the required reduction map is surjective at a cusp. Its injectivity follows from applying the left-exact pushforward to multiplication by a uniformizer on the torsion-free source sheaf. Off the boundary the assertion is immediate from the finite level normalization. Faithfulness of completion for coherent modules gives the assertion on the ordinary open.

The branch decomposition allows the zero lift on every other branch. Finally, $\mathcal{O}[t]/(t^b)$ is a finite free $\mathcal{O}$ -module, so the entire construction applies coefficientwise and gives the stated Artin-ring assertion. This proves the required special case directly; the general ordinary subcanonical vanishing theorem is Lan [55, Theorem 8.2.1.3, with the setup of Section 8.1]. $\square$

7.3 Hasse extension and the shifted weight

The base-change lemma turns vanishing constant terms into a cusp section modulo the working precision on the ordinary open. We now extend that section over the compactification and lift it to characteristic zero, retaining its coefficients and Hecke relations modulo the same precision.

Choose positive integers m_i tending to infinity and tending p -adically to zero on the fixed torsion branch, as in the preceding sections. Choose integers $M_i \rightarrow \infty$ sufficiently slowly along the filter that, after the fixed clearing of denominators, all constant coefficients vanish modulo p^{M_i} through degree $b - 1$. The theta section is then a cusp section on the ordinary open over $\mathcal{O}[t]/(p^{M_i}, t^b)$ by Lemma 7.3. We also require $M_i \leq n_i - c_b$, with c_b a fixed constant large enough for the truncated tame substitution of Section 3. Indeed, the coefficients of $(1 + t)^{p^{n_i}} - 1$ below degree b have valuations at least $n_i - c_b$. Thus the finite group relation is respected at this precision. All subsequent congruences use this one common precision for the full ring.

The ordinary locus is the nonvanishing locus of the total Hasse invariant. A section of a coherent sheaf on this locus extends over $X^{\min}$ after multiplication by a sufficiently high power of Hasse. The assertion follows from localization on a finite affine cover. It applies to the sheaf $\mathcal{M}_\lambda$ and to each of the finitely many t -coefficients simultaneously.

For precision M , a sufficiently large p -power of Hasse lifts modulo p^M . To see this directly, choose local lifts of Hasse. On overlaps their differences are divisible by p ; raising to p^r makes those differences divisible by p^{r+1} , with the bundle transition functions included. They therefore glue for r sufficiently large. In multiplicative differential frames, the resulting lift reads 1 modulo p^M . Its exponent can be increased arbitrarily while retaining this property.

Twisting farther by the ample Hodge line gives, by Serre vanishing [39, Chapter III, Theorem 5.2 and Proposition 5.3], surjectivity of reduction on global sections. We may consequently lift the extended section to characteristic zero, coefficientwise in $\mathcal{O}[t]/(t^b)$. Choose the Hasse weight shift H_i arbitrarily large and divisible by $(p - 1)p^{M_i + c}$, with c any needed fixed constant. The resulting

weight, at each real place, is

$$\lambda'_i = (m_i + H_i, m_i + H_i, 1 + H_i; -1 - H_i). \quad (7.1)$$

These generic sections are genuine cusp forms: the sheaf before pushforward was twisted by minus the pulled-back boundary.

The size of H_i , the point at which Serre vanishing applies, and the dimensions of the resulting spaces may depend on i , m_i and M_i . None of these choices consumes congruence precision. If a finer neat cover was used, descent by its fixed averaging operator costs only the valuation of its fixed index. We absorb this constant by reducing M_i once. There is no averaging at the varying prime q_i .

7.4 The two integral normalized operators

Write U_j for the actual double-coset operator, with the measures and expanding-lattice convention used in the theta construction. For either the original or shifted weight put

$$\tilde{U}_j = p^{-N_j(\lambda')} U_j, \quad N_2(\lambda') = 2 + \lambda'_3 + \lambda'_4, \quad N_3(\lambda') = 3 + \lambda'_4. \quad (7.2)$$

Lemma 7.4 (Integral trace calculation). *The operators in Equation (7.2) preserve the dual-Fock differential lattice on the canonical ordinary branch and its reductions modulo every power of p . After the Hasse shift, both comparison eigenvalues are 1 modulo the prescribed precision.*

Proof. It suffices to work at one distinguished p -adic factor. The split PEL idempotents select the rank-three multiplicative character lattice and the rank-one étale lattice; polarization determines the conjugate factor. Choose Serre–Tate coordinates q_1, q_2, q_3 for their bilinear pairing [47, Theorem 2.1(1),(2)]. Locally their deformation ring is

$$R = \mathcal{O}[[q_1 - 1, q_2 - 1, q_3 - 1]].$$

For U_j , expand the lattice transverse to a multiplicative j -summand by p^{-1} . The expanded multiplicative directions are selected by $p^{j(3-j)}$ integral graphs. Fix one graph. Functoriality of the Serre–Tate pairing [47, Theorem 2.1(4)] for the lattice inclusion and p times the reverse inclusion shows that the remaining extension choices replace the j retained coordinates by their p th roots; the other $3 - j$ coordinates are unchanged. Indeed, expanding the étale generator requires a root when paired with a retained multiplicative character, whereas the simultaneous expansion of a transverse multiplicative direction cancels this change for the other coordinates.

The corresponding finite free extension is the appropriate completion of

$$R' = R[u_1, \dots, u_j] / (u_a^p - q_a : 1 \leq a \leq j).$$

Its basis consists of $\prod_a u_a^{e_a}$ with $0 \leq e_a < p$. The trace of every such basis element except 1 is zero, and $\text{Tr}_{R'/R}(1) = p^j$. Hence

$$\text{Tr}_{R'/R}(R') \subseteq p^j R. \quad (7.3)$$

The same monomial calculation applies on a boundary cone whenever the Hecke correspondence takes a p th root of a torus coordinate. For $j = 2$ there are p^2 graphs and a degree- p^2 trace, and for $j = 3$ there is one graph and a degree- p^3 trace. Thus the total counts are $p^{j(4-j)}$, as required by the double cosets.

The transported multiplicative flags and differential comparison maps depend only on the graph, not on these root choices. They can therefore be put outside the trace. In an integral graph basis,

pullback scales $3 - j$ positive differential directions by p . The opposite differential scales by p^{-1} , by duality to the distinguished etale direction.

For the weights in use, the positive integral coefficient lattice is

$$(\det \omega^+)^{\lambda'_3} \otimes \mathcal{D}_{m-1}(\omega^+),$$

and the opposite line occurs with exponent $-\lambda'_4$. By [Lemma 7.2](#), the integral graph changes preserve this lattice. For $j = 2$, choose a graph basis in which the positive differential map is $\text{diag}(1, 1, p)$. It acts on the three minor directions with entries $1, p, p$; hence on their monomial-dual coefficient of degrees (a_{12}, a_{13}, a_{23}) it acts by $p^{a_{13}+a_{23}}$. This is integral for every m , without a factorial or a change of lattice. The determinant factor supplies $p^{\lambda'_3}$. For $j = 3$, the positive differential map is the identity in a graph basis. The opposite line supplies $p^{\lambda'_4}$ in both cases. Before taking trace, the lower bounds are consequently

$$\lambda'_3 + \lambda'_4 = N_2 - 2 \quad (j = 2), \quad \lambda'_4 = N_3 - 3 \quad (j = 3).$$

Combining these with [Equation \(7.3\)](#) gives p^{N_j} divisibility. The graph sum introduces no denominator. This proves integrality of $p^{-N_j}U_j$ on the stated lattice, including when N_j is negative. The other p -adic factor is a spectator, so the argument applies independently at both places.

In the original weight $(m, m, 1; -1)$, both N_j are 2; the theta eigenvalue p^2 thus gives normalized value 1. The shift $(H, H, H; -H)$ changes N_2 by zero and N_3 by $-H$. Its differential determinant contributes $p^{(2-j)H}$ on the same graph, precisely this change of normalization. The remaining unit frame factors, raised to the chosen divisible H , are 1 modulo p^M ; in canonical multiplicative frames the lifted Hasse section itself is 1. Thus multiplication by the Hasse lift intertwines the normalized operators modulo p^M . Explicitly, in the shifted weight

$$N_2 = 2, \quad N_3 = 2 - H.$$

Prime-to- p Hecke correspondences preserve the canonical branch and its integral lattice as well. Their Hodge determinant changes are p -adic units; the same divisibility of H makes their additional weight factors trivial modulo precision. Hence all prescribed comparison eigenvalues survive the lift. $\square$

7.5 The Hecke order and full Artin-ring factorization

The lifted cusp section now has the theta comparison eigenrelations on the ordinary locus. To obtain a map from the Hecke order, we must show that these values satisfy every relation in its classical action. The nonconstant coefficient retained through the lift will let us cancel the section from such a relation with only a fixed precision loss.

Let S_i be the classical cusp space in the weight given by [Equation \(7.1\)](#) and at the prescribed finite level, with the quotient detected on the canonical ordinary locus taken if necessary. We use its lattice of sections integral in the same dual-Fock differential bundle, expressed in ordinary frames. This is a full bounded lattice. Indeed, finitely many ordinary evaluations detect a finite-dimensional classical space and bound the lattice above; clearing vertical denominators of a basis bounds it below. These statements may be checked over a finite model field depending on i . More explicitly, if $\mathcal{Q}_i^0$ is the module of integral ordinary sections and S_i is its rational classical subspace, the lattice is $S_i \cap \mathcal{Q}_i^0$. Consequently $w \in S_i \cap p^M \mathcal{Q}_i^0$ implies $w \in p^M (S_i \cap \mathcal{Q}_i^0)$. Approximate classical eigenrelations on the ordinary locus therefore give the same precision in this lattice, without a further denominator loss. [Lemma 7.4](#) and the prime-to- p correspondences preserve this lattice.

Define T_i^0 to be the $\mathcal{O}$ -algebra of commuting operators on this space generated by the split good spherical coefficients with Satake normalization $Q_v^{3/2}$, the operators $\tilde{U}_2, \tilde{U}_3$, and the prescribed positive binary controls with their Frobenius scalings. Those latter scalings use only powers of residue characteristics different from p . We retain the specified invertible center expansion and last-block spherical degree-one control at each binary place. Over each finite model field the algebra is contained in the endomorphism ring of the lattice. It is therefore a finite torsion-free $\mathcal{O}$ -algebra: an $\mathcal{O}$ -submodule of that finite endomorphism lattice is finite. Neither its rank nor the size of the model field is required to be bounded as i varies.

Lemma 7.5 (Uniform cancellation in a truncated ring). *Let K'/k be a finite extension, and let $a(t) \in \mathcal{O}_{K'}[t]/(t^b)$ satisfy $v_p(a(0)) \leq C$, where C is a fixed nonnegative integer. If $a(t)r(t)$ belongs to $p^M \mathcal{O}_{K'}[t]/(t^b)$, then*

$$r(t) \in p^{M-bC} \mathcal{O}_{K'}[t]/(t^b).$$

The bound is independent of the extension and of a .

Proof. Write $a = a_0(1 + u)$, with $u \in t\mathcal{O}_{K'}[1/p][t]/(t^b)$. Then

$$a^{-1} = a_0^{-1} \sum_{j=0}^{b-1} (-u)^j \in p^{-bC} \mathcal{O}_{K'}[t]/(t^b).$$

Multiplication by this inverse proves the assertion. The use of the constant coefficient is essential, since it makes a invertible over $K'[t]/(t^b)$. $\square$

Proposition 7.6 (Classical cusp congruence). *After reducing M_i by a fixed constant and relabeling it, there are homomorphisms of $\mathcal{O}$ -algebras*

$$\lambda_i^0 : T_i^0 \longrightarrow (\mathcal{O}/p^{M_i} \mathcal{O})[t]/(t^{b_*}), \quad M_i \longrightarrow \infty, \quad (7.4)$$

which send every indicated operator to its theta comparison eigenvalue, in the full truncated ring. The weights are Equation (7.1), and the two normalized p -operators have value 1 at each distinguished place. One may restrict T_i^0 to the local summand determined by the residual augmentation system. In that summand both normalized p -operators are units. No reduction of the operator algebra is asserted here.

Proof. The comparison eigenvalues lie in the fixed coefficient field k and are integral in the specified normalizations. This follows from the theta parameter, the normalized value 1 at p , and the rational auxiliary block parameters. Away from p , the Frobenius-scaled polynomial coefficients are integral; the character and residual-cardinality factors in these normalizations are p -adic units. The coefficients of $(1+t)^\ell$ are integral for $\ell \in \mathbb{Z}_p$, and the preceding choice $M_i \leq n_i - c_b$ makes the finite tame relations valid. Consider the polynomial $\mathcal{O}$ -algebra on all the chosen Hecke generators and its evaluation at these comparison eigenvalues in the target of Equation (7.4). We must show that every relation in its action on S_i belongs to the kernel of this evaluation.

Let s_i be the lifted cusp section, viewed with its t coefficients. On the canonical ordinary locus it agrees modulo the working precision with Hasse times the integral theta section. Every generator acts there with the prescribed comparison value modulo that precision, by Lemma 7.4. The same is true for any integral polynomial in the generators: multiplication and addition of integral operators consume no precision, regardless of the degree or number of its monomials. If P is a relation on the classical cusp space, it follows that

$$P(\text{comparison values}) s_i \equiv 0 \pmod{p^{M_i}}.$$

Evaluate the nonconstant coefficient furnished by [Proposition 6.5](#), using the same Fock monomial and its literal dual coefficient basis. The definition of $\mathcal{E}_{\lambda'_i}$ retains this integral pairing through the lift. Hasse is 1 in that frame modulo precision, so its value $a_i(t)$ has integral coefficients and $v_p(a_i(0)) \leq C$ for one fixed C . The preceding display and [Lemma 7.5](#) imply

$$P(\text{comparison values}) \equiv 0 \pmod{p^{M_i-bC}, t^b}.$$

The scalar polynomial on the left has coefficients in k ; integrality and divisibility can be descended from the finite evaluation field because $p^n \mathcal{O}_{K'} \cap \mathcal{O} = p^n \mathcal{O}$. The fixed loss bC , together with the earlier fixed clearing and neat-descent losses, is independent of P and i . Reducing and relabeling M_i gives the factorization through the entire operator order T_i^0 .

Finally, a finite algebra over the complete local ring $\mathcal{O}$ decomposes as a product of its factors at the residual maximal ideals. The target ring is local, with residue field obtained by setting $t = 0$ and reducing modulo the uniformizer. Hence λ_i^0 factors through the summand selected by this residual augmentation. The images of $\tilde{U}_2, \tilde{U}_3$ are 1, so these operators lie outside that summand's maximal ideal and are units there. This construction keeps every coefficient through degree $b_* - 1$ and permits nilpotents in T_i^0 , as required for the next section. $\square$

8 Characteristic-zero parameters and local control

We attach Galois representations to the genuine cusp systems of [Proposition 7.6](#), retain the local information supplied by the prescribed operators, and then remove the nilpotents in the operator order with a uniform loss of congruence precision. The local information takes the form of invariant subspaces and polynomial identities in inertia matrices. These identities will also impose the local conditions on the extensions extracted in [Section 9](#).

8.1 Simultaneous automorphic transfer

Put $G = \mathrm{U}(B \oplus \mathbb{H})$, with the Hermitian space and conventions of [Section 4](#). Thus $G(F_v) \simeq \mathrm{U}(3, 1)$ at each real place, G is quasi-split at every finite place, and its proper F -Levi subgroup has real points $\mathbb{C}^\times \times \mathrm{U}(2)$ at either real place. Let π be a cuspidal automorphic representation occurring in [Proposition 7.6](#). Its infinite components are holomorphic discrete series of lowest compact type λ' from [Equation \(7.1\)](#). At the finite places nonsplit in L , its components are hyperspecial spherical.

For a unitary cuspidal representation τ of $\mathrm{GL}_d(\mathbb{A}_L)$, write $\tau[a]$ for its Arthur block of length a and put

$$S_a = \{(a-1)/2, (a-3)/2, \dots, -(a-1)/2\}.$$

The local standard representation attached to this block is the Langlands quotient of its shifted cuspidal data. In particular, for a finite place w of L its Weil–Deligne parameter is

$$\mathrm{rec}_w(\tau_w[a]) = \bigoplus_{s \in S_a} \mathrm{rec}_w(\tau_w) \otimes |\cdot|_w^s. \quad (8.1)$$

The monodromy on the right is the direct sum of the monodromies of the shifted copies. This is the Arthur-block convention, as opposed to a new Steinberg block. Local Langlands and the classification of the unitary dual of general linear groups give [Equation \(8.1\)](#); normalized products of the resulting unitary representations are irreducible. At split places these character identities are the linear-group case of endoscopic transfer, also treated in Kaletha et al. [\[42, Section 1.6.3\]](#).

We obtain one formal sum of these blocks, with its full infinite character and finite local parameters, from Labesse [\[53, Corollary 5.3\]](#). The needed sign property of the individual holomorphic

discrete series is verified below at each real place. Labesse's simple trace-formula argument applies because $[F : \mathbb{Q}] = 2$ and does not use the weighted fundamental lemma [53, Sections 4.8 and 5.1]. In Equation (7.1) we take $H_i \geq 3$, as we may while preserving the specified divisibility and congruences.

Lemma 8.1 (Weak transfer with full infinite characters). *There is a conjugate-dual formal parameter*

$$\Psi = \boxplus_{\beta} \tau_{\beta}[a_{\beta}], \quad \sum_{\beta} d_{\beta} a_{\beta} = 4,$$

with unitary cuspidal factors, the standard transferred full infinitesimal character of π at both real places, and the transferred Satake parameter at every finite place outside a finite set S whose finite places split in L/F . Every cuspidal factor is unramified at every nonsplit finite place.

Proof. First consider one real factor $U(3, 1)$, and omit the subscript i . The Harish–Chandra coordinates of the selected holomorphic discrete series D are

$$k = (m + H + \tfrac{1}{2}, m + H - \tfrac{1}{2}, H - \tfrac{1}{2}, \tfrac{1}{2} - H).$$

Their successive gaps are $1, m, 2H - 1$. They are regular integral, and the unique negative-sign coordinate of D is k_4 . Choose its individual pseudocoefficient f_D , with trace one on D and trace zero on every other basic standard module [28, Proposition 4 and its corollary, pp. 212–213]. In particular it annihilates all properly parabolically induced standard modules. Standard-module expansions preserve infinitesimal character, so an irreducible representation with nonzero trace against f_D must have infinitesimal character k .

Every unitary representation with this regular integral character is an admissible cohomologically induced module $A_{\mathfrak{q}}(\mu)$ [65, Theorem 1.2]. That theorem applies to $SU(3, 1)$; adjoining the compact central character gives the same assertion for $U(3, 1)$. Write the theta-stable Levi of $\mathfrak{q}$ as ordered signature blocks $\prod_j U(p_j, q_j)$, where $\sum p_j = 3$ and $\sum q_j = 1$. The inducing character is constant on each Levi block and dominant across the nilradical roots. Hence the infinitesimal coordinates in each block form a consecutive string, with successive difference one, and the blocks occur in their decreasing order in k . There is exactly one block with $q_j = 1$.

If this Levi is noncompact, its block containing the negative coordinate has size at least two. In the standard-module resolution of $A_{\mathfrak{q}}(\mu)$, the discrete-series terms arise from positive systems extending the nilradical roots [1, Theorem 8.2 and Lemma 8.8]. They may reorder coordinates within a Levi block, but not between blocks. Their negative coordinate must therefore belong to that same block. For D to occur, the block would have to contain k_4 . A consecutive string of size at least two containing k_4 also contains k_3 and forces $k_3 - k_4 = 1$, contrary to $k_3 - k_4 = 2H - 1 \geq 5$. Thus the coefficient of D in this resolution is zero. All other terms are annihilated by f_D . If the Levi is compact, the induced module is itself discrete series, and its trace is one precisely when it is D . We have proved that $\text{tr } X(f_D)$ is zero or one for every irreducible unitary X , with value one only for $X = D$. This argument treats each real factor separately. Their product pseudocoefficient therefore satisfies the sign property $(*)$ of Labesse [53, Section 5.1], including competitors with different types at the two real places. No lower bound on the first gap beyond its actual value one is used.

Apply Labesse [53, Corollary 5.3] to π . It gives one automorphic transfer induced from conjugate-dual discrete general-linear representations. The discrete-spectrum classification of Mœglin and Waldspurger [58] expresses its factors as Arthur blocks and gives the formal parameter

$$\Psi = \boxplus_{\beta} \tau_{\beta}[a_{\beta}], \quad \sum_{\beta} d_{\beta} a_{\beta} = 4,$$

with unitary cuspidal factors. The same corollary gives local base change at every split finite place and at every unramified place. It also retains the full infinitesimal character at both real places:

its proof uses the coefficient systems for the complex group and compatibility with infinitesimal-character multipliers, not just the Casimir eigenvalue. We do not invoke its additional conclusion requiring cuspidality of the transfer.

Choose S to contain the infinite places and the finite ramification of π and the chosen standard embedding. All its finite places can be chosen split in L/F . At a nonsplit finite place, the transferred representation is spherical, so its Weil–Deligne parameter has trivial inertia and zero monodromy. Formula (8.1) and compatibility of local Langlands with the normalized product express this parameter as the direct sum of the shifted parameters of $\tau_{\beta,w}$. Each summand therefore has trivial inertia and zero monodromy, so each $\tau_{\beta,w}$ is unramified. The corollary’s stronger matching at every split place concerns this same Ψ and will also be used below. $\square$

Lemma 8.2 (Algebraicity of the cuspidal factors). *Every cuspidal factor in Lemma 8.1 is conjugate-self-dual and tempered at infinity. After a unitary parity-correcting character twist it is regular algebraic. The formal parameter is elliptic: its conjugate-self-dual blocks occur without repetition.*

Proof. At a distinguished complex embedding the first infinitesimal coordinates are

$$(k_1, k_2, k_3, k_4) = (\lambda'_1 + \tfrac{1}{2}, \lambda'_2 - \tfrac{1}{2}, \lambda'_3 - \tfrac{3}{2}, \lambda'_4 + \tfrac{3}{2}). \quad (8.2)$$

Indeed the half-sums of positive compact and noncompact roots are $\rho_c = (1, 0, -1; 0)$ and $\rho_n = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; -\frac{3}{2})$, and the holomorphic discrete-series formula is $\lambda' + \rho_c - \rho_n$. The coordinates are distinct and belong to $\mathbb{Z} + \frac{3}{2}$; the coordinates at the conjugate embedding form the multiset $\{-k_1, -k_2, -k_3, -k_4\}$.

Consider an archimedean character $z^u \bar{z}^v$ in a cuspidal factor τ before inserting a shift $s \in S_a$. The two coordinates $u + s$ and $v + s$ occur in the two prescribed infinitesimal characters. Hence u, v are real and $u + v \in \mathbb{Z}$. A unitary generic representation of $\mathrm{GL}_d(\mathbb{C})$ has $|\mathrm{Re}(u + v)| < 1$ for every inducing character. Cuspidality supplies genericity, so $u + v = 0$. This proves temperedness at infinity.

The archimedean parameters of τ and $\tau^{c,\vee}$ are now identical. A pair of distinct conjugate-dual cuspidal factors in Ψ would consequently repeat the first infinitesimal coordinates, contrary to regularity. Thus all factors are conjugate-self-dual. Repeated copies of an identical Arthur block are excluded for the same reason. In particular the resulting unitary parameter is elliptic.

Choose a unitary Hecke character χ of L with $\chi^c = \chi^{-1}$, with infinity type $z/|z|$ at both distinguished embeddings, and with conductor supported on split auxiliary primes. Its existence is the character construction of Section 2.3, with the following parity prescription. Require $\chi|_{\mathbb{A}_F^\times} = \omega_{L/F}$, the quadratic character attached to L/F . At infinity this agrees with the restriction of $z/|z|$ to $\mathbb{R}^\times$. At nonsplit finite places use unramified extensions of $\omega_{L/F,v}$, and use a sufficiently deep conductor at split auxiliary primes to remove the remaining ray-class unit obstruction. The resulting Hecke character satisfies

$$\chi(x)\chi(x^c) = \chi(N_{L/F}x) = \omega_{L/F}(N_{L/F}x) = 1,$$

which gives the required global identity $\chi^c = \chi^{-1}$. For a factor τ of rank d and block length a , choose $\delta \in \{0, 1\}$ with

$$\delta \equiv a + d - 1 \pmod{2}.$$

If u is one of its first infinitesimal coordinates, then $u \in \mathbb{Z} + (4 - a)/2$, and therefore $u + \delta/2 \in \mathbb{Z} + (d - 1)/2$. Thus $\Pi = \tau \otimes \chi^\delta$ is regular algebraic in the cohomological normalization for GL_d . Regularity follows because its shifted coordinates form a subset of the distinct coordinates in Equation (8.2). The equality $\Pi^c \simeq \Pi^\vee$ follows from those for τ and χ . $\square$

Proposition 8.3 (Simultaneous transfer). *There is a formal parameter*

$$\Psi = \boxplus_{\beta} \tau_{\beta}[a_{\beta}], \quad \sum_{\beta} d_{\beta} a_{\beta} = 4,$$

with unitary cuspidal factors on $\mathrm{GL}_{d_{\beta}}(\mathbb{A}_L)$, such that:

- (i) *its archimedean infinitesimal character is the standard transfer of that of π ;*
- (ii) *at every split finite place $v = ww^c$, the representation π_v , viewed on $\mathrm{GL}_4(F_v)$ using w , is the representation attached to Ψ_w ;*
- (iii) *at every nonsplit finite place, the standard local parameter is unramified and has zero monodromy.*

Proof. Take the single weak parameter supplied by [Lemma 8.1](#). Its archimedean character is the required one, and its unramified cuspidal factors at nonsplit places give assertion (iii) by [Equation \(8.1\)](#). The application of Labesse’s corollary in that lemma already gives local base change at every split finite place. Under the chosen identification $G(F_v) = \mathrm{GL}_4(F_v)$ this is exactly assertion (ii), including the places above p , the bad places, and the moving levels. All three assertions thus concern the same parameter. This transfer argument places no restriction on the coefficient prime p . $\square$

8.2 Galois representations and their weights

We now attach Galois representations to the algebraic twists of the cuspidal factors in the simultaneous parameter. The resulting four-dimensional representation retains the direct-sum decomposition over the shifted copies of every Arthur block. This decomposition will matter when we control its actual monodromy.

We use the following established Galois input, in geometric reciprocity and with de Rham weight 1 for ϵ^{-1} .

Theorem 8.4 (Regular algebraic conjugate-self-dual Galois input). *Let Π be a regular algebraic conjugate-self-dual cuspidal representation of $\mathrm{GL}_d(\mathbb{A}_L)$ and fix $\iota : \overline{\mathbb{Q}}_p \simeq \mathbb{C}$. There is a continuous semisimple representation $r_p(\Pi)$ of G_L , with*

$$r_p(\Pi)^c \simeq r_p(\Pi)^{\vee} \epsilon^{1-d}.$$

It is de Rham above p . If its first infinitesimal coordinates at an embedding are $b_1, \dots, b_d$, its de Rham weights there are $(d-1)/2 - b_j$. At every finite place w ,

$$\iota \mathrm{WD}(r_p(\Pi)|_{G_{L_w}})^{\mathrm{F-ss}} \preceq \mathrm{rec}_w(\Pi_w \otimes |\det|_w^{(1-d)/2}). \quad (8.3)$$

Here $\preceq$ means equality after semisimplifying the Weil representation and domination of monodromy partitions on each inertial class. At spherical places away from p the representation is unramified, and at spherical places above p it is crystalline.

Existence, polarization and the de Rham assertions are Barnet-Lamb et al. [6, Theorem 1.2]; compatibility away from p in this strength is Chenevier and Harris [27, Theorem 3.2.3]. The assertion above p , including the monodromy bound for arbitrary regular algebraic weight, is Barnet-Lamb et al. [5, Theorem 1.1]. The rank-one case is algebraic global class field theory.

Proposition 8.5 (Galois representations of the cusp systems). *For every characteristic-zero cusp eigensystem in Proposition 7.6, there is a continuous semisimple representation*

$$\rho_\pi : G_L \longrightarrow \mathrm{GL}_4(\overline{\mathbb{Q}}_p), \quad \rho_\pi^c \simeq \rho_\pi^\vee \epsilon^{-3}.$$

It is unramified at every finite place nonsplit over F . At every split finite place $v = ww^c$,

$$\iota \mathrm{WD}(\rho_\pi|_{G_{Lw}})^{\mathrm{F-ss}} \preceq \mathrm{rec}_w(\pi_v \otimes |\det|_w^{-3/2}). \quad (8.4)$$

The representation is de Rham above p , and its increasing weights at either distinguished $w \mid p$ are

$$(h_1, h_2, h_3, h_4) = (1 - \lambda'_1, 2 - \lambda'_2, 3 - \lambda'_3, -\lambda'_4). \quad (8.5)$$

The gaps $h_3 - h_2$ and $h_4 - h_3$ tend to infinity with the choices in Equation (7.1).

Proof. Apply Theorem 8.4 to the representations $\Pi_\beta = \tau_\beta \chi^{\delta_\beta}$ supplied by Lemma 8.2. For a factor of rank d , length a , and $s \in S_a$, put

$$t_s = s + \frac{d-4}{2}, \quad \eta_s = \chi^{-\delta} | \cdot |_L^{t_s}.$$

At a distinguished complex embedding its exponents are $t_s - \delta/2$ and $t_s + \delta/2$. Both are integers: $2t_s \equiv a + d - 5 \equiv \delta \pmod{2}$. Thus η_s is an algebraic Hecke character even when its two displayed factors are not separately algebraic. Let $r_p(\eta_s)$ be its Galois character and define

$$\rho_\pi = \bigoplus_{\beta} \bigoplus_{s \in S_{a_\beta}} r_p(\Pi_\beta) \otimes r_p(\eta_{\beta,s}). \quad (8.6)$$

Its dimension is $\sum_{\beta} d_{\beta} a_{\beta} = 4$. On the automorphic side, the normalization in each summand is

$$(\tau \chi^{\delta}) | \det |^{(1-d)/2} \eta_s = \tau | \det |^{s-3/2}.$$

Equations (8.1) and (8.3) now give Equation (8.4). The same construction at the unramified standard parameters in Proposition 8.3(iii) gives unramifiedness: the bound forces the monodromy to vanish, and the semisimplified Weil inertia is trivial. Twisting and direct sums preserve the de Rham property.

For completeness, the polarization follows directly with the stated normalization. Since $\chi^c = \chi^{-1}$,

$$\eta_s^c \eta_{-s} = | \cdot |_L^{d-4}.$$

Geometric reciprocity sends $| \cdot |_L$ to ϵ . Consequently the summand $R_s = r_p(\Pi) r_p(\eta_s)$ satisfies

$$R_s^c \simeq R_{-s}^{\vee} \epsilon^{1-d+d-4} = R_{-s}^{\vee} \epsilon^{-3}.$$

The set S_a is invariant under $s \mapsto -s$, proving the polarization of Equation (8.6).

If u is a first infinitesimal coordinate of τ and $k = u + s$ is the corresponding coordinate of the full parameter, the associated summand has de Rham weight

$$\frac{d-1}{2} - u - \frac{\delta}{2} - \left(t_s - \frac{\delta}{2} \right) = \frac{3}{2} - k.$$

Substitution from Equation (8.2) proves Equation (8.5). In particular, for $\lambda' = (m + H, m + H, 1 + H; -1 - H)$ these weights are

$$1 - m - H, \quad 2 - m - H, \quad 2 - H, \quad 1 + H,$$

with the asserted gaps m and $2H - 1$. □

8.3 Local flags from the two unit operators

All the operator rings in this section are the residual local summands specified in [Proposition 7.6](#). In particular the two normalized operators $\tilde{U}_2, \tilde{U}_3$ are already units. The normalization is the one on the integral ordinary lattice, namely $N_2 = 2$ and $N_3 = 2 - H$ in the shifted weight. It is not a condition on the reduction of A_0 .

Lemma 8.6 (The two Hodge–Newton contacts). *At each distinguished $w \mid p$, every ρ_π under consideration has a G_{L_w} -stable flag of dimensions 2, 3, 4. Its last quotient is*

$$X_w = \epsilon^{-h_4} \text{unr}(\tilde{U}_3^{-1}), \quad (8.7)$$

where $\text{unr}(a)$ takes geometric Frobenius to a . At the conjugate place it has a first subline of character

$$Y_{\bar{w}} = X_w^{-c} \epsilon^{-3}. \quad (8.8)$$

Under [Equation \(7.4\)](#) their values tend, uniformly on local Galois elements and coefficientwise in t , to $x = \epsilon^{-1}$ and $y = \epsilon^{-2}$, respectively.

Proof. Positive finite-slope Jacquet restriction at w [[20](#), Section 4.1] allocates two unramified characters to the final two slots of the Levi $\text{GL}_2 \times \text{GL}_1 \times \text{GL}_1$. Write α_3, α_4 for their geometric Frobenius eigenvalues with the cohomological norm shift of [Proposition 8.5](#). The half-modulus in normalized Jacquet restriction and the integral normalization of the two operators give

$$\tilde{U}_2 = \frac{p^{h_3+h_4}}{\alpha_3\alpha_4}, \quad \tilde{U}_3 = \frac{p^{h_4}}{\alpha_4}. \quad (8.9)$$

Thus $v_p(\alpha_3) = h_3$ and $v_p(\alpha_4) = h_4$. The sum of the other two slopes is $h_1 + h_2$, since the whole potentially semistable module is weakly admissible. Its Newton polygon lies above its Hodge polygon. The sum of the lowest two slopes is at most the sum of the other two slopes and at least $h_1 + h_2$; hence equality holds. The strict Hodge gap at position two forces a strict Newton gap there. Similarly the lowest three slopes sum to $h_1 + h_2 + h_3$, with a strict gap at position three. In particular the two final roots are uniquely selected by their individual slopes.

After a finite semistable extension, the low-slope spaces are stable under inertia and monodromy. On either such space its induced Hodge number is at least the sum of the corresponding lowest Hodge weights, whereas weak admissibility bounds it above by its Newton number. Both inequalities are equalities. The subspaces are therefore weakly admissible. The Colmez–Fontaine equivalence and its subobject correspondence [[31](#), Theorem A, Proposition 4.2 and the corollary following Theorem 4.3] give semistable subrepresentations over this finite extension. The argument also applies to the descended module: the slope spaces are canonical and hence stable under descent. All embeddings above the fixed embedding have the same Hodge multiset, so the same inequalities hold after this extension.

The last quotient is unramified in its Weil part, has de Rham weight h_4 , and its Frobenius value is α_4 . This proves [Equation \(8.7\)](#); the polarization proves [Equation \(8.8\)](#). Their values belong to the integral operator ring. To check uniform convergence on local elements, recall that $h_4 = 1 + H_i$ and that H_i is divisible by $(p-1)p^{M_i+c}$. Hence

$$u^{H_i} \equiv 1 \pmod{p^{M_i}} \quad (u \in \mathbb{Z}_p^\times)$$

uniformly in u , after the fixed choice of c . This controls the cyclotomic factor on the whole local Galois group. For the unramified factor, continuous powers of the unit $\tilde{U}_3^{-1}$ define the character on the procyclic unramified quotient. Since $\lambda_i^0(\tilde{U}_3) = 1$ in the full truncated ring, every such power also has image 1 there. Formula (8.7) therefore gives $X_w \equiv \epsilon^{-1}$ at the working precision, uniformly on local elements and coefficientwise in t ; the polarization gives $Y_{\bar{w}} \equiv \epsilon^{-2}$. These congruences hold on every sequence of local elements. $\square$

8.4 Binary projections and small monodromy

Away from p , the required local conditions come from inertia on the final binary allocation. We first separate its Frobenius roots from the head, and then use the construction of ρ_π to control actual monodromy on that allocation.

At each split $(2, 2)$ control place v the positive central expansion is a unit. Jacquet lifting therefore supplies an unramified spherical final binary block. Let $A_v(Z)$ be its monic Frobenius polynomial, obtained from the determinant and degree-one binary Hecke operators. The full degree-four polynomial $P_v(Z)$ also has coefficients in the integral operator ring generated by the split good spherical Hecke operators. Indeed, for the finitely many cusp systems at a fixed stage, choose good split Frobenius elements tending simultaneously to the chosen local Frobenius. Their characteristic coefficients converge in the finite integral matrix lattice. The good operator subalgebra is already semisimple, finite over $\mathcal{O}$, and closed in that lattice; the limits consequently belong to that subalgebra itself. No normalization of a Hecke order is used here. The same approximation, applied to Equation (7.4), gives the comparison polynomial at every local element.

Perform monic division of P_v by A_v in the operator ring and write C_v for the monic quadratic quotient. For forming an operator, the possible remainder can be ignored. On every characteristic-zero system $A_v C_v = P_v$. Set

$$\Delta_v = \prod_{d=-4}^4 \text{Res}_Z(A_v(Z), Q_v^{2d} C_v(Q_v^{-d} Z)), \quad \Delta = \prod_v \Delta_v, \quad (8.10)$$

where the product is over the bounded number of binary control places, including the moving place. All powers of Q_v are p -adic units.

Lemma 8.7 (Separation and inertia). *The augmentation values of Δ are nonzero with uniformly bounded valuation. On any characteristic-zero system where $\Delta \neq 0$, the Frobenius primary projection P_v^{bin} onto the roots of A_v is defined and satisfies*

$$(\rho_\pi(\iota) - 1)P_v^{\text{bin}} = 0 \quad (\iota \in I_{L_v}). \quad (8.11)$$

The assertion concerns actual inertia, without a semisimplicity assumption on Frobenius.

Proof. At the moving place $Q_v = q_i$ tends to 1 and the head Frobenius matrix tends to $V(\gamma)$, while the binary roots tend to 1. The invertibility of $V(\gamma) - 1$ in Proposition 3.2 gives the required bound on its nonzero valuation. At a fixed potentially good place the head roots have the elliptic purity size, namely $Q_v^{3/2}$ after the cohomological twist, whereas the final roots are Q_v, Q_v^2 up to the specified fixed unitary character. The half-integral difference in weights prevents equality after an integral power of Q_v . At a ramified potentially multiplicative place choose the Frobenius lift whose quadratic sign is negative; its head roots cannot equal the positive final roots after such a power. These are finitely many fixed nonzero comparisons. The weight and tame specializations converge to them, so their valuations are uniformly bounded.

The nonzero resultants exclude both coincidences and norm linkages between the two allocations. The segment classification of Bernstein and Zelevinsky [8] and Zelevinsky [79, Theorem 6.1 and Proposition 8.5] then factors the local representation into the unlinked head and binary representations. A segment crossing the two allocations would yield a forbidden ratio Q_v^d with $|d| \leq 4$. The binary Jacquet allocation is therefore the whole binary representation; its prescribed hyperspecial invariants make it spherical, with zero predicted monodromy.

Separation from the head still allows norm linkages within the binary pair. Since Equation (8.4) gives only monodromy domination, we use the direct-sum construction of ρ_π to exclude those links

as well. In each individual shifted copy of a cuspidal factor τ in Equation (8.6), a selected root has a length-one local segment. A longer segment either crosses the forbidden allocation boundary or lies wholly in the binary part, contradicting sphericity. Two length-one unramified data inside one unitary generic τ_v cannot differ by Q_v : their real exponents lie in an interval of width strictly less than one [72, Theorem 7.5] and [79, Theorem 9.7]. Thus the actual monodromy cannot link two selected roots within that copy. It cannot link different shifted copies, since the Galois construction is their direct sum; it cannot link to the complement by the resultants. This proves that actual monodromy vanishes on the selected part even though Equation (8.4) only gives an upper bound.

The semisimple Weil inertia is finite. Its fixed subspace is stable under Frobenius, and its selected generalized Frobenius primary spaces have inertia one and monodromy zero. Monic coprime factorization gives the projection onto their sum, including any Frobenius Jordan blocks. This proves Equation (8.11). $\square$

Lemma 8.8 (The Steinberg control). *At a split place with the specified $(3, 1)$ parahoric level, $\rho_\pi|_I$ is unipotent and*

$$\text{rank}(\rho_\pi(\iota) - 1) \leq 1 \quad (\iota \in I).$$

Proof. Iwahori invariants make semisimple Weil inertia trivial. Write the local Langlands parameter as unramified Steinberg blocks of lengths s_j . The standard induction of these blocks surjects onto the local representation. Compact invariants are exact in characteristic zero, so the standard induction must have $(3, 1)$ parahoric invariants. Restriction to the maximal compact and its double-coset decomposition reduces this to invariants of each Steinberg factor under the induced partial flag. The finite Weyl sign representation of a Steinberg factor has such invariants only when all its positions are separated. A single-step flag can therefore accommodate at most one block of length two, all others having length one. The predicted monodromy has rank at most one and square zero. The monodromy bound in Proposition 8.5 gives the same for the actual representation; its inertia matrices are $\exp(t_p(\iota)N)$. $\square$

8.5 Removing the operator nilpotents

The cusp systems now have the required local subspaces and inertia identities. The same separated Jacquet allocations also control the nilpotents in the operator order. We use this fact to descend the congruence of Proposition 7.6 to the algebra detected by characteristic-zero eigensystems, retaining every coefficient of the truncated tame variable.

Proposition 8.9 (Uniform nilpotent control). *Let T_i be the image of T_i^0 in its characteristic-zero cusp eigensystems. There is an integer C , independent of i , the weights and q_i , such that*

$$\Delta^C \ker(T_i^0 \rightarrow T_i) = 0. \quad (8.12)$$

After decreasing the precisions by a fixed constant, Equation (7.4) descends to

$$\lambda_i : T_i \rightarrow \mathcal{O}/(p^{M_i})[t]/(t^{b_*}), \quad M_i \rightarrow \infty. \quad (8.13)$$

Proof. Work first on one irreducible local representation over an algebraically closed field of characteristic zero. Realize it as a subquotient of induction from its full supercuspidal support. The geometric lemma filters the relevant block Jacquet module by at most $4! = 24$ allocation terms [8, Lemma 2.12]. Taking the required compact Levi invariants is exact. On the resulting finite-dimensional space, taking a joint generalized eigenspace for the commuting controlled operators is also exact. These operations therefore select a filtration of the actual operator module.

At p , the unit summand was chosen before taking any reduced quotient. On every eigensystem its final slopes are h_3, h_4 by [Lemma 8.6](#), and the other slopes are at most h_2 . The two gaps and [Equation \(8.9\)](#) select a unique allocation to the $(2, 1, 1)$ Levi. On its surviving grade the two final GL_1 center operators are scalars. At a binary control place where $\Delta_v \neq 0$, the determinant and the degree-one spherical operator determine the full quadratic A_v . Separation from the head similarly selects one allocation. The hyperspecial invariants of the full unramified binary induction are one-dimensional, so all the binary operators are scalars on that grade, including when the two binary inducing characters coincide.

Repeated factors within the head create no second allocation: permutations inside that block already belong to the Levi Weyl group in the geometric-lemma double-coset indexing. No supercuspidal factor splits internally across an allocation. Thus there is exactly one nonzero grade in the selected generalized eigenspace. It equals that grade, and the controlled operators act by actual scalars there. In particular there is no residual Jordan extension on the separated locus. The good spherical operators are already scalar on each automorphic representation.

For comparison, on a joint generalized component where Δ has eigenvalue zero, it acts by zero on every associated allocation grade and therefore lowers the filtration. A fixed power kills that component. Tensoring the filtrations at the bounded number r of control places gives, for example, the coarse bound 24^r . On the complementary components the kernel of eigensystem detection is zero by the preceding scalar-action argument. This proves [Equation \(8.12\)](#) as a characteristic-zero operator identity. The integral order T_i^0 is torsion-free, so the same identity holds integrally.

Finally the constant coefficient of $\lambda_i^0(\Delta)$ has uniformly bounded nonzero valuation by [Lemma 8.7](#), and all of its other coefficients are integral. Successive coefficient comparison in the truncated polynomial ring shows that multiplication by its fixed power can lose only a bounded amount of p -adic precision. More explicitly, inversion over $k[t]/(t^{b_*})$ uses at most b_* powers of the inverse constant coefficient; the resulting denominator bound depends only on C, b_* and that valuation bound. Applying this to [Equation \(8.12\)](#) kills the kernel modulo $p^{M_i - O(1)}$. Renaming these still increasing precisions proves [Equation \(8.13\)](#). $\square$

9 Extraction of the extension space

Set $b = b_*$ and $R_b = k[t]/(t^b)$. We use the admissible cohomology of [Definition 3.3](#). The aim is to derive from [Equation \(8.13\)](#) an admissible Greenberg extension space of length at least b . This will contradict [Proposition 3.7](#). The construction uses an algebra represented by 4×4 matrices, with block sizes 2, 1, 1 throughout.

9.1 A finite limiting matrix algebra

For each i , take the product of the Galois representations of [Proposition 8.5](#) over all characteristic-zero systems of T_i , and denote it by ρ_i . Choose stable integral lattices in the finitely many factors. Put $B_i = T_i[\rho_i]$. This is a submodule of a finite product of integral matrix lattices, and hence is finite over $\mathcal{O}$ and over T_i . Chebotarev approximation, as in [Section 8.4](#), shows that the trace and all characteristic coefficients of every group element belong to T_i .

We need a bound on the number of T_i -module generators of B_i , even though their ranks over $\mathcal{O}$ may grow. After base change, the images of these generators under an algebra map into $M_4(R_b)$ will have one common denominator. Expressing group matrices as linear combinations of the generators will then give bounded finite-stage representatives, as proved in [Lemma 9.2](#).

Lemma 9.1 (Uniformly many module generators). *There is an integer N independent of i such that the T_i -module B_i is generated by N monomials in images of Galois elements.*

Proof. The residual characteristic polynomials are fixed by the comparison system. After a fixed finite extension of L killing their semisimplification, the residual image in every factor is unipotent, and the full integral image is pro- p . All factors are unramified outside the fixed set S and the bounded number of places over q_i . The maximal pro- p Galois group of this extension with such ramification has a bounded number of topological generators: its minimal number is the dimension of $H^1(-, \mathbb{F}_p)$, which the class field sequence bounds by a fixed contribution from S , the units and class group, plus one bounded contribution per new place. This is the usual global Galois finiteness theorem [59, Chapter X]. Add fixed coset representatives for the finite extension. It follows that uniformly many elements generate the matrix image topologically.

Their algebraic span is already closed, because any $\mathcal{O}$ -submodule of the finite integral matrix lattice is finite and closed. Inverses are polynomial expressions with integral coefficients by the characteristic equation and the unit determinant. Hence the same bounded list generates B_i as an algebra.

Reduce modulo the maximal ideal of T_i . Matrix algebras of size four satisfy the degree-eight standard polynomial identity [2], and this identity passes to the quotient. Shirshov's height theorem bounds the number of power factors needed to span the quotient, with base words of length less than eight; the bound depends only on the identity and the number of generators [69, Section 2, Theorem 1 and its proof]. Each base word before reduction is a group image and satisfies its degree-four characteristic polynomial over T_i . Reducing its powers to exponents at most three leaves a uniformly bounded list of monomials. These span B_i modulo the maximal ideal of T_i . Since B_i is finite over T_i , Nakayama's lemma proves the assertion. $\square$

Choose the nonprincipal ultrafilter $\mathcal{U}$ used in Section 3, and form the algebraic ultraproduct $A = \prod_{\mathcal{U}} T_i$. The constant coefficients of Equation (8.13) have compact p -adic ultralimits, giving a map $A \rightarrow \mathcal{O} \subset k$; let $\mathfrak{q}$ be its kernel. Taking the ultralimit of every coefficient of t gives a map $A \rightarrow R_b$ with this constant coefficient. Every element outside $\mathfrak{q}$ has unit image in R_b , so this map extends to $A_{\mathfrak{q}}$. Since R_b is henselian, it extends uniquely to the henselization. Put

$$T = (A_{\mathfrak{q}})^h, \quad \lambda : T \longrightarrow R_b, \quad B = \left(\prod_{\mathcal{U}} B_i \right) \otimes_A T. \quad (9.1)$$

The residue field of T is k : the image before localization contains $\mathcal{O}$, and localizing at the kernel inverts every nonzero constant. Write $\mathfrak{m} = \ker(T \rightarrow k)$.

There is a useful faithful matrix realization. Put

$$C = \prod_{\mathcal{U}} \prod_{\pi \text{ at } i} \overline{\mathbb{Q}}_p, \quad \mathcal{R} = C \otimes_A T. \quad (9.2)$$

The injections $A \hookrightarrow C$ and $\prod_{\mathcal{U}} B_i \hookrightarrow M_4(C)$ remain injective after tensoring with the flat localization and henselization T . Thus $T \hookrightarrow \mathcal{R}$ and $B \hookrightarrow M_4(\mathcal{R})$ are faithful. The ring C is absolutely flat. Its residue fields are algebraically closed ultraproducts obtained from ultrafilters on the total set of systems which prolong $\mathcal{U}$. Base change to T is a filtered combination of localization and etale operations. These preserve absolute flatness and reducedness, so $\mathcal{R}$ is again absolutely flat and reduced. Its field tests can be made in these algebraically closed ultraproducts; finite etale choices split over them. This observation will allow finite matrix identities to be tested on actual arrays of characteristic-zero systems.

Let Γ denote the ultraproduct of the Galois groups unramified outside Sq_i , with the prescribed local subgroups. The bounded monomials in [Lemma 9.1](#) still generate B after both base changes. The monomials may vary with i ; their number is fixed. They are products of group images, so B is generated by $\rho_*(\Gamma)$, and its trace is T -valued. The limiting trace under λ is

$$\mathrm{tr}(f \oplus x \oplus y), \quad f = V(-2)\psi^{\pm 1}, \quad x = \epsilon^{-1}, \quad y = \epsilon^{-2}. \quad (9.3)$$

Here the sign follows the reciprocity choice in [Section 2.3](#), and either sign is permitted in [Proposition 3.7](#). Indeed, the split good polynomials are those of the theta comparison. The factor $\mathcal{C}^m$ tends uniformly to one on the fixed torsion branch, and Frobenius approximation at each finite stage may be made to arbitrary increasing precision. This proves [Equation \(9.3\)](#) on all sequences of elements.

The corresponding diagonal representation factors through B . Here reduction modulo $\mathfrak{m}$ has coefficient field k , of characteristic zero. Its diagonal blocks are absolutely irreducible and pairwise distinct: $V|_{G_L}$ is absolutely irreducible by the choice of L in [Lemma 2.3](#). Their group images span $M_2(k) \times k \times k$, and hence, by Nakayama, span $M_2(R_b) \times R_b \times R_b$. If a relation vanishes in B , its product with every group-algebra element has trace zero. The trace identity [Equation \(9.3\)](#) and the perfect trace pairing on this matrix product show that its image in the diagonal representation is zero as well.

9.2 Corners and admissible representations

We spell out the elementary matrix-algebra construction to keep track of finiteness and of the coefficient ring; compare the general framework of Bellaïche and Chenevier [[7](#), Sections 1.3–1.5]. Choose an element of B acting with four distinct prescribed eigenvalues in the residual diagonal representation. Its characteristic coefficients belong to T : Newton identities express them through traces, and the integers 1, 2, 3, 4 are units in this local ring with characteristic-zero residue field. Henselianity splits its characteristic polynomial into four linear factors with pairwise unit differences. The associated polynomial projectors are orthogonal idempotents e_1, e_2, e_x, e_y summing to one. In every ambient field their ranks are one, since the split characteristic polynomial has four simple roots. The first two lie over the f block.

Lift the two off-diagonal matrix units in that residual 2×2 block. Their products in the rank-one corners are unit scalars; rescaling gives inverse matrix units between e_1 and e_2 . A diagonal corner is precisely Te_j : in the faithful rank-one realization each of its elements equals its trace times e_j . Using the matrix units to identify the first two positions, let D_{uv} , for $u, v \in \{f, x, y\}$, be a single-entry corner from v to u . These finite T -modules have the usual composition products, and $D_{uu} = T$. All crossing diagonal cycles map to zero under λ , since the diagonal representation annihilates every crossing entry. In particular

$$\lambda(D_{uv}D_{vu}) = 0 \quad (u \neq v). \quad (9.4)$$

Our eventual extension module will be $D_{fy} \otimes_{T, \lambda} R_b$. A functional on it should retain the extension of y by f and discard the other crossing entries. To respect multiplication, it must also kill the discarded path from y through x to f . We will first prove $D_{xy} = D_{xf}D_{fy}$, so that this path contains a crossing diagonal cycle and vanishes under [Equation \(9.4\)](#). We will then show that D_{fy} is nonzero in every ambient field, giving the vanishing Fitting ideal from which the length bound follows.

Lemma 9.2 (Finite-value admissibility). *If $\Phi : B \rightarrow M_4(R_b)$ is a T -linear algebra homomorphism with scalar map λ , then $\Phi\rho_*$ has admissible matrix coefficients. The same holds with k in place of R_b . Algebraic local identities holding on all local sequences have the uniform increasing-precision meaning in [Definition 3.3](#).*

Proof. Let b_{ij} , $1 \leq j \leq N$, be the monomial generators from Lemma 9.1, and b_j their images in B . Only the fixed finite list of matrices $C_j = \Phi(b_j)$ is needed. One denominator clears all coefficients of all C_j in the fixed basis of R_b .

For any increasing precision s_i , the compact surjection $T_i^N \rightarrow B_i$ induces a surjection on the finite quotients modulo p^{s_i} . Choose a section on that finite set. Composing with ρ_i produces locally constant coefficient functions $a_{ij}(g) \in T_i$ such that

$$\rho_i(g) \equiv \sum_{j=1}^N a_{ij}(g)b_{ij} \pmod{p^{s_i}B_i}.$$

Apply Equation (8.13) and multiply the coefficients by successively accurate approximations to the fixed C_j . The resulting continuous matrix functions have the one denominator already chosen, and their limit is $\Phi\rho_*$ by T -linearity. The precision may be slowed to the minimum of s_i and M_i minus this fixed loss; the bound $M_i \leq n_i - c_b$ already respects the tame relation. The variable element $(p^{s_i})_i$ has image zero under λ in every coefficient of t , so the remainders vanish. Although the constant element p is a unit in T , this does not change the assertion about the variable sequence.

If a matrix identity holds for all sequences but fails uniformly to increasing precision, choose a violating local element at each of a filter-large set of stages. That sequence contradicts the identity in the limit. The same argument applies to local splittings after choosing the finitely many splitting coefficients. In particular no coordinatewise lifting of the henselian idempotents is required. $\square$

The local subspaces of the characteristic-zero representations need not themselves survive a corner quotient. We transport their rank conditions as polynomial identities in corner entries instead. The following lemma makes these identities intrinsic to B and hence preserves them under every algebra map used below.

Lemma 9.3 (Corner determinant identities). *Choose four rows of a stack of 4×4 matrices in B , with column indices $1, 2, x, y$. If the row indices consist of the column indices with y replaced by u , then its signed determinant is a well-defined element of D_{uy} . The same assertion holds for a square minor whenever canceling common row and column indices leaves precisely one surplus row and one surplus column; if no indices remain, its value lies in T . If the ordinary determinant is zero in every ambient field, this corner element is zero. Its identity is preserved by algebra maps out of B .*

Proof. Each signed determinant monomial has entries directed from its column index to its row index. At every vertex the incoming and outgoing counts agree except for a path from y to u . Decompose the directed graph into that path and cycles. Compose the path in D_{uy} ; each cycle is a scalar in its rank-one diagonal corner and hence in T . The inverse matrix units identify the two f positions whenever needed. This gives the claimed element. In a rank-one matrix realization it is exactly the usual product of entries, independent of the chosen bases, since the intermediate basis changes cancel. Faithfulness in $M_4(\mathcal{R})$ and reducedness of $\mathcal{R}$ prove the vanishing assertion. The construction uses sums and corner products, so is preserved by algebra homomorphisms. $\square$

9.3 Eliminating the cyclotomic extension corner

Proposition 9.4. *The corner modules satisfy*

$$D_{xy} = D_{xf}D_{fy}. \tag{9.5}$$

Proof. Let

$$Q_{xy} = D_{xy}/(D_{xf}D_{fy} + \mathfrak{m}D_{xy}).$$

A k -linear functional $Q_{xy} \rightarrow k$ defines a triangular representation, retaining the separate f block and an extension of y by x . Products through f have been killed, and other crossing products vanish by Equation (9.4); thus the triangular assignment is an algebra map out of B . It yields an injection

$$\mathrm{Hom}_k(Q_{xy}, k) \hookrightarrow H_{\mathrm{adm}}^1(\Gamma, k(1)). \quad (9.6)$$

Indeed, a split Galois extension has a splitting preserved by the algebra generated by the group, hence by B . The idempotents then force this splitting to be the diagonal y line. Its invariance annihilates the whole functional. Admissibility follows from Lemma 9.2.

We verify the strict local conditions on every class in Equation (9.6). At a conjugate place $\bar{w} \mid p$, set $S_g = \rho_*(g) - Y_{\bar{w}}(g)$ for local sequences g . In every characteristic-zero system these matrices have a common kernel by Lemma 8.6. Take four rows of their stack with types f, f, x, x . The four-row determinant is an element of D_{xy} by Lemma 9.3, and it is zero by the common kernel. Its identity therefore holds in the triangular quotient.

On the three competing columns f_1, f_2, x , one may choose two f rows and one x row with an invertible 3×3 minor. Here the inverse matrix units of the f block transport the selected coordinate rows to the two required output positions. Writing these units as E_{ab} , the rows can be represented by $e_1 E_{1a} S_g$ and $e_2 E_{2b} S_h$, with $a, b \in \{1, 2\}$ allowed to coincide. These operations preserve the common kernel and the corner identities of Lemma 9.3. For the f part this follows from

$$\mathrm{Hom}_{G_{L_{\bar{w}}}}(y, f) = H^0(L_{\bar{w}}, V) = 0;$$

there is no common kernel, so finitely many rows span the two-column dual. The tame character is trivial locally at these fixed places. For the x row use $x \neq y$ on the local group. Normalize the y coordinate to one and solve these three equations for the other coordinates. Each additional x row is consistent with this solution because its four-row determinant is zero. The f coordinates are zero, all the remaining f equations already hold, and the y rows are zero. Thus a common y -eigenline exists in the triangular extension: its restriction at $\bar{w}$ splits.

At $w \in \Sigma_p$, use the common quotient of character X_w in Lemma 8.6 and apply the transposed minor argument. The competing-column condition becomes

$$\mathrm{Hom}_{G_{L_w}}(f, x) = H^0(L_w, V) = 0,$$

using $V \simeq V^*(1)$, together with $x \neq y$. The resulting quotient of character x splits the same extension of y by x . Equations (8.7) and (8.8) supply the required characters on all local elements. This proves strictness at every place above p .

Away from p the extension is unramified, as follows. Outside the controlled places this holds for the original representations. At a binary control place, the image of Δ_v is a unit in T . The coprime factorization of the full Frobenius polynomial therefore defines P_v^{bin} over B , and the identity Equation (8.11) holds there by faithful field testing. In the triangular representation this projection is the identity on the x, y block: both of its Frobenius roots are in A_v and are separated from the f roots. Hence inertia acts trivially on that extension. This includes the moving places q_i .

At a Steinberg control place, the rank bound of Lemma 8.8 gives two-by-two minors of $\rho_*(\iota) - 1$ with rows f, x and columns f, y , which are zero in D_{xy} . In the triangular representation they say

$$(f(\iota) - 1)_{ab} u(\iota) = 0$$

for every matrix entry and every inertia element, where u is the extension entry. The comparison elliptic Steinberg representation has a nonzero actual inertia difference for some element h . Thus

$u(\iota) = 0$ whenever $f(\iota) - 1 \neq 0$, in particular for h . If $f(\iota) = 1$, apply the same assertion to ιh and use the cocycle equation and the unramifiedness of x, y to obtain $u(\iota) = u(\iota h) - u(h) = 0$.

By admissibility and unramifiedness at the moving places, the class now descends to ordinary continuous cohomology over L , by [Lemma 3.4](#). Its coefficients are $k(1)$, it is unramified away from p , and it is zero locally at all p -places. Kummer theory identifies such classes with p -units tensored with k , subject first to zero valuations at the p -places and then to zero local logarithms. The valuation condition leaves global units. The CM biquadratic field L has unit rank one, and a nontrivial fundamental unit has nonzero p -adic logarithm at a split p -place: if its image in $\mathbb{Q}_p^\times$ had logarithm zero, being a unit it would be a root of unity, and injectivity of the field embedding would make the global unit torsion. Thus the local logarithm is injective on this one-dimensional space, and the class vanishes. We have shown $Q_{xy} = 0$. Nakayama's lemma for the finite module $D_{xy}/D_{xf}D_{fy}$ proves [Equation \(9.5\)](#). $\square$

9.4 A nonzero corner in every characteristic-zero field

Lemma 9.5 (Excluding a stable last line). *The image of D_{fy} is nonzero at every field evaluation of $\mathcal{R}$.*

Proof. Suppose its image vanished at such a field. By [Equation \(9.5\)](#), D_{xy} also vanishes there, so the one-dimensional e_y -space is globally stable. We will obtain a rank-one geometric character with conjugate weight sum three. The residual corner character $y = \epsilon^{-2}$ has weight sum four. Evaluating on a fixed pair of conjugate local units will turn this discrepancy into a scalar unit that would have to vanish in the ambient field.

We first identify the stable e_y -space at a conjugate place $\bar{w} \mid p$ with the first line in [Lemma 8.6](#). Choose the two f rows and one x row used above from the stack $\rho_*(g) - Y_{\bar{w}}(g)$, and project their columns to the complement of e_y . Their 3×3 determinant is a scalar in T and is a unit: its reduction is the invertible minor in $f \oplus x \oplus y$. On a filter-large set of actual systems, the common kernel is consequently a unique line with nonzero e_y coordinate. Once e_y is stable, all competing entries in its column are zero. The invertible complementary minor therefore forces the existing common kernel of the matrices $\rho_*(g) - Y_{\bar{w}}(g)$ to be the e_y line. Its character is thus $Y_{\bar{w}}$.

Here these assertions really do refer to arrays of actual characteristic-zero systems. Stability need only be checked on the uniformly bounded list of module generators in [Lemma 9.1](#). The rank, inverse matrix-unit relations and the competing-column condition also use finitely many matrices. Include also $\rho_*(g)$ and $\rho_*(g^c)$ for the fixed unit element selected below; this only enlarges that finite list. Elements introduced by henselization can be represented in the algebraically closed residue ultraproduct of [Equation \(9.2\)](#), with their finite relations. Vanishing or nonvanishing of each of these finitely many matrix expressions then holds on a filter-large set of actual systems. Thus at almost every such system its semisimple global representation has a one-dimensional constituent whose local character is the prescribed $Y_{\bar{w}}$. This passage uses finite matrix equations, not an assertion about infinitely many coordinatewise projectors.

Denote this actual character by ξ . It is de Rham above p , hence Hodge–Tate with semisimple rank-one inertia, and therefore locally algebraic [[67](#), Chapter III, Section 1.2 and Appendix A7, Theorem 3]. Global class field theory and the Serre-group construction then identify it with the p -adic realization of an algebraic Hecke character; see Serre [[67](#), Chapter III, Sections 2.2–2.3, including the proof, and Chapter II, Section 2.7]. Here the construction is over the coefficient field. The sums of its weights at conjugate embeddings are parallel: apply its infinity type to the global units of the real subfield. Write $w(\xi) \in \mathbb{Z}$ for this common sum.

The character occurs in a shifted copy of a cuspidal factor $\tau[a]$ in [Equation \(8.6\)](#). If $a > 1$, the same rank-one constituent occurs in all its consecutive shifts. Its weight $3 - h_4$ at $\bar{w}$ would then

have an adjacent integer weight in the full representation. This is impossible: it is the lowest weight there, separated from the next by $h_4 - h_3$, which was chosen arbitrarily large. Hence $a = 1$. At every unramified place, the roots of a unitary generic cuspidal local representation have absolute values strictly between $Q^{-1/2}$ and $Q^{1/2}$ [72, Theorem 7.5] and [79, Theorem 9.7]. After the cohomological shift in Equation (8.4), the value of ξ has absolute size strictly between Q and Q^2 . On the other hand, for an algebraic Hecke character with parallel conjugate weight sum $w(\xi)$ its absolute value is $Q^{w(\xi)/2}$: dividing out this norm power leaves a unitary character. Consequently

$$w(\xi) = 3. \quad (9.7)$$

There is a uniform open subgroup of the local units above p on which the smooth factors of all these characters are trivial. Indeed the compact open level J at p is fixed. Bernstein's decomposition theorem gives finitely many inertial cuspidal-support classes among representations with J -invariants [9, Section 2.3]; local Langlands therefore gives finitely many semisimple inertia types. The smooth factor of a rank-one constituent belongs to this finite list. This bounds its conductor independently of the growing algebraic weights. Choose in this fixed open subgroup an element g with $\epsilon(g) \neq 1$, deep enough that $\epsilon(g)$ is not torsion, and use its conjugate g^c at the opposite place. Equation (9.7) implies at every actual constituent under consideration

$$\xi(g)\xi(g^c) = \epsilon(g)^{-3}. \quad (9.8)$$

Thus it implies the same identity in the evaluated ambient field.

But the scalar corner element

$$(e_y \rho_*(g) e_y)(e_y \rho_*(g^c) e_y) - \epsilon(g)^{-3} e_y$$

is a unit in Te_y . Its reduction in the residual diagonal representation is $(\epsilon(g)^{-4} - \epsilon(g)^{-3})e_y \neq 0$, since the residual y is ϵ^{-2} and cyclotomic is invariant under conjugation. A unit cannot vanish in a field evaluation, contradicting Equation (9.8). The assumed vanishing of D_{fy} is impossible. $\square$

Corollary 9.6. *One has*

$$\text{Fitt}_{0,T}(D_{fy}) = 0. \quad (9.9)$$

Proof. If $a \in T$ annihilates D_{fy} , then its image at every ambient field is zero by Lemma 9.5. Since T embeds in the reduced ring $\mathcal{R}$, this gives $a = 0$. For a finite module over an arbitrary commutative ring its zeroth Fitting ideal is contained in its annihilator: choose finitely many generators and apply the adjugate identity to each square matrix of relations. This does not require a finite presentation. It proves Equation (9.9). $\square$

9.5 The Greenberg extension module and the contradiction

Proposition 9.7 (Extension extraction). *There is an injective R_b -linear map*

$$\text{Hom}_{R_b}(D_{fy} \otimes_{T,\lambda} R_b, R_b) \hookrightarrow H_{\text{Gr}}^1(L, V \otimes \psi^{\pm 1}). \quad (9.10)$$

Its source has R_b -length at least b .

Proof. Put $M = D_{fy} \otimes_{T,\lambda} R_b$. A functional $M \rightarrow R_b$ gives a triangular extension of y by f while retaining the separate x block. We check the only additional product relation. The path from y through x to f has image in

$$D_{fx}D_{xy} = D_{fx}D_{xf}D_{fy}$$

by Equation (9.5). Its first two factors form a crossing diagonal cycle, killed by Equation (9.4) after applying λ . All other discarded products are killed by the same cycle relation or by the diagonal representation. Hence the triangular assignment is an algebra homomorphism out of B . As in Equation (9.6), a split extension must have its splitting preserved by B and then by its idempotents; it forces the functional to be zero. This proves injectivity into extension cohomology. Moreover

$$\mathrm{Hom}(y, f) = V \otimes \psi^{\pm 1},$$

and Lemma 9.2 proves admissibility with coefficients in the full ring R_b .

At a conjugate p -place, use the same common-kernel stack $S_g = \rho_*(g) - Y_{\bar{w}}(g)$. Choose four rows of types f, f, x, f , with either of the two f positions as the final row. By Lemma 9.3 the vanishing four-row determinants are identities in the corresponding entry of D_{fy} , and they survive in the triangular extension. The first two f rows and one x row give a 3×3 minor on the competing columns whose residue is invertible by $H^0(L_{\bar{w}}, V) = 0$ and $x \neq y$. It is therefore a unit over R_b . Solve these equations with y coordinate one. Varying the final f row proves that every other f equation is satisfied. The x coordinate is zero and the y rows vanish. This produces a common y -eigenline and splits the local extension over the full Artin ring, including its nilpotent coefficients.

The Greenberg condition is full at Σ_p , so no splitting is required there; it is relaxed at the remaining fixed bad places. Outside those places and the moving places the representations are unramified, as required for admissible classes. Thus the preceding local splitting is precisely the one-sided condition of Proposition 3.7, and gives Equation (9.10).

Fitting ideals commute with base change for finite modules, so Equation (9.9) gives $\mathrm{Fitt}_{0, R_b}(M) = 0$. The finite module M over the principal Artin ring R_b has an elementary-divisor decomposition

$$M \simeq \bigoplus_j R_b / (t^{a_j}), \quad 1 \leq a_j \leq b.$$

Its zeroth Fitting ideal is $(t^{\sum_j a_j})$, with the evident interpretation for free summands $a_j = b$. Vanishing is equivalent to $\sum_j a_j \geq b$. Each summand and its R_b -dual have length a_j , since $\mathrm{Hom}_{R_b}(R_b / (t^{a_j}), R_b) = (t^{b-a_j})$. Thus the source in Equation (9.10) has length at least b . $\square$

We can now finish the arithmetic argument. Outside the strict case, $b = b_* = 1$, whereas Proposition 3.7 gives a Greenberg module of length zero. In the strict case, $b = b_* = 3$, whereas the same proposition bounds its length by two. Both contradict Proposition 9.7. Consequently the torsion alternative for the conductor-one Heegner point in Equation (2.3) cannot occur. The arithmetic reduction of Proposition 2.4, Gross–Zagier and Kolyvagin then give the analytic order, the Mordell–Weil rank and the finiteness of the whole Tate–Shafarevich group asserted in Theorem 1.1.

10 Sylvester’s positive prime cube-sum cases

Sylvester’s positive prime cases ask whether every prime $\ell \equiv 4, 7, 8 \pmod{9}$ is a sum of two rational cubes. Dasgupta–Voight proved the 4, 7 cases when 3 is not a cube modulo ℓ [32, Theorem 1.2.1]. Yin’s recent construction removes this restriction, with a companion Gross–Zagier formula giving analytic rank one [77, 78]. Burungale–Tian prove the remaining 8 case and state the resulting full prime cube-sum theorem [14, Theorems 1.1 and 1.2]. The ramified-prime CM converse and its Sylvester consequence are also stated in the earlier preprint of Kriz [51, Theorems 1.1 and 1.3].

The proof below instead deduces these cases uniformly from Theorem 1.1 at $p = 3$. This is a prime of additive reduction for the cube-sum curves. The role of a rank-one converse at this prime is also explained by Burungale and Tian [14, Section 1.2.2]. Our input is Satgé’s classical paired

isogeny descent: retaining its Tate–Shafarevich terms bounds the full Selmer corank, rather than only the Mordell–Weil rank.

Corollary 10.1 (Sylvester’s positive prime cases). *Let ℓ be a rational prime with $\ell \equiv 4, 7, 8 \pmod{9}$, and equip the projective cubic*

$$E_\ell : X^3 + Y^3 = \ell Z^3$$

with origin $(1 : -1 : 0)$. Then

$$\text{ord}_{s=1} L(E_\ell, s) = \text{rank}_{\mathbb{Z}} E_\ell(\mathbb{Q}) = 1, \quad \# \text{Sha}(E_\ell/\mathbb{Q}) < \infty.$$

In particular, there are $x, y \in \mathbb{Q}$ such that $x^3 + y^3 = \ell$.

Proof. Use the $\mathbb{Q}$ -isomorphic Weierstrass model $E_\ell : y^2 = x^3 - 27(4\ell)^2$ and its isogenous companion $E'_\ell : y^2 = x^3 + (4\ell)^2$. Satgé’s degree-three isogeny and its dual are

$$\lambda : E_\ell \longrightarrow E'_\ell, \quad \lambda' : E'_\ell \longrightarrow E_\ell, \quad \lambda' \circ \lambda = [3].$$

These are the models and directions of Section I, page 336, of Satgé [66]. Let S and S' be the Selmer groups for λ and λ' , respectively, with their local Kummer conditions. Write $\text{Sha}(E_\ell/\mathbb{Q})[\lambda]$ for the kernel of the map on Sha induced by λ , and similarly for λ' , and put

$$a = \dim_{\mathbb{F}_3} \text{Sha}(E_\ell/\mathbb{Q})[\lambda], \quad b = \dim_{\mathbb{F}_3} \text{Sha}(E'_\ell/\mathbb{Q})[\lambda'].$$

These dimensions are finite: Satgé’s descent sequences (1) and (1') exhibit the two kernels as quotients of S and S' . For $D = \ell$, Proposition 1.2 of Satgé [66] gives

$$\dim_{\mathbb{F}_3} S = \dim_{\mathbb{F}_3} S' = 1.$$

Since $\ell > 3$, his equation (2), obtained from Lemma 1.1 and the two descent sequences, applies and yields the exact equality

$$\text{rank}_{\mathbb{Z}} E_\ell(\mathbb{Q}) = \dim_{\mathbb{F}_3} S + \dim_{\mathbb{F}_3} S' - 1 - a - b = 1 - a - b.$$

The isogeny maps on Weil–Châtelet groups commute with localization, so they induce functorial maps on Sha with $\lambda'_* \circ \lambda_* = [3]$. Thus λ_* maps $\text{Sha}(E_\ell/\mathbb{Q})[3]$ into $\text{Sha}(E'_\ell/\mathbb{Q})[\lambda']$, and the kernel of this restriction is $\text{Sha}(E_\ell/\mathbb{Q})[\lambda]$. Consequently

$$\dim_{\mathbb{F}_3} \text{Sha}(E_\ell/\mathbb{Q})[3] \leq a + b.$$

The full Kummer sequence is

$$0 \longrightarrow E_\ell(\mathbb{Q}) \otimes \mathbb{Q}_3/\mathbb{Z}_3 \longrightarrow \text{Sel}_{3^\infty}(E_\ell/\mathbb{Q}) \longrightarrow \text{Sha}(E_\ell/\mathbb{Q})[3^\infty] \longrightarrow 0.$$

The last group is a cofinitely generated $\mathbb{Z}_3$ -module, as a quotient of the full Selmer group. The structure theorem for its Pontryagin dual gives

$$\text{corank}_{\mathbb{Z}_3} \text{Sha}(E_\ell/\mathbb{Q})[3^\infty] \leq \dim_{\mathbb{F}_3} \text{Sha}(E_\ell/\mathbb{Q})[3].$$

Combining these facts gives the required full-corank bound

$$s_3(E_\ell) = \text{rank}_{\mathbb{Z}} E_\ell(\mathbb{Q}) + \text{corank}_{\mathbb{Z}_3} \text{Sha}(E_\ell/\mathbb{Q})[3^\infty] \leq (1 - a - b) + (a + b) = 1.$$

No finiteness of Sha has been assumed in obtaining this bound.

Now $s_3(E_\ell) \in \{0, 1\}$, so Theorem 1.1 at $p = 3$ gives equality of analytic rank, Mordell–Weil rank, and this corank, as well as finiteness of the entire group $\text{Sha}(E_\ell/\mathbb{Q})$. For the displayed prime classes, the functional-equation sign is -1 by Dasgupta and Voight [32, Section 1.1, equation (1.1.3)]. The functional equation therefore forces $L(E_\ell, 1) = 0$, excluding analytic rank zero. The common rank is consequently one. This step uses no separate parity hypothesis.

A non-torsion rational point on the projective cubic is not the origin. The origin is its only rational point with $Z = 0$, so such a point has $Z \neq 0$. Dividing the cubic equation by Z^3 gives the asserted rational cube-sum representation. $\square$

A Measure comparison and change of characteristic

We give two comparisons for nonstandard weighted orbital integrals, for fixed root data and outside a finite set of residue characteristics. First, we compare the normalized global centralizer factors in the equal-characteristic argument of Halleck-Dubé [38, Theorem 17.7]. The factors agree even though the centralizer tori need only be isogenous. Second, we express the paired local orbital-integral identity as the vanishing of a constructible function, with common parameters and compatible measures. The transfer principle then gives the required mixed-characteristic identity.

A.1 The global centralizer factor under an isogeny

A rational identification of character spaces need not identify the corresponding tori. In the global nonstandard comparison, the factor that is invariant is the quotient of the centralizer constant by the component-group order. The following lemma includes the degree-lattice term needed over a function field.

Lemma A.1 (Comparison of global torus factors). *Let $K = \mathbb{F}_q(X)$, where X is a smooth projective geometrically connected curve of genus g , and let Γ_K be its absolute Galois group. Let T_1, T_2 be K -tori with isomorphic rational character modules as Galois representations, and let $\ell : \text{Lie } T_1 \simeq \text{Lie } T_2$ be a K -linear isomorphism. Choose local Haar measures $\eta_{i,v}$ compatible under ℓ on the Lie algebras, and give the smooth integral compact subgroups volume one at almost all places. Write η_i for the resulting adelic measures, using counting measure on $T_i(K)$.*

For a torus T , put $Y = X^(T)$, $r = \text{rank } Y^{\Gamma_K}$, and*

$$\deg_T(t)(\chi) = \sum_v [k(v) : \mathbb{F}_q] v(\chi(t_v)), \quad I(T) = [\text{Hom}(Y^{\Gamma_K}, \mathbb{Z}) : \deg_T T(\mathbb{A}_K)].$$

Let $T(\mathbb{A}_K)^1 = \ker \deg_T$, and set

$$\text{Sha}^1(K, T) = \ker \left(H^1(K, T) \longrightarrow \prod_v H^1(K_v, T) \right), \quad b(T) = |\pi_0(\widehat{T}^{\Gamma_K})|.$$

If

$$c_\eta(T) = \text{vol}_\eta(T(K) \backslash T(\mathbb{A}_K)^1) |\text{Sha}^1(K, T)|,$$

then

$$\frac{c_{\eta_1}(T_1)/b(T_1)}{c_{\eta_2}(T_2)/b(T_2)} = \frac{I(T_1)}{I(T_2)}. \quad (\text{A.1})$$

In particular the two normalized factors agree if each torus splits at a place of degree one. The places may be different.

Proof. We will express each normalized factor as the same scalar times $I(T_i)$. First, the exponential sequence for the complex dual torus is

$$0 \longrightarrow Y \longrightarrow Y \otimes_{\mathbb{Z}} \mathbb{C} \longrightarrow \hat{T}(\mathbb{C}) \longrightarrow 1.$$

The Galois action factors through a finite group. Taking its invariants, and averaging on the complex vector space, identifies the component group of $\hat{T}^{\Gamma_K}$ with $H^1(K, Y)$. Indeed, the image of the invariant vector space is precisely the identity component of the fixed torus. Thus

$$b(T) = |H^1(K, Y)|. \quad (\text{A.2})$$

We specify the measure comparison. Choose invariant top differential forms ω_i over K with $\ell^* \omega_2 = \omega_1$. Compatibility of the local Haar measures means

$$\eta_{i,v} = a_v |\omega_i|_v$$

with the same positive constant a_v for $i = 1, 2$. Put $d = \dim T_i$. Let $P_v(T, t)$ be the local Artin polynomial and $\rho(T) = \lim_{s \rightarrow 1} (s-1)^r L(T, s)$, in the conventions of Geisser and Suzuki [36, Section 6]. The rational Galois-module isomorphism gives the same d, r, P_v , and ρ for both tori. In these conventions the Tamagawa measure is

$$\mu_i = \frac{1}{\rho q^{(g-1)d}} \prod_v P_v(q_v^{-1})^{-1} |\omega_i|_v, \quad q_v = |k(v)|. \quad (\text{A.3})$$

Consequently $\eta_i = \alpha \mu_i$ with the same scalar

$$\alpha = \rho q^{(g-1)d} \prod_v a_v P_v(q_v^{-1}).$$

At almost all places the forms are integral generators and the tori have good reduction; there the volume of the integral compact under $|\omega_i|_v$ is $P_v(q_v^{-1})$. The displayed product therefore has only finitely many factors different from 1. Replacing a global form by a $K^\times$ -multiple does not change the global measure, by the product formula. This also explains why clearing denominators in a rational torus correspondence adds no unrecorded global scalar.

For clarity, the function-field Tamagawa number includes a degree correction. The definition in Geisser and Suzuki [36, Section 6] and their Proposition 6.3 give

$$\text{vol}_{\mu_i}(T_i(K) \backslash T_i(\mathbb{A}_K)^1) = (\log q)^r I(T_i) \tau(T_i), \quad \tau(T_i) = \frac{|H^1(K, X^*(T_i))|}{|\text{Sha}^1(K, T_i)|}.$$

Here their discriminant $\text{Disc}(h_{T_i})$ is exactly $I(T_i)$: the degree pairing identifies the torsion-free adelic class group with the image lattice of $\deg_{T_i}$ in $\text{Hom}(X^*(T_i)^{\Gamma_K}, \mathbb{Z})$. Together with (A.2), this proves the more explicit formula

$$\frac{c_{\eta_i}(T_i)}{b(T_i)} = \alpha (\log q)^r I(T_i), \quad (\text{A.4})$$

and hence (A.1).

Finally suppose T splits at a degree-one place v_0 . The invariant sublattice Y^{Γ_K} is saturated in Y : if ny is invariant for $n \neq 0$, torsion-freeness of Y makes y invariant. Therefore restriction gives a surjection

$$X_*(T_{K_{v_0}}) = \text{Hom}(Y, \mathbb{Z}) \longrightarrow \text{Hom}(Y^{\Gamma_K}, \mathbb{Z}).$$

For a local cocharacter λ and uniformizer ϖ_{v_0} , the adele equal to $\lambda(\varpi_{v_0})$ at v_0 and to 1 elsewhere has degree $\chi \mapsto \langle \chi, \lambda \rangle$. Thus the degree map is onto and $I(T) = 1$. $\square$

Application to the global nonstandard comparison. In Halleck-Dubé [38, Corollary 17.11, equation (17.2.2)], take $K = F_0$ and let T_i be the generic fiber of the regular centralizer J_{i,a_M} . The rational root-ray correspondence is equivariant for the common cameral monodromy. It gives isomorphic rational character modules; after clearing the fixed denominators it gives a torus isogeny. At the good characteristics under consideration its differential is invertible and induces the Lie-algebra identification used for the local measures, as in Ngô [61, Section 1.12, before Theorem 1.12.7]. There is no requirement that the integral character lattices be isomorphic.

The marked point in the global construction is rational, and the rigidification makes each T_i split there [38, Section 14.1 and the proof of Lemma 15.11]. Hence $I(T_1) = I(T_2) = 1$. Moreover, Chaudouard and Laumon [26, Lemma 12.2.5, pp. 1735–1736] identify the Frobenius-coinvariant kernel used in that draft with $\text{Sha}^1(F_0, T_i)$:

$$\ker(T_i(F)_\tau \longrightarrow T_i(\mathbb{A})_\tau) \simeq \text{Sha}^1(F_0, T_i).$$

Here $F = \overline{\mathbb{F}}_q(X)$, $\mathbb{A}$ is its geometric adèle ring in that construction, and τ is arithmetic Frobenius. This statement concerns coinvariants for the infinite cyclic Weil group; it does not replace them silently by continuous Galois cohomology. Also $\widehat{T}_i^{\Gamma_{F_0}}$ is the fixed dual torus denoted $\widehat{T}_i^{W_a}$ in the trace formula. To see this even when W_a is defined geometrically, use the section of the arithmetic fundamental group supplied by the rational marked point. Its image acts trivially on the character lattice because the torus is split there. The arithmetic and geometric monodromy images on that lattice are consequently equal. The lemma therefore gives the exact cancellation needed in equation (17.2.2):

$$\frac{c(J_{1,a_M})}{|\pi_0(\widehat{T}_1^{W_a})|} = \frac{c(J_{2,a_M})}{|\pi_0(\widehat{T}_2^{W_a})|}. \quad (\text{A.5})$$

The two numerators and the two denominators need not agree separately. Equation (A.5) gives their combined cancellation in place of the generic-torus identifications and separate cancellations on page 169 of that draft.

The remaining coroot covolumes. It remains to identify the factors in its equation (17.2.3) with Waldspurger’s coefficient. This calculation uses the coroot lattices within each group, rather than an identification of integral torus lattices across the pair. Let S_i be the pinned maximal torus of G_i , and let Γ be the based-root-datum monodromy group. Denote the rational root-ray correspondence by ψ , and write

$$\psi_* : X_*(S_1) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\sim} X_*(S_2) \otimes_{\mathbb{Z}} \mathbb{Q}$$

for its Γ -equivariant map on geometric cocharacters. For $H_i = M_i, G_i$, put $\Lambda_{H_i} = \mathbb{Z}\langle \Phi_{H_i}^\vee \rangle$ inside the geometric cocharacter lattice $X_*(S_i)$. The map ψ_* carries $\Lambda_{H_1} \otimes \mathbb{Q}$ onto $\Lambda_{H_2} \otimes \mathbb{Q}$ for $H = M, G$, and therefore induces isomorphisms on the corresponding rational coinvariant quotients. The lattice and real space in Halleck-Dubé [38, Lemma 17.10] are

$$\mathcal{X}_*(H_i) = ((X_*(S_i)/\Lambda_{H_i})_\Gamma)_{\text{tf}}, \quad \mathfrak{a}_{H_i} = \mathcal{X}_*(H_i) \otimes_{\mathbb{Z}} \mathbb{R},$$

where the subscript tf means quotient by torsion. For S_i the same definition uses the zero coroot lattice. Write $\mathfrak{a}_{S_i}^{H_i}$ for the kernel of $\mathfrak{a}_{S_i} \rightarrow \mathfrak{a}_{H_i}$, with its induced measure. We also write ψ_* for the induced real isomorphisms $\mathfrak{a}_{H_1} \simeq \mathfrak{a}_{H_2}$ and $\mathfrak{a}_{S_1}^{H_1} \simeq \mathfrak{a}_{S_2}^{H_2}$. Set

$$V_{H_i} = \text{vol}(\mathfrak{a}_{S_i}^{H_i}/(\Lambda_{H_i})_\Gamma).$$

First suppose $\mathfrak{a}_{G_i} = 0$. Dividing the formula of that lemma for $H_i = G_i$ by the formula for $H_i = M_i$ gives

$$\frac{V_{G_i}}{V_{M_i}} = \frac{|\pi_0(Z_{\widehat{G}_i}^\Gamma)|}{|\pi_0(Z_{\widehat{M}_i}^\Gamma)|} \text{vol}(\mathfrak{a}_{M_i}/\mathcal{X}_*(M_i)).$$

The torus covolume and the torus component factor cancel within each fixed i . Transporting the real measures through ψ_* gives

$$\frac{V_{H_1}}{V_{H_2}} = [(\Lambda_{H_2})_\Gamma : \psi_*((\Lambda_{H_1})_\Gamma)], \quad H = M, G.$$

Here $[A : B] = [A : A \cap B]/[B : A \cap B]$ is the relative index of commensurable lattices: it is the covolume of B divided by that of A in their common real space. Thus the remaining ratio is Waldspurger's coefficient

$$c_\psi = \frac{V_{G_1} V_{M_2}}{V_{G_2} V_{M_1}} = \frac{[(\Lambda_{G_2})^\Gamma : \psi_*((\Lambda_{G_1})^\Gamma)]}{[(\Lambda_{M_2})^\Gamma : \psi_*((\Lambda_{M_1})^\Gamma)]}.$$

The simple coroots form permutation bases for Γ , and ψ_* scales their rays by scalars constant on each Γ -orbit. It acts with the same scalars on orbit sums in the invariants and orbit classes in the coinvariants. The relative indices therefore agree, as in the proof of Halleck-Dubé [38, Corollary 17.11]. For general G_i , that proof reduces to this case by quotienting by the maximal split central tori. Its reduction, using Theorem 15.16, preserves the weighted orbital integrals, the stable recursions, and the coroot-lattice coefficient.

This proves the required global centralizer normalization. The geometric inputs remain the isogeny statement for neutral Picard components and the cohomological comparison on the anisotropic locus [61, Proposition 4.18.1 and Theorem 8.8.2], together with the support statement used in the draft. The split rational marked point identifies arithmetic and geometric monodromy [38, Proposition 2.21 and Corollary 2.22]; thus, after the split-center reduction, the center satisfies the geometric anisotropy hypothesis of that comparison. The calculation does not assert a generic isomorphism $J_1^0 \simeq J_2^0$ or constitute a new proof of those geometric results.

A.2 Change of characteristic

We now transfer the local nonstandard identity, with the compatible normalizations just described. The uniformity is over a fixed finite collection of root data and all its integral generically regular matching parameters.

Lemma A.2 (Change of characteristic for fixed nonstandard data). *Fix a finite collection of paired based root data with rational root-ray maps ψ , matching Levi data, and compatible finite cyclic outer actions. Suppose the nonstandard weighted fundamental lemma holds over local fields of sufficiently large positive characteristic for their unramified forms, including all cyclic-generator variants. Then it holds over characteristic-zero local fields of sufficiently large residue characteristic, uniformly for this collection and for all integral generically regular matching parameters.*

Here the ambient hyperspecial subgroups have volume one; the Arthur weight-space measures are transported through ψ ; and the paired regular-centralizer measures are transported through the induced Lie-algebra isomorphism.

Proof. We apply the general transfer principle for parameterized integrals [29, Theorems 2.8.1–2.8.3]. To do so, we construct a common parameter space for the paired integrals, express their measures by differential forms, and remove the volume and discriminant normalizations. The resulting identity will be the vanishing of one constructible function.

Common parameters and the identity. Choose an integer N excluding the bad primes for the finitely many root data, Chevalley and Kostant constructions, and outer actions, and making ψ and its inverse integral over $R = \mathbb{Z}[1/N]$. Include the ordinary endoscopic descendants in this finite list. Write G_i and M_i for the groups and Levi subgroups attached to the two paired data, and $\mathfrak{t}_i$ for their chosen Cartan Lie algebras. The Weyl-equivariant Cartan identifications of Ngô [61, Sections 1.12.4–1.12.6] give

$$C_M = \mathfrak{t}_1/W_{M_1} \simeq \mathfrak{t}_2/W_{M_2}, \quad C_G = \mathfrak{t}_1/W_{G_1} \simeq \mathfrak{t}_2/W_{G_2}$$

over R , compatibly with $C_M \rightarrow C_G$ and with the common outer action. They identify the integral models and the generically G -regular loci. Consequently the same statements hold after a common unramified outer twist.

For a local field F and an integral generically regular parameter b in this common base, let $Y_{i,b}$ be its Kostant lift to $\mathrm{Lie}(M_i)$. Write $S_{M_i}^{G_i}(b)$ for the stable weighted orbital integral at $Y_{i,b}$ with test function $1_{\mathrm{Lie}(G_i)(\mathcal{O}_F)}$. With the measures in the statement, the identity to be transferred is

$$S_{M_1}^{G_1}(b) = c_\psi S_{M_2}^{G_2}(b), \tag{A.6}$$

in the convention of Halleck-Dubé [38, Definition 16.4 and Theorem 17.7]. The coroot-lattice coefficient c_ψ is fixed on each cyclic-action stratum. We next parameterize the unramified splitting extensions and outer twists so that both sides of (A.6) are defined on the same space.

Encode an unramified splitting extension of fixed degree d by a monic polynomial with integral coefficients and irreducible separable reduction, together with the matrix of an algebra automorphism τ of exact order d . All these conditions are first-order conditions of fixed degree, and every unramified extension has such a presentation. We use the general coefficient encoding of Cluckers et al. [29, Section 3.2]; restriction to binomials is unnecessary. Use the same extension and generator on both sides. The resulting common definable space is

$$\Lambda_F = \{(\mathbf{a}, \tau, b) : b \in C_{M, \mathbf{a}, \tau}(\mathcal{O}_F), \quad b \text{ is generically } G\text{-regular}\}.$$

The subscript denotes the descended integral model: in extension coordinates, its points satisfy the common semilinear fixed-point equations. The map ψ intertwines those equations. Thus every required matching pair occurs over Λ_F . Varying τ only adds the specified cyclic-generator variants.

Compatible measures. Over this regular locus, the centralizers J_i in M_i are the twists of the pinned tori along the same cameral Weyl torsor. Multiplying ψ by an integer produces an equivariant torus isogeny. Descending it and dividing its differential by that integer gives an algebraic isomorphism

$$\lambda : \mathrm{Lie}(J_1) \simeq \mathrm{Lie}(J_2).$$

Enlarge Λ by a nonzero top covector ω_1 on $\mathrm{Lie}(J_1)$, and set $\omega_2 = (\lambda^{-1})^*\omega_1$. Their invariant extensions give compatible Haar measures, as in Ngô [61, Section 1.12, before Theorem 1.12.7]. Every field-valued parameter has such a frame. Any other compatible pair differs by a common positive scalar, which rescales both sides of the desired identity by its inverse.

For every ambient group H in either finite stable recursion, also include a nonzero invariant top-form frame η_H . On a regular stable orbit X_b in $\mathrm{Lie}(H)$, the exact sequence

$$0 \longrightarrow \mathrm{Lie}(J_b) \longrightarrow \mathrm{Lie}(H) \xrightarrow{Z \mapsto [Z, Y]} T_Y X_b \longrightarrow 0$$

gives the quotient orbital form from η_H and ω_J . On finite smooth charts its comparison with the motivic measure is the absolute value of a rational determinant, hence a constructible factor

$\mathbb{L}^{-\text{ord}(r)}$. This is the differential-form realization of quotient measures used in Cluckers et al. [29, Sections 7 and 8.6]. If K_H is the hyperspecial subgroup, put

$$V_H = \int_{K_H} |\eta_H| > 0.$$

This is a constructible integral. The normalized ambient measure is $|\eta_H|/V_H$; below we clear the finitely many factors V_H .

Discriminants and stable recursion. We must also account for the discriminant normalization $|D_H(Y)|^{1/2}$ in the weighted orbital integrals. In either stable recursion, write M for its fixed Levi. Every ordinary endoscopic descendant has this same actual F -Levi. Choose an F -parabolic $P_H = MU_H$ and set

$$R_{H,M}(Y) = \det(\text{ad}(Y) | \text{Lie}(U_H)).$$

Opposite roots give

$$D_H(Y) = (-1)^{\dim U_H} D_M(Y) R_{H,M}(Y)^2.$$

Thus division of the specialized stable recursion by $|D_M(Y)|^{1/2}$ replaces every discriminant factor by $|R_{H,M}(Y)|$, an integral valuation power of the residue cardinality. Under the pairing, the two Levi discriminants differ by a fixed rational scalar u_ψ . Enlarge N by its numerator and denominator. Then $|u_\psi| = 1$, so the two removed Levi-discriminant factors agree at every matched parameter.

The ordinary recursion is

$$S_M^H = J_{M,M}^H - \sum_{H' \neq H} \iota_M(H, H') S_M^{H'}.$$

Its index sets are finite and its rational coefficients depend only on the fixed root data and cyclic action; each H' has fewer roots and the same Levi M [29, Section 9.1]. The weighted integrands and hyperspecial-lattice indicators are the definable, relatively integrable families constructed in Cluckers et al. [29, Sections 6 and 8.6]. Use their integer-valuation weights with the fixed compatible weight-space volumes; real constants are permitted by Section 2.10 of that reference. The form comparisons and resultant factors above preserve constructibility and relative integrability. Indeed, the intrinsic quotient form and the Lie-algebra/Chevalley quotient form are invariant on each geometrically transitive regular stable orbit. Their ratio comes from the base, including the frames, and can be evaluated at the Kostant lift. Its absolute value is a constructible valuation power, so the projection formula preserves the relative integrability of the CHL families.

Define the modified functions by this recursion using $|R_{H,M}|$ times the quotient orbital integrals. Induction shows that their specializations equal $S_M^H/|D_M|^{1/2}$. No division by a square root in the motivic ring is being made. With this common discriminant factor removed, expand the finite recursions and multiply (A.6) by the product of the ambient volumes V_H , with multiplicities if necessary. The result is the vanishing of one constructible function Φ on the enlarged common parameter space. Each specialized multiplier is positive.

Transfer and recovery of the normalized identity. By hypothesis $\Phi_{k((t))}$ vanishes identically for every finite residue field k of sufficiently large characteristic, including all permitted generator variants. The abstract transfer principle therefore makes Φ_F identically zero for every sufficiently large-residue-characteristic local field F with residue field k , in particular every characteristic-zero such F . Undo the positive ambient-volume multiplier and the common nonzero Levi-discriminant factor. The surjectivity of the extension and frame parameterizations yields the required identity for every matching parameter. Taking the maximum of the finitely many bounds proves the assertion. $\square$

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